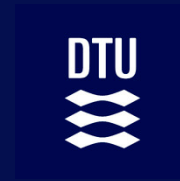
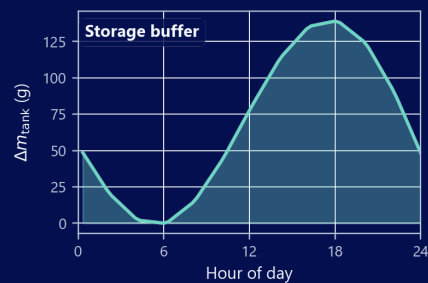
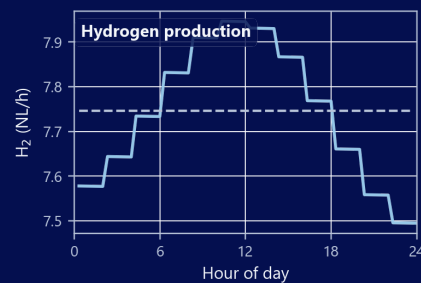
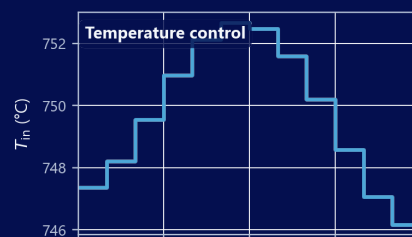
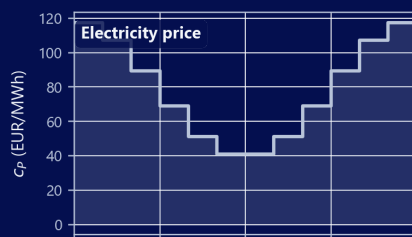


# Optimal Control of SOEC's and sensitivity towards degradation



Analysis of the optimal control of SOEC's with physically modelled degradation in a one-dimensional flow model.



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Master Thesis

July 14, 2026

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# List of Symbols

| Symbol                                    | Unit                                                                              | Description                                                  |
|-------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------|
| <i>Geometry, discretization, and time</i> |                                                                                   |                                                              |
| $A_{\text{cell}}$                         | $\text{m}^2$                                                                      | Active electrode area                                        |
| $A_{\text{int}}$                          | $\text{m}^2$                                                                      | Flow-channel cross-sectional area                            |
| $d$                                       | $\text{m}$                                                                        | Cell thickness (channel height)                              |
| $L$                                       | $\text{m}$                                                                        | Channel length                                               |
| $w$                                       | $\text{m}$                                                                        | Channel width                                                |
| $z$                                       | $\text{m}$                                                                        | Axial coordinate along the channel                           |
| $N$                                       | —                                                                                 | Number of axial channel segments                             |
| $N_{\text{cell}}$                         | —                                                                                 | Number of FVM control volumes along $z$                      |
| $N_t$                                     | —                                                                                 | Number of control / time segments in the NLP                 |
| $\Delta z$                                | $\text{m}$                                                                        | FVM mesh spacing                                             |
| $z_k$                                     | $\text{m}$                                                                        | Centre of FVM cell $k$                                       |
| $\Omega_k, V_k$                           | $\text{m}^3$                                                                      | FVM cell $k$ and its volume                                  |
| $t$                                       | $\text{s}$                                                                        | Time                                                         |
| $T$                                       | $\text{s}$ or days                                                                | Optimization horizon                                         |
| $\Delta t_k$                              | $\text{s}$                                                                        | Duration of control interval $k$                             |
| <i>Species, flow, and thermodynamics</i>  |                                                                                   |                                                              |
| $\mathcal{S}$                             | —                                                                                 | Gas species $\{\text{H}_2\text{O}, \text{H}_2, \text{O}_2\}$ |
| $\mathcal{S}_\zeta$                       | —                                                                                 | Species on side $\zeta \in \{a, f\}$ (anode / fuel)          |
| $\mathcal{M}$                             | —                                                                                 | Species index set for mass fractions $y_k$                   |
| $c_i, \mu_i$                              | $\text{mol}/\text{m}^3$                                                           | Molar concentration; $\mu_i$ along channel (state)           |
| $x_i, y_k$                                | —                                                                                 | Molar / mass fraction of species $i$ or $k$                  |
| $N_i, N_{c,i}$                            | $\text{mol}/(\text{m}^2 \cdot \text{s})$                                          | Molar flux; electrochemical source flux                      |
| $\dot{n}_i, \dot{V}_\zeta$                | $\text{mol}/\text{s}, \text{m}^3/\text{s}$                                        | Molar flow rate; volumetric flow on side $\zeta$             |
| $v_{\text{fuel}}, v_{\text{air}}$         | $\text{m}/\text{s}$                                                               | Gas velocity on fuel / air side                              |
| $p_i, P$                                  | $\text{Pa}$                                                                       | Partial / total pressure                                     |
| $p_{\text{ref}}, p_{\text{air}}$          | $\text{Pa}$                                                                       | Reference and air-side pressures                             |
| $\rho$                                    | $\text{kg}/\text{m}^3$                                                            | Gas density                                                  |
| $c_{p,i}, c_p, C_p$                       | $\text{J}/(\text{mol} \cdot \text{K}),$<br>$\text{J}/(\text{K} \cdot \text{m}^3)$ | Species, mixture, and volumetric heat capacity               |
| $h_i(T), \Delta h$                        | $\text{J}/\text{mol}$                                                             | Species enthalpy; reaction enthalpy change                   |
| $Q_T$                                     | $\text{W}/\text{m}^3$                                                             | Volumetric heat source                                       |
| <i>Electrochemistry</i>                   |                                                                                   |                                                              |
| $F$                                       | $\text{C}/\text{mol}$                                                             | Faraday constant                                             |
| $n_e$                                     | —                                                                                 | Electrons per reaction ( $n_e = 2$ )                         |
| $\nu_i$                                   | —                                                                                 | Stoichiometric coefficient of species $i$                    |
| $i$                                       | $\text{A}/\text{m}^2$                                                             | Current density                                              |
| $I, I_{\text{cell}}$                      | $\text{A}$                                                                        | Total cell current                                           |
| $U$                                       | $\text{V}$                                                                        | Applied cell voltage (control)                               |
| $U_{\text{th}}$                           | $\text{V}$                                                                        | Thermal (reversible) cell voltage                            |

| Symbol                                                                                    | Unit                               | Description                                                                   |
|-------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------|
| $\eta_{\text{cell}}$                                                                      | V                                  | Cell overpotential ( $\eta_{\text{cell}} = U$ )                               |
| $\eta_{\text{Nernst}}, \eta_{\text{act}}, \eta_{\text{ohm}}$                              | V                                  | Nernst, activation, and ohmic overpotentials                                  |
| $\eta_{\text{act}}^{\zeta}$                                                               | V                                  | Activation overpotential on side $\zeta$                                      |
| $i_0^{\zeta}$                                                                             | A/m <sup>2</sup>                   | Exchange current density on side $\zeta$                                      |
| $R_{\text{deg}}, R_{\text{ohm}}$                                                          | $\Omega \cdot \text{m}^2$          | Degradation and total ohmic resistance                                        |
| $R_{\text{contact}}$                                                                      | $\Omega \cdot \text{m}^2$          | Contact resistance                                                            |
| $B_{\text{ohm}}$                                                                          | $\Omega \cdot \text{m}^2/\text{K}$ | Base factor in intrinsic ohmic term                                           |
| $E_{\text{act}}^{\text{ohm}}$                                                             | J/mol                              | Activation energy in intrinsic ohmic term                                     |
| $\Delta G$                                                                                | J/mol                              | Gibbs free energy change                                                      |
| <i>Degradation</i>                                                                        |                                    |                                                                               |
| $\mathcal{D}$                                                                             | —                                  | Set of slow degradation mechanisms                                            |
| $\mathcal{D}_{\text{ohm}}$                                                                | —                                  | Subset in $R_{\text{ohm}}$ (Ni-mi, IC-ox)                                     |
| $\mathbf{d}, d_i$                                                                         | —                                  | Degradation state vector and component $i$                                    |
| $h_{\text{Ni-mi}}, h_{\text{IC-ox}}$                                                      | m                                  | Ni-migration gap; interconnect oxide thickness                                |
| $s_{\text{IC-ox}}$                                                                        | m <sup>2</sup>                     | Squared IC-oxide thickness (ODE state)                                        |
| $d_{i,\text{critical}}$                                                                   | m                                  | End-of-life limit for mechanism $i$                                           |
| $R_{\text{Ni-mi}}, R_{\text{IC-ox}}$                                                      | $\Omega \cdot \text{m}^2$          | Ohmic contributions from Ni-mi and IC-ox                                      |
| $\sigma_{\text{elec}}, \sigma_{\text{elec},0}$                                            | S/m                                | Electrode conductivity and pre-factor                                         |
| $\sigma_{\text{IC-ox}}$                                                                   | S/m                                | Oxide-layer conductivity                                                      |
| $E_{a,\text{elec}}, E_{a,\text{ox}}, E_{\text{el}}$                                       | J/mol                              | Activation energies (electrode, IC ox., IC cond.)                             |
| $c_1, \dots, c_4$                                                                         | various                            | Fitted Ni-mi growth-rate coefficients                                         |
| $k_{\text{mg}}, k_{\text{ox}}$                                                            | various                            | IC-oxidation rate and conductivity prefactors                                 |
| $\varepsilon, \tau$                                                                       | —                                  | Electrode porosity and tortuosity                                             |
| <i>Optimal control and economics</i>                                                      |                                    |                                                                               |
| $\mathbf{u}(t), \mathcal{U}_{\text{ad}}$                                                  | —                                  | Control vector ( $U, T_{\text{in}}$ ); admissible (constrained) control space |
| $T_{\text{in}}, T_{\text{out}}, T_{\text{loss}}$                                          | K                                  | Inlet (control), outlet, and thermal recovery loss                            |
| $R(\mathbf{u}, \mathbf{d}), R_{\text{tank}}$                                              | EUR/s                              | Revenue rate; contract rate with tank                                         |
| $P_{\text{cell}}, P_{\text{heat}}$                                                        | W                                  | Stack power; auxiliary sensible heating                                       |
| $\varphi, \Psi(\mathbf{d}), \Psi_{\text{tank}}$                                           | EUR                                | Integrated revenue; salvage; tank liquidation                                 |
| $c_{\text{H}_2}, c_P, c_{P,k}$                                                            | EUR/kg, EUR/kWh                    | Hydrogen and power prices                                                     |
| $\bar{c}_P, a$                                                                            | —                                  | Mean daily power price and relative swing                                     |
| $c_{\text{cell}}$                                                                         | EUR/m <sup>2</sup>                 | Cell capital cost in $\Psi$                                                   |
| $\alpha(t), \alpha_{\Psi}$                                                                | —                                  | Schematic weight; salvage exponent                                            |
| $\kappa_{\text{buy}}, \kappa_{\text{sell}}$                                               | —                                  | Spot-purchase penalty; liquidation multiplier                                 |
| $m_{\text{tank}}, n_{\text{purchase},k}$                                                  | mol, kg                            | Tank inventory; shortfall purchase                                            |
| $\dot{n}_{\text{H}_2,\text{agreement}}, \dot{n}_{\text{prod}}, \dot{n}_{\text{purchase}}$ | mol/s                              | Delivery, production, and purchase rates                                      |
| $\Delta \dot{n}_{\text{H}_2}$                                                             | mol/s                              | Net H <sub>2</sub> production in $R$                                          |
| <i>PDE / FVM notation</i>                                                                 |                                    |                                                                               |
| $u, Q_u, F_u$                                                                             | various                            | Generic FVM field; source and flux terms                                      |
| $m_k$                                                                                     | kg                                 | Gas mass in FVM volume $k$                                                    |
| <i>Greek symbols</i>                                                                      |                                    |                                                                               |
| $R$                                                                                       | J/(mol·K)                          | Universal gas constant                                                        |
| $\alpha_{\zeta}$                                                                          | —                                  | Butler–Volmer symmetry factor; $\alpha_a = 0.65, \alpha_f = 0.59$             |

| Symbol                 | Unit             | Description                                                                                                              |
|------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------|
| $\gamma_\zeta$         | A/m <sup>2</sup> | Exchange-current pre-factor; $\gamma_a = 1.515 \times 10^8$ ,<br>$\gamma_f = 7.666 \times 10^5$                          |
| $E_{\text{act}}^\zeta$ | J/mol            | Electrode activation energy; $E_{\text{act}}^a = 1.40 \times 10^5$ ,<br>$E_{\text{act}}^f = 1.05 \times 10^5$            |
| $m_i$                  | —                | Species exponent in $i_0^\zeta$ ; $m_{\text{H}_2\text{O}} = 0.33$ , $m_{\text{H}_2} = -0.1$ ,<br>$m_{\text{O}_2} = 0.22$ |
| $\lambda_a, \lambda_c$ | —                | Exchange-current multipliers ( $\lambda_a = 1$ ; $\lambda_c \approx 0.55$ )                                              |
| $\lambda_T$            | W/(m·K)          | Thermal conductivity (neglected)                                                                                         |
| $\Omega$               | m                | Spatial domain                                                                                                           |
| $x_0$                  | —                | Reference molar-fraction scale ( $p_{\text{ref}}/p_{\text{air}}$ )                                                       |

*Greek parameter values follow table 3.1 and table 3.2 unless stated otherwise.*

# 1 Abstract

This thesis studies the sensitivity of optimal control for a solid oxide electrolysis cell (SOEC) over its operating lifetime, with degradation included as an economic parameter. A physics-based unit-cell model couples steady-state electrochemical and thermal fields along the gas channels (solved with a finite volume method) to differential degradation laws for nickel migration and interconnect oxidation. The model uses implicit Euler for forwards time integration. In addition to this, an optimal control problem is formulated to maximize integrated revenue (hydrogen sales minus electricity cost) plus a terminal *salvage* term penalizing degradation. Cell voltage  $U$  and inlet temperature  $T_{\text{in}}$  are the primary controls; the resulting nonlinear program is solved by gradient-based optimization (SLSQP/Trust-Constr) using single-shooting forward replay.

Three scenarios examine how much temporal control freedom matters and what drives the optimum. In an unrestricted 200-day baseline, the optimizer selects constant potentiostatic-isothermal (pot-iso) operation at an exothermal set-point ( $U^* \approx 1.32$  V,  $T_{\text{in}}^* \approx 751$  °C); granting additional segment-wise freedom in  $U$  and  $T_{\text{in}}$  does not change the optimal trajectory. Nickel migration reaches the breaking condition at the horizon end, breaking the cell, while interconnect oxidation remains secondary. When a fixed hydrogen delivery contract and buffer tank are introduced, inventory-aware operation becomes relevant: The optimizer alternates gentle pot-iso periods with galvanostatic intervals dictated by tank inventory and penalty prices, at the cost of lower average net economy. A final 24-hour case with a cyclic power price and cyclic tank constraint shifts production toward low-price midday intervals mainly by modulating  $T_{\text{in}}$  while voltage stays essentially fixed.

Current-voltage validation, comparison of degradation trajectories with published reference simulations, and gradient checks support using the toolchain for qualitative optimal-control studies at engineering fidelity. The main conclusion is that optimal SOEC operation is scenario-dependent: unrestricted operation favors steady potentiostatic-isothermal controls at exothermal voltage; a fixed delivery contract introduces tank-mediated galvanostatic intervals, with thermoneutral; and a cyclic power price is met mainly by modulating inlet temperature while voltage stays near the thermoneutral value.

## 2 Introduction

Most hydrogen produced today is derived from fossil fuels. This so-called grey or black hydrogen remains the cheapest route in direct production cost, but carries a large climate penalty through fossil carbon emissions [6]. Solid oxide electrolysis (SOE) offers an alternative: High-temperature electrolysis in a solid oxide electrolysis cell (SOEC) can convert steam into hydrogen using electricity with high efficiency [10]; when the electricity is low-carbon, the product qualifies as green hydrogen. The technology is not new, but capital and operating costs have kept it less competitive than fossil routes on open markets. Substantial effort is therefore directed at lowering the levelized cost of hydrogen through improved materials, scale-up, and smarter operation [4, 13].

A practical consequence is that many early green-hydrogen projects are structured around long-term delivery contracts between buyers and producers rather than spot sales alone. This setting matters for operation: The cell must meet an agreed hydrogen flow while electricity prices, inventory, and stack health evolve over time. Degradation is central to that trade-off. Prolonged high-temperature operation increases area-specific resistance, raises power demand for the same output, and eventually forces stack replacement [3, 5]. Recent work has started to couple degradation models with economic objectives and optimal control. At unit-cell scale, it remains open how much temporal freedom in voltage and temperature actually changes the optimum, and how contractual delivery or time-varying prices reshape it.

### 2.1 Literature review

Preceding this thesis, degradation-aware modelling and economic optimization of SOEC operation have been studied mainly at stack or plant scale; the main results are summarized below.

Beyrami et al. [5] find that galvanostatic operation turns exothermic as resistance grows, potentiostatic operation stays nearer thermoneutral, and potentiostatic-galvanostatic modes limit area-specific resistance increase over long horizons. In follow-on work they report more than 50,000 h lifetime at  $0.6 \text{ A/cm}^2$  when optimizing against component failure criteria [3]. At plant scale, degradation accounts for roughly 8% of hydrogen production cost, with economically optimal current densities of  $0.6\text{-}0.7 \text{ A/cm}^2$  and 4-5 year stack replacement [4].

Giridhar et al. [13] show that dynamic optimisation over extended horizons lowers levelized cost of hydrogen relative to single-objective strategies. Flexible schedules outperform fixed galvanostatic or potentiostatic operation, with replacement intervals of roughly 2-5 years depending on electricity price and mode. Cai et al. [8] maximize hydrogen output or minimize energy use under temperature constraints when pairing SOEC operation with intermittent renewables.

For model calibration, Rizvandi et al. [23] provide commercial-cell polarization data; Larrain [16] and Brylewski et al. [7] quantify interconnect oxidation kinetics; Skafte et al. [24] define electrothermally balanced reference conditions (figure 4.3, figure 4.5).

### 2.2 Introduction to the thesis

This thesis examines how optimal operating conditions for a solid oxide electrolysis cell (SOEC) change when cell voltage and inlet temperature are free to vary independently over time, a fixed hydrogen delivery contract

applies, and electricity prices vary in time. Unlike Giridhar et al. [13], who optimize stack-scale schedules by switching between galvanostatic and potentiostatic modes for levelized hydrogen cost, the present formulation allows joint voltage–temperature control at unit-cell scale. It further resolves nickel migration and interconnect oxidation as slow states with break limits, and adds buffer storage to meet offtake obligations. The work is organized around the following question:

*How does physically modelled degradation influence the optimal control of an SOEC when electrochemistry and temperature of the cell are captured in a one-dimensional flow model? To what extent do external operating constraints and time-varying inputs reshape that optimum?*

The analysis is carried out at unit-cell scale. A steady-state electrochemical and thermal model along the gas channel, largely following the formulation of Beyrami et al. [5], is solved with a finite volume method and coupled to slow degradation states for nickel migration and interconnect oxidation.

The primary controls are cell voltage  $U$  and inlet temperature  $T_{\text{in}}$ . The central question is whether the optimizer uses both degrees of freedom, or whether a restricted potentiostatic-isothermal mode remains optimal. Three scenarios are studied:

1. **Unrestricted baseline:** 200-day horizon without delivery or price constraints.
2. **Constant delivery contract:** fixed hydrogen offtake with buffer storage.
3. **Daily cyclic operation:** 24-hour horizon with a time-varying power-price profile.

The governing physics and degradation models are presented in table 3.1 and section 3.2.1. The optimal-control formulation is developed in section 4.1.1 (equation (4.1)). The numerical results are reported in section 5.1 to section 5.4.

The discussion in section 6.1 to section 6.7 interprets how much control freedom is actually used, whether the implied lifetimes are plausible, and which modelling choices bound the conclusions. Together, the thesis provides a cell-scale toolchain for comparing baseline, delivery-constrained, and price-driven operation under a common degradation-aware objective.

# 3 Physical Modelling

## 3.1 Modelling governing SOEC equations

### 3.1.1 Introduction

In this chapter the aspects of solid oxide electrolysis cell's (SEOC's) will be introduced. Briefly visiting different scales of operation, before a more detailed description of how the mole flow is modelled, completing the section with energy balances that determines the behavior of temperature flow.

The basics are presented with the assumption that the readers have a background in applied mathematics. Derivations throughout the chapter therefore assume familiarity with standard intermediate steps; lengthier or less routine derivations are given in the main text or in an appendix. Readers who wish to proceed directly to the governing equations will find them summarized in table 3.1.

The three different scales of SOEC discussed in this paper are defined as follows. The first and smallest scale is the unit cell, where the focus is only on the dynamics occurring within a single SOEC cell. The major focus of this thesis, will be at this scale.

Next scale is the stack. At this level, multiple unit cells are stacked together, which naturally introduces more complexity. Figure 3.1 is a quick visualization of the different scales.

Finally, the largest scale considered is at facility scale. At this level, multiple SOEC stacks are connected together; the facility operates much like an electrical circuit, with each stack behaving as its own component with distinct characteristics.

In this thesis, every stack is taken to be the same and every cell within every stack is identical. The full facility is therefore not modelled stack by stack; instead, one cell is studied as an **average cell** that stands in for any cell in the system (figure 3.1, left). If the plant contains, for example, 5,000 such cells, a facility-scale quantity is obtained by multiplying the per-cell result by 5,000. This chapter develops that single-cell formulation step by step, from channel-scale transport and electrochemistry to the energy balance that sets the temperature field.

### 3.1.2 Basics of SOEC

**Modelling scope.** The physical system represented in the model is summarized in section 3.1.1; the notation *unit cell* below refers to the channel geometry of that average cell model.

The cell can, in simple terms, be described by figure 3.2, which shows the processes at the smallest scale [25].

The cell is composed of three main components: The **cathode**, the **electrolyte**, and the **anode**. At the cathode, water molecules are split into dihydrogen ( $H_2$ ) and oxide ions ( $O^{--}$ ). This process is only possible

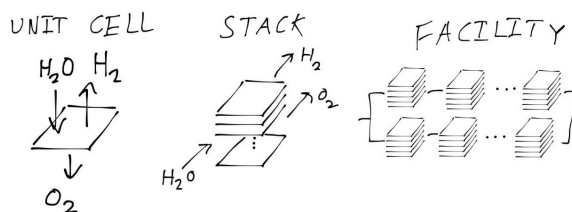


Figure 3.1: A sketch of the scales: The three different scales mentioned in this paper from smallest (left) to largest (right)

due to the presence of electrons provided by an external electric current: These electrons participate directly in the reaction at the cathode, facilitating the reduction of water.

The newly-formed oxide ions then migrate through the solid **electrolyte**, propelled by the electric field generated by the applied voltage, highlighting the central role of electrons and electrical energy in driving the overall process. Once the oxide ions reach the **anode**, they release their excess electrons and recombine to form dioxygen ( $O_2$ ). The electrons released travel through the external circuit, closing the electrical loop.

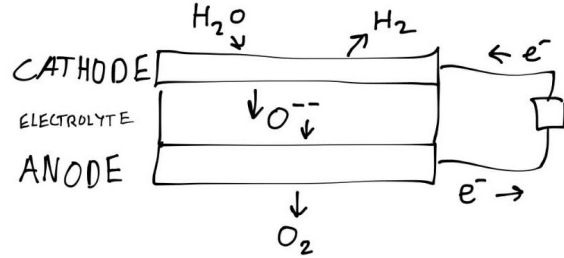


Figure 3.2: A sketch of a unit cell: The basic principles of an SOEC cell are illustrated. The cathode pulls apart the hydrogen and oxygen in the water. The oxygen is transferred through the electrolyte, then formed to dioxygen in the anode, all due to an applied current.

Summing up, the operation of the SOEC is fundamentally tied to the movement of electrons, both in the chemical reactions at the electrodes and in sustaining the ionic and electronic currents through the device. This is captured in the overall reaction,  $2H_2O + 4e^- \rightarrow 2H_2 + 2O^{--} \rightarrow 2H_2 + O_2 + 4e^-$ , where the electrons facilitate water splitting at the cathode and are recovered at the anode. For modeling purposes, and since the focus will generally not be on the internal mechanisms, the model simplifies to the absolute global equation (eq. 3.1).



To clarify the terminology used in SOEC modeling: The upper side of the cell, where water is introduced and dihydrogen ( $H_2$ ) is produced, is referred to as the **fuel side**. This name reflects the role of water as the “fuel,” acting as the reactant that is split during electrolysis. The opposite side—where dioxygen ( $O_2$ ) is evolved, is known as the **air side**, as atmospheric air is typically supplied here. The designations “fuel side” and “air side” are standard throughout the SOEC literature [5, 21] and are used to distinguish these two main compartments of the cell.

### The chemical flow in a cell

The equation in 3.1 treats the reaction as strictly one-way, but in reality, some produced hydrogen and oxygen can recombine to form water again, especially as the cell approaches equilibrium based on its electrical state. This reverse reaction is rare at the start but becomes more significant near equilibrium. Yielding equation 3.2.



Here,  $p_l$  stands for the transition probability that two hydrogen and one oxygen molecule will combine and turn back into water (moving from right state  $\rightarrow$  left state), while  $p_r$  is the probability for splitting water into hydrogen and oxygen (left state  $\rightarrow$  right state).



From these probabilities, the production rate of  $\text{H}_2\text{O}$  at every point in the cell can be determined (assuming concentrations and probabilities are known everywhere). The difference between how much  $\text{H}_2\text{O}$  is produced versus how much is consumed is taken. Keeping in mind that this is calculated per unit volume of the cell, it has the unit  $\frac{\text{mol}}{\text{s}\cdot\text{m}^3}$ :

$$R_{\text{H}_2\text{O}} = p_r c_{\text{H}_2\text{O}}^2 - p_l c_{\text{H}_2}^2 c_{\text{O}_2}$$

Where  $c_x$  is the concentration ( $\frac{\text{mol}}{\text{s}\cdot\text{m}^3}$ ) of molecule  $x$

To obtain the production rate for the whole cell (the global rate), this quantity is integrated over the thickness (height) of the cell. Which gives the total rate  $N_{c,\text{H}_2\text{O}}$  (later this will be treated as a local source, but for now it is kept global):

$$N_{c,\text{H}_2\text{O}} = \int_0^d R_{\text{H}_2\text{O}}$$

This gives the average production rate per area of the cell, which is a useful parameter to work with. But realistically, calculating this from scratch could get a bit complicated. Thankfully, Faraday already did the heavy lifting here. By Faraday's law [25], the rate per cell area can be written as in equation 3.3:

$$N_{c,\text{H}_2\text{O}} = \frac{i}{2F} \quad (3.3)$$

Where  $i$  is the current,  $F$  is Faraday's constant, and the 2 comes from the number of electrons involved in the chemical equation.

For now, equation 3.3 is kept as it is, since the formulation is still at the global level; hence the current in the equation is the global current. Later, when making all this local, the current must be computed using the Nernst equation. More about that later. First, a form that describes the cell locally is introduced.

### Behavior along the cell channels

To describe the cell locally, figure 3.3 is used. The fuel side is modeled as a channel (the upper section in 3.3), where the steam ( $\text{H}_2\text{O}$ ) flows over the cathode. The width of the channel is  $w$  and the length is  $L$ , with the channel extending from the inlet at  $z = 0$  to the outlet at  $z = L$  (where  $L$  is the length of the cell). The gas flows through a constant cross-sectional area  $A_{int}$ .

For any control volume within this channel, mass conservation dictates that the accumulation of water over time equals the flow in plus the production of the cell reaction minus the flow out. Following the formulation by Primdahl and Mogensen [21] for a continuously stirred tank reactor (CSTR), the mass balance for the water molar fraction  $x_{\text{H}_2\text{O}}$  is:

$$\frac{PV}{RT} \frac{\partial x_{\text{H}_2\text{O}}}{\partial t} = A_{int} N_{in,\text{H}_2\text{O}} - A_{int} N_{out,\text{H}_2\text{O}} + A_{cell} N_{c,\text{H}_2\text{O}} \quad (3.4)$$

Where  $V$  is the volume of the gas in the reactor,  $N_i$  is the molar flux for species  $i$ , and  $A_{cell}$  is the active electrode area.

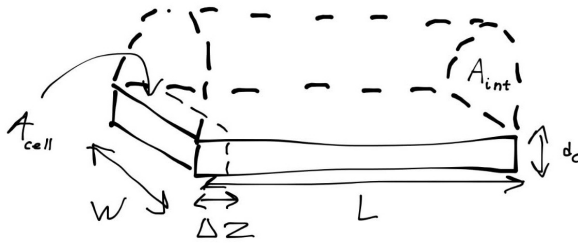


Figure 3.3: Schematic illustration of channel discretization for modeling local behavior along the fuel side of a SOEC. The channel is divided into  $N$  segments of length  $\Delta z$ , showing the flow of steam over the cathode.

To obtain the local behavior along the channel, discretization along the channel length is required, meaning the length is split into sections. This is also illustrated in figure 3.3. Hence  $z_i - z_{i-1} = \Delta z$  for all  $i = 1, \dots, N$ , with  $z_0 = 0$  and  $z_N = L$ , giving a sequence:

$$z_0 < z_1 < \dots < z_N$$

Each section  $z \in (z_{i-1}, z_i)$  can then be viewed as its own cell, where the inlet flow at  $z_{i-1}$  is the outlet flow of the previous interval. Letting  $\Delta z \rightarrow 0$  and inserting the definition of  $N_{c,H_2O}$  (equation

3.3), and the PDE in equation 3.5 pops right out. (Appendix 1)

$$\frac{\partial \mu_{H_2O}}{\partial t} = \frac{w}{A_{int}} N_{c,H_2O} - \frac{\partial N_{H_2O}}{\partial z} \quad (3.5)$$

Where  $N_{H_2O} = v_{fuel} \mu_{H_2O}$  and  $N_{c,H_2O} = \nu_{H_2O} \frac{i}{n_e F}$ , with  $n_e$  the number of electrons in the reaction being 2.

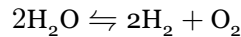
By similar computations for  $H_2$  and  $O_2$  the following equations will appear:

$$\frac{\partial \mu_{H_2}}{\partial t} = \frac{w}{A_{int}} N_{c,H_2} - \frac{\partial N_{H_2}}{\partial z} \quad (3.6)$$

$$\frac{\partial \mu_{O_2}}{\partial t} = \frac{w}{A_{int}} N_{c,O_2} - \frac{\partial N_{O_2}}{\partial z} \quad (3.7)$$

Where  $N_{H_2} = v_{fuel} \mu_{H_2}$  and  $N_{O_2} = v_{air} \mu_{O_2}$ , with  $\nu_i$  the stoichiometric coefficient for species  $i$ ,  $\mu_i$  the molar concentration of species  $i$ , and  $v_{fuel/air}$  the corresponding velocity. And  $N_{c,i} = \nu_i \frac{i}{n_e F}$ .

From the chemical equation of the reaction:



The stoichiometric coefficients are determined as  $\nu_{H_2O} = -\nu_{H_2} = -2\nu_{O_2} = 1$ .

This equation describes the local evolution of the species concentration. As noted earlier, the current  $i$  is no longer a global current. To determine it, the Nernst equation and more is required; this will be explained shortly.

### Introducing the overpotentials

From the equation stated in 3.5 only the current density  $i$  remains to be determined. The control variable is the voltage  $U$ , which equals the overpotential of the whole cell ( $\eta_{cell}$ ) or cell voltage.

The cell voltage can be written as the sum of all overpotentials over the cell as in equation 3.8. Concentration overpotentials that appear when accounting for diffusion are neglected here, and the concentration is assumed to be the same everywhere on each side of the cell. (section 3.3)

$$\eta_{cell} = \eta_{nernst} + \eta_{act} + \eta_{ohm} \quad (3.8)$$

$$\eta_{nernst} = \frac{\Delta G}{n_e F} + \frac{RT}{n_e F} \log \left( \frac{x_{H_2} \sqrt{x_{O_2}}}{x_{H_2O} \sqrt{x_0}} \right), \quad (3.9)$$

Where  $x_0 = p_{ref}/p_{air} \approx 1$ ,  $\Delta G$  is Gibbs free energy approximated by the polynomial  $\Delta G = -0.0031 T^2 - 49.119 T + 244778$ . [5]

$$\eta_{ohm} = i R_{deg} \quad (3.10)$$

Where  $R_{deg}$  is the resistance induce from degradation mechanisms.

$$\eta_{act} = \eta_{act}^a + \eta_{act}^c \quad (3.11)$$

Where  $\eta_{nernst}$  is the Nernst overpotential, representing the voltage generated by the electrochemical reaction, and  $\eta_{act}$  is the activation overpotential, which accounts for the energy barrier that must be overcome for the reaction to proceed,  $\eta_{ohm}$  is the Ohmic resistance, which accounts for resistance that is active due to degradative mechanisms. [5]

The Ohmic overpotential is essentially Ohm's law with an effective resistance; it is through this resistance that degradation enters the voltage balance. The Nernst overpotential describes the willingness of the reaction to proceed. When there is a lot of  $H_2O$  and very little  $H_2$  and  $O_2$ , the Nernst overpotential is large in absolute value but negative. This drives the reaction away from the high concentration in  $H_2O$  and closer to equilibrium. Away from the Gibbs term and the other overpotentials, the *equilibrium* occurs where the fraction is 1. The Nernst overpotential can therefore be viewed as the *force* that pulls the concentrations towards equilibrium.

The activation overpotential is probably the most difficult part to compute. The governing equations are listed first; their meaning is explained afterwards.

The overpotentials for the anode and cathode arise by solving equation (3.12), also known as the Butler-Volmer equation.

$$i = i_0^\zeta \left( \exp \left[ \alpha_\zeta \frac{n_e F}{RT} \eta_{act}^\zeta \right] - \exp \left[ -(1 - \alpha_\zeta) \frac{n_e F}{RT} \eta_{act}^\zeta \right] \right), \quad \zeta \in \{a, f\}. \quad (3.12)$$

Where  $\alpha_\zeta$  is the symmetry factor for the corresponding side, and  $i_0^\zeta$  is the exchange current for the respective side, with  $\alpha_a = 0.65$  and  $\alpha_c = 0.59$ . [17]

The exchange current is computed as in equation (3.13).

$$i_0^\zeta = \lambda_\zeta \gamma_\zeta \exp \left[ -\frac{E_{act}^\zeta}{RT} \right] \prod_{i \in \text{Species}_\zeta} \left( \frac{x_i}{x_0} \right)^{m_i}, \quad \zeta \in \{a, f\}. \quad (3.13)$$

Where  $\text{Species}_a = \{\text{O}_2\}$  and  $\text{Species}_c = \{\text{H}_2\text{O}, \text{H}_2\}$ .  $\lambda_a = 1$  (in principle it does not appear in the equation) and  $\lambda_c$  is also determined by degradation mechanisms, discussed later.  $\gamma_\zeta$  is the pre-exponential factor for the respective side with values  $\gamma_a = 1.515 \cdot 10^8$  and  $\gamma_f = 7.666 \cdot 10^5$ .  $E_{act}^\zeta$  is the activation energy, with  $E_{act}^a = 1.40 \cdot 10^5$  and  $E_{act}^f = 1.05 \cdot 10^5$ . And  $m_{\text{H}_2\text{O}} = 0.33$ ,  $m_{\text{H}_2} = -0.1$  and  $m_{\text{O}_2} = 0.22$ . [17]

This describes the electrical energy required to overcome the energy barrier for the reaction. Somewhat similar to how water does not necessarily freeze at 0 °C. There is some discussion in the field about whether the Butler-Volmer equations are adequate for this phenomenon; but for this project the Butler-Volmer model is taken as a sufficient approximation.

### Computing the current $i$

With the overpotentials introduced, it is clear that some of them depend on the current  $i$ , which is the missing piece in 3.3 needed to complete the coupled differential equations 3.5, 3.6 and 3.7.

The current is obtained by solving equation (3.8), with  $\eta_{cell} = U$ , and  $i$  defined as in equation (3.14)

$$i = \text{Solve}_i [U - \eta_{nernst} = \eta_{ohm}(i) + \eta_{act}(i)] \quad (3.14)$$

Using various simplifications this equation can be solved in different ways; for now it is solved numerically, using a root solver from the python library Scipy [27].

With the equations for mole transport in place, the temperature behavior is treated next, before all equations are summarized in a table.

### 3.1.3 Energy Balance

To derive the energy balances, the derivation starts at the thermal energy equation given by equation (3.15) [15]. A number of simplifications are then applied to obtain equation (3.16).

$$\rho \frac{\partial h}{\partial t} = \frac{\partial p}{\partial t} + \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in S} \nabla \cdot (h_i N_i) + Q_T \quad (3.15)$$

where  $\rho$  is density,  $h$  specific enthalpy,  $\lambda_T$  thermal conductivity,  $h_k$  and  $N_k$  the specific enthalpy and flux of species  $k$ , and  $Q_T$  the volumetric heat source (e.g. electrochemical heating).

Starting with equation (3.15), essentially stating how much energy a volume accumulates, in enthalpy per unit volume (left-hand side). Is equal to the compression (the change in pressure) plus the energy added by heat conduction (second order differential term in temperature) and the transportation of enthalpy by species ( $\text{H}_2\text{O}$ ,  $\text{H}_2$ ,  $\text{O}_2$ ), and lastly adding the reaction heat  $\Phi$ .

The first simplifications is to neglect the energy accumulation due to heat conduction ( $\nabla \cdot (\lambda \nabla T) = 0$ ). Also assuming constant pressure,  $h \approx c_p T$  and  $\partial_t p = 0$ , which yields [5]:

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i \frac{\partial T \mu_i}{\partial z} c_{p,i} + Q_T \quad (3.16)$$

Where  $c_p = \sum_{e \in \mathcal{M}} c_{p,k} y_k$ , with  $y_k$  being the mass fractions (Not the molar fraction), for further information look at Appendix 2

It remains to define the source term, which follows from the power expression in equation (3.17) [5]. Power is the energy per unit time, hence the energy added to the cell.

$$Q_T = \frac{i w}{A_{int}} U \quad (3.17)$$

### At steady state

At steady state the expression for the molar progression can be used to simplify the equations further, yielding equation (3.18).

$$0 = i \frac{w}{A_{int}} (U - U_{th}) - C_p \frac{\partial T}{\partial z} \quad (3.18)$$

Where  $U_{th} = \frac{\Delta h}{n_e F}$  with  $\Delta h = h_{H_2O} - h_{H_2} - \frac{h_{O_2}}{2}$ , (Appendix 2) [5].

With the energy balance in place, the quasi-steady cell model is complete. The following chapter treats the degradation mechanisms that evolve on the slow time scale (section 3.2.1).

## 3.2 Modelling degradation

### 3.2.1 Introduction

This section examines the two components  $R_{ohm}$  that appears in equation (3.10) and  $\lambda_c$  which appears in equation (3.13). The latter is more complicated, so the Ohmic resistance is treated first. Below a equation is set up, which describes how the degradation plays a role on the ohmic resistance.

$$R_{ohm} = \frac{T}{B_{ohm}} \exp \left( \frac{E_{act}^{ohm}}{RT} \right) + R_{contact} + \sum_{d \in \mathcal{D}_{ohm}} R_d \quad (3.19)$$

Here, the first two terms are time-invariant and depend only on the present conditions of the cell. In contrast, the degradation terms, represented by the summation, evolve and increase as the cell ages. The first degradation mechanism considered is the resistance arising from nickel migration.

### 3.2.2 Nickel migration

Nickel migration is a simple phenomenon to understand. Essentially the electrode is saturated with nickel, but as time goes by the saturated part of the cell will shrink, hence we will have a gap in the electrode where there are no nickel. Since the nickel plays a huge part in transporting the oxygen ions fast and smoothly, the absence of nickel will induce an added travel resistance for the ions. Now the resistance can be represented by equation (3.20).

| Description             | Equation                                                                                                                                          | Ref.               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Faraday's law           | $N_{c,i} = \nu_i \frac{i}{2F}$                                                                                                                    | equation (3.3)     |
| Molar concentration DEs | $\frac{\partial \mu_i}{\partial t} = \frac{w}{A_{int}} N_{c,i} - \frac{\partial N_i}{\partial z}$                                                 | Eqs. 3.5, 3.6, 3.7 |
| Cell voltage            | $\eta_{cell} = \eta_{nernst} + \eta_{ohm} + \eta_{act}^a + \eta_{act}^c$                                                                          | equation (3.8)     |
| Ohmic overpotential     | $\eta_{ohm} = i R_{deg}$                                                                                                                          | equation (3.10)    |
| Nernst overpotential    | $\eta_{nernst} = \frac{\Delta G}{n_e F} + \frac{RT}{n_e F} \log \left( \frac{x_{H_2} \sqrt{x_{O_2}}}{x_{H_2O} \sqrt{x_0}} \right)$                | equation (3.9)     |
| Butler–Volmer           | $i = i_0^{a/f} \left( \exp[\alpha_{a/f} \frac{n_e F}{RT} \eta_{act}^{a/f}] - \exp[-(1 - \alpha_{a/f}) \frac{n_e F}{RT} \eta_{act}^{a/f}] \right)$ | equation (3.12)    |
| Exchange current        | $i_0^{a/f} = \lambda_{a/f} \gamma_{a/f} T \exp \left[ -\frac{E_{act}^{a/f}}{RT} \right] \prod_{i \in \mathcal{S}_{a/f}} (x_i/x_0)^{m_i}$          | equation (3.13)    |
| Gibbs free energy       | $\Delta G \approx -0.0031T^2 - 49.119T + 244778$                                                                                                  | equation (3.9)     |
| Energy balance          | $\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i \frac{\partial T \mu_i}{\partial z} c_{p,i} + \frac{i w}{A_{int}} U$     | Eqs. 3.16, 3.17    |

Table 3.1: Overview of model equations (Equation covering the development for ohmic resistance and degradation can be found in section 3.2.1).

$$R_{Ni-mi} = \frac{h_{Ni-mi}}{\left( \frac{\epsilon}{\tau} \sigma_{elec} \right)} \quad (3.20)$$

Where  $d_{Ni-mi}$  is the thickness of nickel absence layer, which develops over time,  $\sigma_{elec}$  is the electrode conductivity,  $\epsilon$  is the zirconia fraction and  $\tau$  is the tortuosity of the zirconia fraction [5]

The sketch in figure 3.4 illustrates how the nickel migration develops over time, and the impact it has.

The conductivity of the electrode is given by equation (3.21). This conductivity is in reality the reduction in conductivity due to the creep of the nickel part of the electrode.

$$\sigma_{elec} = \frac{\sigma_{elec,0}}{T} \exp \left( -\frac{E_{a,elec}}{RT} \right) \quad (3.21)$$

Javid Beyrami et al. [5] describe the development of nickel migration by the following differential equation, where  $c_1$  and  $c_2$  are negative and  $c_3$  and  $c_4$  are positive. For high values of current and temperature, migration increases; if either  $T$  or  $i$  is too small or too large, migration also increases.

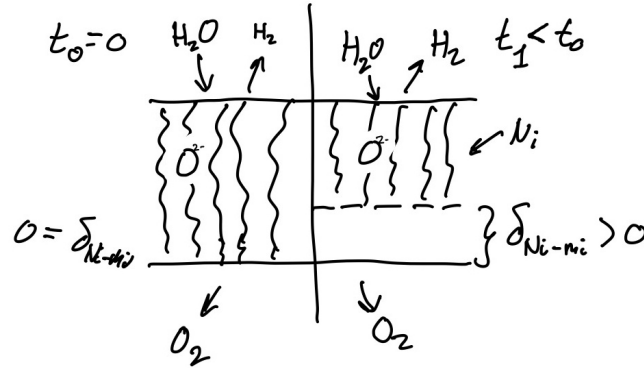


Figure 3.4: For a cell without degradation (left), nickel migration has not occurred and  $d_{\text{Ni-mi}} = 0$  at the layer between the nickel-filled electrode and the electrolyte. Over time the nickel-filled part of the electrode creeps away from the electrolyte (right), giving rise to resistance induced by the raw electrode on the traveling species.

$$\frac{\partial h_{\text{Ni-mi}}}{\partial t} = c_1 T + c_2 i + c_3 iT + c_0$$

Where  $c_k$  is fitted to experimental data.

### 3.2.3 Interconnect Oxidation

Another important degradation mechanism to consider is interconnect oxidation. This process also contributes to the overall ohmic resistance, and is relatively straightforward to understand: It occurs when the metallic interconnect undergoes oxidation, in other words, when the metal effectively “rusts.”

The resistance resulting from interconnect oxidation is in structure much like that of nickel migration. Specifically, it is determined by the thickness of the oxidized layer divided by its (reduced) electrical conductivity, as shown in equation (3.22) [5], [16], [7].

$$R_{\text{IC-ox}} = \frac{h_{\text{IC-ox}}}{\sigma_{\text{IC-ox}}} \quad (3.22)$$

Where the oxidized interconnect conductivity is given as a function of temperature in equation (3.23).

$$\sigma_{\text{IC-ox}} = \frac{k_{\text{ox}}}{T} \exp \left\{ -\frac{E_{\text{el}}}{RT} \right\} \quad (3.23)$$

Beyrami et al. [5] describe the growth of the oxide layer by the following rate law.

$$\frac{dh_{\text{IC-ox}}^2}{dt} = k_{\text{mg}} \exp \left\{ -\frac{E_{a,\text{ox}}}{RT} \right\} \quad (3.24)$$

The squared-thickness form in equation (3.24) reflects parabolic oxidation kinetics: Layer growth follows a

| Description                       | Equation                                                                                                                                                                  | Ref.            |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Total ohmic resistance            | $R_{\text{ohm}} = \frac{T}{B_{\text{ohm}}} \exp \left\{ \frac{E_{\text{act}}^{\text{ohm}}}{RT} \right\} + R_{\text{contact}} + \sum_{k \in \mathcal{D}_{\text{ohm}}} R_k$ |                 |
| Nickel-migration resistance       | $R_{\text{Ni-mi}} = \frac{d_{\text{Ni-mi}}}{\left( \frac{\varepsilon}{\tau} \sigma_{\text{elec}} \right)}$                                                                | equation (3.20) |
| Electrode conductivity            | $\sigma_{\text{elec}} = \frac{\sigma_{\text{elec},0}}{T} \exp \left( -\frac{E_{a,\text{elec}}}{RT} \right)$                                                               | equation (3.21) |
| Ni-mi thickness rate              | $\frac{\partial d_{\text{Ni-mi}}}{\partial t} = c_1 T + c_2 i + c_3 i T + c_4$                                                                                            | [2]             |
| Interconnect oxidation resistance | $R_{\text{IC-ox}} = \frac{d_{\text{IC-ox}}}{\sigma_{\text{IC-ox}}}$                                                                                                       | equation (3.22) |
| IC oxide conductivity             | $\sigma_{\text{IC-ox}} = \frac{k_{\text{ox}}}{T} \exp \left\{ -\frac{E_{\text{el}}}{RT} \right\}$                                                                         | equation (3.23) |
| IC oxidation rate                 | $\frac{dd_{\text{IC-ox}}^2}{dt} = k_{\text{mg}} \exp \left\{ -\frac{E_{a,\text{ox}}}{RT} \right\}$                                                                        | equation (3.24) |
| Normalized triple phase boundary  | $\lambda_c = 0.55$                                                                                                                                                        | [2]             |

Table 3.2: Overview of degradation mechanisms (ohmic contributions and slow-state dynamics). The base cell equations are summarized in table 3.1.

square-root-of-time law, and the rate depends on local temperature alone through the Arrhenius term. Unlike nickel migration, interconnect oxidation is therefore not sensitive to current density. As  $h_{\text{IC-ox}}$  increases, the contribution  $R_{\text{IC-ox}}$  in equation (3.22) rises and adds to the degradation summation in the total ohmic resistance. The constants  $k_{\text{mg}}$  and  $E_{a,\text{ox}}$  are fitted to experimental data [5], [7]. Together with nickel migration, this completes the ohmic degradation terms; the remaining mechanism acts on the cathode exchange current and is discussed next.

### 3.2.4 Nickel Agglomeration

A further degradation mechanism affecting the cathode exchange current is nickel agglomeration. Unlike the ohmic contributions discussed above, it is typically modelled not as a resistance but as a multiplier on  $\lambda_c$ , the normalized triple phase boundary [5]. Whether the underlying process is nickel migration or another phenomenon remains debated, and the mechanistic details are therefore not pursued here. Because agglomeration develops on a relatively short timescale,  $\lambda_c$  is assumed to have reached its asymptotic value of approximately 0.55 [5].

## 3.3 Assumptions

The average cell model in the preceding sections couples molar transport, electrochemistry, and an energy balance along a one-dimensional channel. The resulting relations are collected in table 3.1; degradation is summarized in table 3.2. The closed formulation rests on the simplifications listed below. Each item



states the assumption and why it is reasonable for the average cell model; how stacks, cells, and balance-of-plant equipment enter that representation is summarized in section 3.1.1. How the assumptions shape the optimization results is discussed in section 6.5.

### 3.3.1 Geometry, flow, and thermodynamics

1. **Constant channel area.** The cross-sectional area  $A_{\text{int}}$  is uniform along the channel. Commercial repeat-element geometries use nearly constant channel dimensions over the flow length [23], so spatial variation in  $A_{\text{int}}$  is small relative to other uncertainties.
2. **One-dimensional flow.** Properties vary only in the axial direction  $z$ ; width  $w$  and height  $d$  enter as geometric scalars. The channel length is much larger than its width, and repeat-unit models therefore resolve composition and temperature along  $z$  only [5, 21].
3. **Ideal gas.** Gas mixtures obey the ideal-gas law. Operating temperatures (700–900 °C) and near-atmospheric pressure give low compressibility, which is standard in channel-resolved SOEC models [5, 21].
4. **Negligible concentration overpotential.** Mass-transport limitations at the electrodes are not resolved; molar fractions on each electrode side are taken as uniform (equation (3.8)). The inlet is steam-rich and the fitted Butler-Volmer parameters describe the IV curve well in the operating window (figure 4.3) [17]; explicit concentration losses matter mainly at higher current or lower steam supply [18].
5. **Negligible heat conduction.** The conductive term  $\nabla \cdot (\lambda_T \nabla T)$  is dropped in the energy balance (equation (3.16)). At the prescribed inlet flows, advection and electrochemical heating dominate the along-channel temperature rise in repeat-unit models [5, 15].
6. **Constant pressure.**  $\partial_t p \approx 0$  and  $h \approx c_p T$  in the enthalpy formulation. Gas velocities are low (large Mach number not reached), so pressure work and compressibility effects are secondary to enthalpy transport [15].
7. **Prescribed inlet flows.** Fuel and air volumetric flow rates are inputs, not optimized. The study targets optimal  $U$  and  $T_{\text{in}}$ ; balance-of-plant flow scheduling is outside scope. Flows are set from commercial-cell data [23] and applied uniformly in the average cell model (section 3.1.1, items 14–15).

### 3.3.2 Electrochemistry and degradation

8. **Butler-Volmer kinetics.** Anodic and fuel-side (cathode) activation overpotentials follow equation (3.12) with parameters from [17]. This form is standard for high-temperature solid-oxide cells and matches the experimental polarization used for validation (figure 4.3).
9. **Two dynamic ohmic degradation mechanisms.** Nickel migration and interconnect oxidation evolve via table 3.2; other ageing paths are not modelled dynamically. These two mechanisms dominate long-term ASR growth in the Beyrami framework and set the failure criteria used here [3, 5].
10. **Nickel agglomeration at steady state.** Because agglomeration develops on a short timescale relative to the optimization horizon, the cathode multiplier  $\lambda_c$  is fixed at  $\approx 0.55$  [5] rather than solved as an additional state. Beyrami et al. treat agglomeration as fast compared with nickel migration and interconnect oxidation over thousands of hours [5].

11. **Spatially resolved degradation.** Slow states are stored per FVM cell ( $\mathbf{d}_{\text{Ni-mi},i}, \mathbf{s}_{\text{IC-ox},i} \in \mathbb{R}^{N_{\text{cell}}}$ ); interconnect oxidation is advanced in squared-thickness form (equation (4.18)). Current and temperature vary along  $z$ , so migration and oxidation rates must be resolved spatially to capture along-channel profiles [5].

### 3.3.3 Quasi-steady coupling and time integration

12. **Quasi-steady cell physics.** At each time step the FVM steady-state problem (molar fractions and temperature along  $z$ ) is solved for the current controls and degradation state before the slow ODEs are updated. Thermal and mass-transport transients in the channel are fast (seconds to minutes) compared with the optimization horizon (days to months), so the cell can be treated as quasi-steady while degradation evolves [5, 13].
13. **Implicit Euler in time.** Degradation states are advanced with the implicit Euler scheme on a uniform grid (equation (4.16)). The degradation ODEs are stiff at high temperature and current; implicit Euler gives stable marching on the coarse time grids used in the scenarios without restricting the step size to the fastest local rate.

### 3.3.4 Average cell model

14. **Average cell formulation.** The governing equations describe one repeat element, a single layer with coupled fuel and air channels, resolved along  $z$  only (section 3.1.1). That channel is taken to represent stack-average behaviour; electrical series connection of multiple layers and spatial structure at stack or facility level are not modelled.
15. **Uniform cells across stacks.** All cells are assumed to share the same geometry, inlet flows, and controls ( $U, T_{\text{in}}$ ). Cell-to-cell maldistribution and stack-to-stack coupling are neglected; facility-scale hydrogen delivery is obtained by scaling per-cell production [23].
16. **Lumped preheat and heat recovery.** Sensible duty to reach  $T_{\text{in}}$  and a fixed recoverable outlet enthalpy ( $T_{\text{out}} - T_{\text{loss}}$ ) enter through  $P_{\text{heat}}$  in equation (4.2). No heat-exchanger geometry, effectiveness, or transient is resolved; compressors and other auxiliary loads are likewise omitted.

# 4 Optimal Control and Solution Method

## 4.1 Optimal Control Problem

### 4.1.1 Introduction

With the cell model and degradation laws in place, operation can be simulated for any admissible inlet conditions and cell voltage within a reasonable range. What remains is to specify an economic objective and formulate the corresponding optimal control problem.

Following discussions with Dynelectro and DTU Energy, and a review of how the objective is formulated in most literature [13], [4], [8], the general structure adopted is to maximize running revenue—expressed as  $R(\mathbf{u}, \mathbf{d})$  (i.e., hydrogen production minus production costs), while also minimizing degradation.

$$\begin{aligned} \max_{\mathbf{u} \in \mathcal{U}_{\text{ad}}} \quad & \int_0^T R \, dt + \Psi \\ \text{s.t.} \quad & \frac{\partial n_i}{\partial t} \text{ satisfies table 3.1, for } i \in \mathcal{S} \\ & \frac{\partial d_i}{\partial t} \text{ satisfies table 3.2, for } i \in \mathcal{D} \\ & d_i \leq d_{i,\text{break}}, \quad \text{for } i \in \mathcal{D} \end{aligned} \tag{4.1}$$

Here  $\mathbf{u}(t) = [U(t), T_{\text{in}}(t)]^\top$  collects the control inputs. The set  $\mathcal{U}_{\text{ad}}$  is the **admissible control space**: the constrained set of trajectories for which  $U$  and  $T_{\text{in}}$  remain within their physical bounds (e.g.  $U \in [U_{\text{min}}, U_{\text{max}}]$  and  $T_{\text{in}} \in [T_{\text{in,min}}, T_{\text{in,max}}]$ ; table 5.1) and satisfy any scenario-specific restrictions imposed on the controls (such as piecewise-constant segments or fixed voltage).  $\mathbf{d}$  collects the degradation state variables;

$$\mathbf{d} = [d_i(t)]_{i \in \mathcal{D}}.$$

Viewing equation (4.1), the integrand  $R(\mathbf{u}, \mathbf{d})$  is the **revenue rate**; maximizing its time integral corresponds to maximizing operating income over  $[0, T]$ . The term  $\Psi(\mathbf{d})$  is the **salvage** contribution: A scalar that rewards the optimizer for how much useful life remains in the cell at the end of the horizon.

It is worth noting that minimizing degradation is in its core, a part of minimizing production cost: When a cell degrades the resistance rises, leading to higher power for the same hydrogen output, and in the worst case; degradation leads to a malfunctioning cell, requiring replacement of the cell or, at facility scale: A stack. It should also be noted that using terminal degradation is a modeling choice; using the running degradation rate inside the integral would have been equally valid and may be simpler to implement.

### 4.1.2 Revenue

The revenue rate is taken in the explicit form shown in equation (4.2).

$$\begin{aligned}
R &= c_{H_2} \Delta \dot{n}_{H_2} - c_P P \\
\Delta \frac{\partial n_{H_2}}{\partial t} &= A_{int} N_{out, H_2} - \frac{\partial n_{in, H_2}}{\partial t} \\
P &= P_{cell} + P_{heat}
\end{aligned} \tag{4.2}$$

Here  $\frac{\partial n_{in, H_2}}{\partial t}$  is specified as an input (part of  $\mathbf{u}$ ),  $P$  is the total electrical power,  $P_{cell}$  the power supplied to the stack, and  $P_{heat}$  the auxiliary sensible heating power.

Thus,  $R$  is the value of net hydrogen production (outlet minus inlet, in appropriate units) minus the cost of power. All quantities in equation (4.2) follow from the cell model summarized in table 3.1. The split between  $P_{cell}$  and  $P_{heat}$  reflects the average cell model in section 3.1.1: preheat and heat recovery are lumped, not resolved as a heat-exchanger model.

From the model date the power of the cell is computed by  $P_{cell} = I_{cell} U$  with:

$$I_{cell} \approx \frac{A_{cell}}{N_{cell}} \sum_{k=1}^{N_{cell}} i_k.$$

The heating power is the **sensible** auxiliary preheat duty: For each stream, the enthalpy difference between the **inlet** composition at the target inlet temperature  $T_{in}$  and the same composition at the recovered outlet temperature  $T_{out} - T_{loss}$ . Composition enthalpy of the electrochemical conversion is excluded from  $P_{heat}$  and is accounted for via  $P_{cell}$ . Positive  $P_{heat}$  means external preheat is required ( $T_{in} > T_{out} - T_{loss}$ ); negative values are recovery credits.

$$P_{heat} = \sum_{\zeta \in \{a, f\}} \dot{V}_{\zeta} A_{int} \left( \sum_{i \in S_{\zeta}} \mu_{i, in} h_i(T_{in}) - \sum_{i \in S_{\zeta}} \mu_{i, in} h_i(T_{out} - T_{loss}) \right) \tag{4.3}$$

Where outlet thermal energy is assumed recoverable toward the inlet stream with an absolute loss  $T_{loss}$ , and  $\mu_{i, in}$  denotes inlet molar concentrations.

With that the running revenue or integrant is settled, hence we can look at the salvage.

### 4.1.3 Salvage

After operating up to time  $T$ , the running profit captures cash flow from hydrogen sales net of power cost. A salvage term is included, with the objective in mind, that the cell should not be overused leading to, a cell which cannot gain its own cost in revenue. The idea is to measure, in a simple aggregate way, how close the cell is to end-of-life limits across degradation mechanisms that can cause failure.

Equal weight is assigned to each index in  $\mathcal{D}$ , leading to

$$\Psi = \frac{c_{cell}}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \left( 1 - \frac{d_i}{d_{i, critical}} \right)^{\alpha_{\Psi}} \tag{4.4}$$

Here  $c_{cell}$  scales the reward to the capital cost of the cell, and  $\mathcal{D}$  is the set of degradation. Initially  $\alpha_{\Psi} = 1$

For each  $i$ , the factor  $\max\{0, 1 - d_i/d_{i,\text{critical}}\}$  is the unused fraction of the margin to the critical level at stop time  $T$ . Averaging over  $\mathcal{D}_{\text{fail}}$  and scaling by  $c_{\text{cell}}$  yields a terminal value that increases when the cell stops farther from those limits.

### Extra note on Salvage

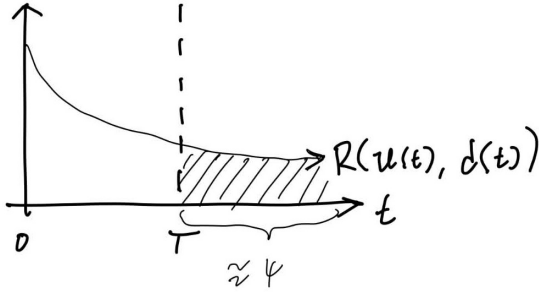


Figure 4.1: Sketch for the lifetime revenue, and the aim of A salvage term: The shaded area under the running revenue rate  $R$  is the part of the lifetime revenue that is not a direct part of the optimized horizon. This part of the lifetime revenue is hence the hardest part to incorporate. The aim is to have a model that approximates the maximal revenue left after the optimized horizon. Hence, being able to balance what is gained now against the potential future revenue.

The ideal aim would be to extract the most value from a cell over its lifetime, which would correspond to optimizing the integral in equation (4.1) with  $\Psi = 0$  and  $T$  a conditioned variable equal to the lifetime. To optimize over the lifetime is not practical, since degradation and prices cannot be predicted that far ahead. Instead, the objective optimizes value over a short future horizon while retaining potential revenue as illustrated in figure 4.1.

The objective is essentially the area under the revenue-rate curve. The formulation used here approximates maximization of that area over the full lifetime, but only up to the finite horizon  $T$ .

#### 4.1.4 H<sub>2</sub> storage and delivery agreement

Beyond the simple revenue model in equation (4.2), a contract-style extension adds buffer storage and a fixed hydrogen delivery obligation. The cell and degradation equations are unchanged; only the economic layer is extended. The contractual offtake is represented by a constant delivery rate  $\dot{n}_{\text{H}_2, \text{agreement}}$ .

In practice, delivery may occur in batches; here it is approximated as a continuous flow over  $[0, T]$ . Shortfalls relative to production are met from inventory or, if the tank is empty, from spot purchases at rate  $\dot{n}_{\text{purchase}}$ , penalized in the revenue at  $\kappa_{\text{buy}} c_{\text{H}_2}$ . The resulting revenue rate is given in equation (4.5).

$$R_{\text{tank}} = c_{\text{H}_2} \dot{n}_{\text{H}_2, \text{agreement}} - c_P P - \kappa_{\text{buy}} c_{\text{H}_2} \dot{n}_{\text{purchase}}. \quad (4.5)$$

Inventory is modelled as a lumped storage tank. The net production surplus  $\Delta \dot{n}$  is the difference between stack output and the agreement rate. Inventory increases only when production exceeds delivery; spot purchases activate only when production falls short. This gives equation (4.6) and equation (4.7).

$$\Delta \dot{n} = \dot{n}_{\text{prod}} - \dot{n}_{\text{H}_2, \text{agreement}}$$

$$\dot{n}_{\text{tank}} = \max(0, \Delta \dot{n}) \quad (4.6)$$

$$\dot{n}_{\text{purchase}} = \max(0, -\Delta \dot{n}). \quad (4.7)$$

At the end of the horizon the tank may still hold inventory. Two terminal treatments are used in the results. In the first, remaining hydrogen is valued as liquidation on the open market, giving the additional term in equation (4.8).

$$\Psi_{\text{tank}} = \kappa_{\text{sell}} c_{\text{H}_2} m_{\text{tank}}(T), \quad (4.8)$$

where  $\kappa_{\text{sell}} \approx \frac{1}{8}$  when end-of-horizon stock must compete with *black* hydrogen. In the second, a cyclic inventory constraint replaces terminal liquidation:

$$m_{\text{tank}}(0) = m_{\text{tank}}(T), \quad (4.9)$$

which is natural when the operating window is meant to repeat periodically.

Under equation (4.9), the objective is invariant to a uniform upward shift of the inventory trajectory: If  $m_{\text{tank}}(0) = m_{\text{tank}}(T) = m_0$  is feasible, so is  $m_{\text{tank}}(0) = m_{\text{tank}}(T) = m_0 + a$  for any  $a > 0$ , provided the shifted trajectory remains admissible. To select the trajectory with the smallest inventory, a penalty term is subtracted from the objective:

$$- \min_t m_{\text{tank}}(t).$$

When the inventory stays strictly positive over  $[0, T]$ , this term equals the minimum inventory level and removes the shift degeneracy.

### Full objective (tank mode)

When tank mode is active, the optimizer solves

$$\begin{aligned} \max_u \quad & \int_0^T R_{\text{tank}} dt + \Psi(\mathbf{d}(T)) + \Psi_{\text{tank}}(m_{\text{tank}}(T)) \\ \text{s.t.} \quad & m_{\text{tank}}(0) = 0 \end{aligned} \quad (4.10)$$

or

$$\begin{aligned} \max_u \quad & \int_0^T R_{\text{tank}} dt + \Psi(\mathbf{d}(T)) - \min_t m_{\text{tank}}(t) \\ \text{s.t.} \quad & m_{\text{tank}}(0) = m_{\text{tank}}(T) \end{aligned} \quad (4.11)$$

instead of equation (4.1) with equation (4.2). Fixed delivery revenue decouples cash flow from instantaneous production: The optimizer minimizes power cost and spot purchases while building inventory when production is favourable, subject to the usual salvage trade-off on degradation.

An overview of all relevant equations for the optimal control problem is stated in table 4.1.

| Description                                    | Equation                                                                                                                                                                               | Ref.            |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| General objective function                     | $\int_0^T R, dt + \Psi$                                                                                                                                                                | equation (4.1)  |
| Revenue                                        | $R = c_{H_2} \Delta \dot{n}_{H_2} - c_P P$                                                                                                                                             | equation (4.2)  |
| Salvage                                        | $\Psi = \frac{c_{cell}}{ \mathcal{D} } \sum_{i \in \mathcal{D}} \left(1 - \frac{d_i}{d_{i,critical}}\right)^{\alpha_\Psi}$                                                             | equation (4.4)  |
| Tank balance                                   | $\dot{n}_{\text{tank}} = \max(0, \dot{n}_{\text{prod}} - \dot{n}_{H_2, \text{agreement}})$                                                                                             | equation (4.6)  |
| Spot purchase                                  | $\dot{n}_{\text{purchase}} = \max(0, \dot{n}_{H_2, \text{agreement}} - \dot{n}_{\text{prod}})$                                                                                         | equation (4.7)  |
| Contract revenue                               | $R_{\text{tank}} = c_{H_2} \dot{n}_{H_2, \text{agreement}} - c_P P - \kappa_{\text{buy}} c_{H_2} \dot{n}_{\text{purchase}}$                                                            | equation (4.5)  |
| Terminal liquidation (alternative A)           | $\Psi_{\text{tank}} = \kappa_{\text{sell}} c_{H_2} m_{\text{tank}}(T)$                                                                                                                 | equation (4.8)  |
| Cyclic inventory constraint (alternative B)    | $m_{\text{tank}}(0) = m_{\text{tank}}(T)$                                                                                                                                              | equation (4.9)  |
| Minimum-inventory penalty (with alternative B) | $-\min_t m_{\text{tank}}(t)$                                                                                                                                                           | equation (4.11) |
| Cell Power                                     | $P_{cell} = I_{cell} U$                                                                                                                                                                |                 |
| Heating Power                                  | $P_{\text{heat}} = \sum_{\zeta \in \{a, f\}} \dot{n}_\zeta \left( \sum_{i \in \mathcal{S}_\zeta} x_i h_i(T_{in}) - \sum_{i \in \mathcal{S}_\zeta} x_i h_i(T_{out} - T_{loss}) \right)$ |                 |

Table 4.1: Overview of the optimal control formulation (objective, revenue, terminal value, and power bookkeeping). The underlying cell equations are summarized in table 3.1; degradation is treated in section 3.2.1. Optional tank economics (section 4.1.4) add contract revenue and inventory dynamics; end-of-horizon inventory is closed either by terminal liquidation (alternative A) or by a cyclic constraint with a minimum-inventory penalty (alternative B).

## 4.2 Building the numerical solver

### 4.2.1 Introduction

This chapter presents the construction of numerical tools required to determine the optimal control  $\mathbf{u}^*(t)$  for the unit-cell model. Unlike simple textbook examples, where a Wikipedia lookup was sufficient, the problem considered here is more complex and necessitates a structured, step-by-step methodology. The central question addressed is: What tools and methods are needed to solve the optimal control problem outlined in section 4.1.1?

The overall strategy is as follows:

1. **Steady-state cell equations:**

As the focus is on long-term behavior, it is assumed that the dynamics described by the differential equations in table 3.1 occur much faster than the overall optimization timescale, treating them as instant. This allows for replacement with a set of steady-state equations, which are simpler to solve. The Finite Volume Method (FVM) is used to solve these equations numerically.

2. **Degradation dynamics:**

With the steady-state cell solver established, its outputs serve as algebraic inputs to the degradation model (table 3.2). This model captures unit-cell degradation over time through differential equations, which are solved numerically using the implicit Euler method.

3. **Objective and optimal control:**

The optimal control objective (table 4.1) is addressed by discretizing the control input with the single shooting method. In this approach, both the revenue and degradation equations are integrated forward in time as a system of differential equations.

Structuring the problem in these three parts (steady-state solution, degradation modeling, and control optimization) enables a systematic approach to the full optimal control challenge.

### 4.2.2 Finite volume method (FVM)

With the descriptive equations in place (table 3.1, section 3.2.1, and section 4.1.1), a numerical solution method is required. All equations can be written in the general form

$$\frac{\partial u}{\partial t} = Q_u - \frac{\partial F_u}{\partial z} \quad (4.12)$$

where  $Q_u$  denotes the source term and  $F_u$  the flow term; both can be computed directly from the state. For now only the steady-state solution is of interest, which simplifies the problem considerably and yields the following *boundary value problem*:

$$0 = Q_u + \frac{\partial F_u}{\partial z}$$

The *finite volume method* is used to solve this steady-state BVP [12].

As shown in figure 4.2, the domain is split into smaller subdomains coupled by flux from one volume into the next. Integration of the density  $u$  over a small volume  $\Omega_k$  of volume  $V_k$  proceeds as follows.



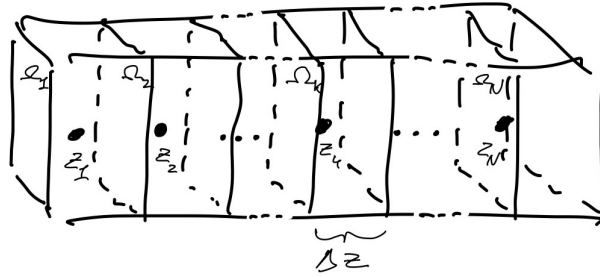


Figure 4.2: A sketch of the finite volume method. A large volume, corresponding to the air and gas channel, is split into smaller volumes. The greater volume is divided into  $N$  volumes, each represented by the midpoint, yielding a 1D system instead of 3D. Integration over the densities in each small volume gives the mass in that volume. The figure also introduces  $z_k$  (midpoint),  $\Omega_k$  (small volume), and  $\Delta z$  (length of the volume).

$$\int_{\Omega_k} \frac{\partial u}{\partial t} = \int_{\Omega_k} Q_u dV - \int_{\Omega_k} \frac{\partial F_u}{\partial z} dV$$

The source term is approximated by a rectangle (midpoint rule):  $\int_{\Omega_k} Q_u \approx Q_u(z_k) A_{int} \Delta z$ . For the flux term, the fundamental theorem of calculus is used to evaluate the antiderivative in the  $z$ -direction, with the assumption that the concentration is constant perpendicular to  $z$ , giving:

$$\int_{\Omega_k} \frac{\partial F_u}{\partial z} dV = A_{int} F_u \Big|_{\partial \Omega_k} = A_{int} \left( F_u(z_{k+\frac{1}{2}}) - F_u(z_{k-\frac{1}{2}}) \right)$$

The boundary values  $F_u(z_{k-\frac{1}{2}})$  are approximated by the value in the previous FVM cell, consistent with the flow direction along the unit-cell channel (in cell  $k$  there is no information from cell  $k+1$ , but there is from cell  $k-1$ ). The integral is therefore approximated as:

$$\int_{\Omega_k} \frac{\partial F_u}{\partial z} \approx \left( F_u(z_k) - F_u(z_{k-1}) \right) A_{int}$$

Combining the source and flux approximations gives the discrete equation equation (4.13).

$$\frac{\partial}{\partial t} \int_{\Omega_k} u dV = Q_u(z_k) A_{int} \Delta z - \left( F_u(z_k) - F_u(z_{k-1}) \right) A_{int} \quad (4.13)$$

or at steady state:

$$0 = Q_u(z_k) A_{int} \Delta z - \left( F_u(z_k) - F_u(z_{k-1}) \right) A_{int}$$

### Applying boundary conditions

The problem has two boundary conditions. The first is a Dirichlet condition at the inlet, imposed by defining the first flux as:

$$F_u(z_{-1}) = F_u(z_{inlet})$$

For the Neumann condition at the outlet, no change is made to the equations, since the upstream flux in the FVM makes it possible to neglect a Neumann boundary term at the end.

### FVM on specific equations

Applying the FVM to the governing equations yields the following discretization; see appendix 3 for derivations.

$$|\Omega_k| \frac{\partial \mu_i}{\partial t}(z_k) \approx w N_{c,i} \Delta z - A_{int} (N_i(z_k) - N_i(z_{k-1})), \quad (4.14)$$

And

$$m_k c_p \frac{\partial T}{\partial t}(z_k) \approx Q_T A_{int} \Delta z - A_{int} \sum_{i \in \mathcal{S}} v_i c_{p,i} \left( T(z_k) \mu_i(z_k) - T(z_{k-1}) \mu_i(z_{k-1}) \right) \quad (4.15)$$

## 4.2.3 FVM model validation and Grid independence

### IV-curve validation

After implementing all the equations given in table 3.1, the model was validated against an IV-curve from experimental data [23]. The error is below  $10^{-2}$  except at quite high voltage/current, where the error is larger. Figure 4.3 compares the model with the data.

This level of agreement is sufficient for the intended operating range, as very high voltages are not relevant for typical operation. The main variable of interest is the change in moles across the cell, which exhibits similar trends; at high voltage or current, the error in molar change decreases further, since the molar fractions approach their extrema (0 or 1), making relative errors negligible.

### Grid independence

Grid refinement was used to assess how much the states change when the mesh is refined, and thereby to estimate the resolution required for reliable simulations later. The steady-state solver was run while doubling the number of grid points each time. Figure 4.4 indicates that a relative error target of approximately  $10^{-3}$  would require on the order of 200 FVM cells. The optimization runs in table 5.1 instead adopt  $N_{\text{cell}} = 20$ , for which the study indicates spatial error on the order of  $10^{-2}$ —adequate for comparing control modes, as discussed in section 6.6.

But nonetheless the Steady-state equations can be solved to a rather fine precision. Leading to the next step: Solving the ODE's using the outputs of the output of the FVM.

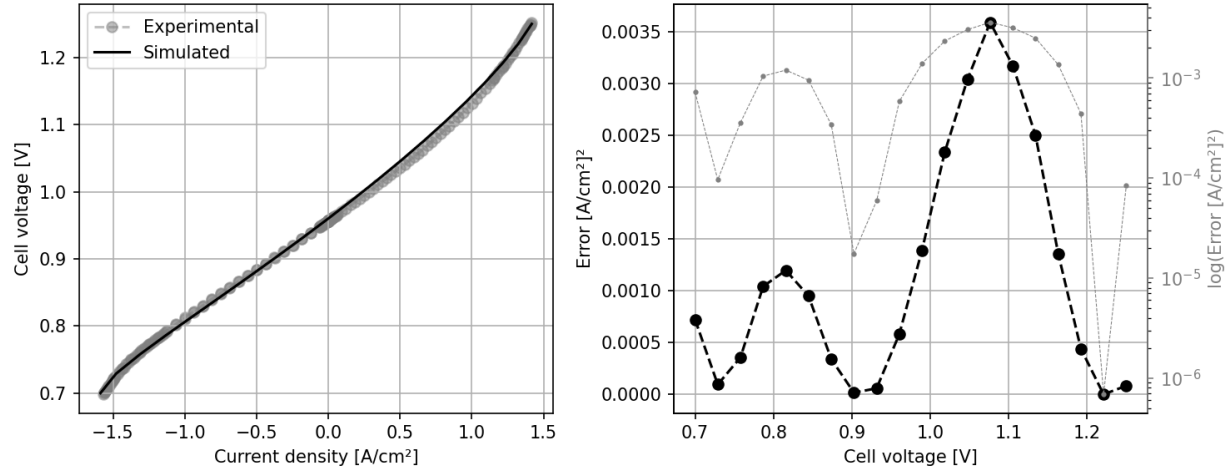


Figure 4.3: Model validation by comparing to an IV-curve from experimental data. The simulated current–voltage curve is compared to the experimental curve on the left. On the right, the error in current is shown as a function of voltage, on both logarithmic and linear scales. The experimental data is from figure. 2. in Rizvandi, Beyrami, and Hauch (2025) ?.

#### 4.2.4 Implicit Euler method (ODE solver)

As stated in the introduction to this chapter, the FVM gives the algebraic outputs needed to solve the differential equations that define degradation (table 3.2). Those ODEs are advanced with the implicit Euler method on a uniform time grid [14].

##### General form

Take the ODE:  $\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$ . Discretize time on  $t \in [0, T]$  into  $N_t$  steps with  $\mathbf{x}_i = \mathbf{x}(t_i)$ ,  $t_i = t_{i-1} + \Delta t_i$ ,  $t_0 = 0$ , and  $t_{N_t} = T$ . Given an initial state  $\mathbf{x}_0$ , the implicit Euler update is

$$\mathbf{x}_i = \mathbf{x}_{i-1} + \Delta t_i \mathbf{f}(t_i, \mathbf{x}_i), \quad i = 1, \dots, N_t. \quad (4.16)$$

Because  $\mathbf{x}_i$  appears on both sides, equation (4.16) is a fixed-point equation at each step. Damped fixed-point iteration is therefore a natural solver. The update is also Markovian: Step  $i$  depends only on  $\mathbf{x}_{i-1}$  and the control on  $[t_{i-1}, t_i)$ , so the integration can proceed one time step at a time.

##### Specific ODE setup

For this thesis the vector field  $\mathbf{f}$  is expensive: Evaluating  $\dot{\mathbf{d}}$  requires converged FVM fields (temperature and current density along the stack), which themselves depend on the degradation state. The degradation state advanced at each control interval is

$$\mathbf{d}_i = \begin{bmatrix} \mathbf{d}_{\text{Ni-mi},i} \\ \mathbf{s}_{\text{IC-ox},i} \end{bmatrix}, \quad (4.17)$$

where  $\mathbf{d}_{\text{Ni-mi},i}, \mathbf{s}_{\text{IC-ox},i} \in \mathbb{R}^{N_{\text{cell}}}$  collect the per-cell slow states at time  $t_i$ , with  $\mathbf{s}_{\text{IC-ox},i} = \mathbf{d}_{\text{IC-ox},i}^2$ . Nickel migration stores layer thickness  $d_{\text{Ni-mi}}$  directly; interconnect oxidation stores squared oxide thickness

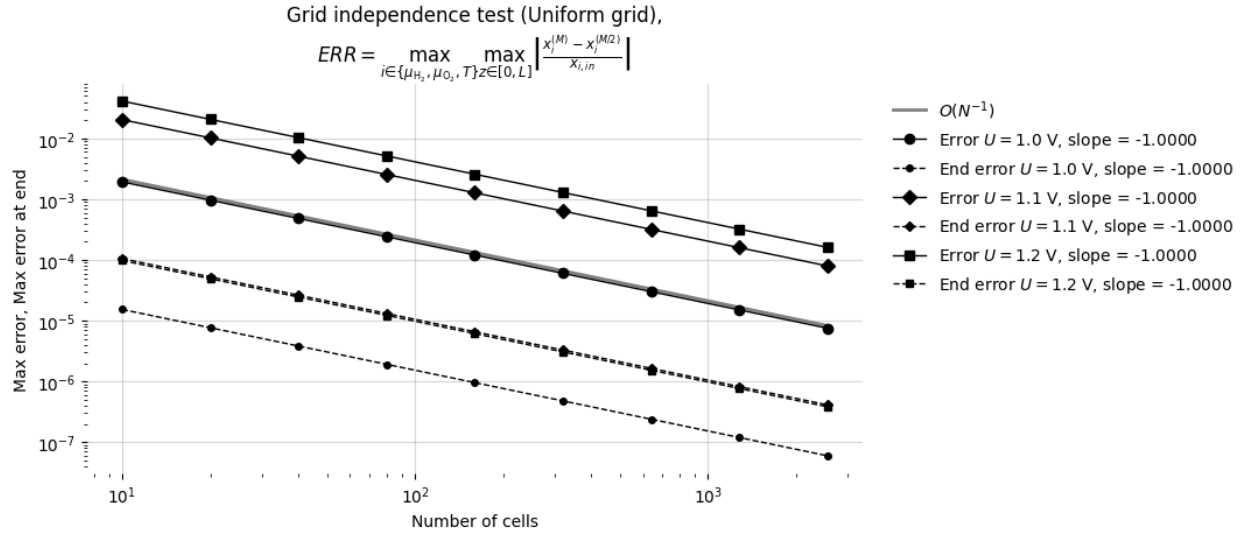


Figure 4.4: Grid-independence study of the steady-state FVM solver. The number of cells  $N$  is doubled repeatedly (5, 10, ...) at fixed inlet conditions and three cell voltages ( $U = 1.0$ , 1.1, and 1.2 V). Solid markers show the maximum relative change in hydrogen and oxygen chemical potentials and temperature between successive refinements; dashed markers show the same quantity at the channel outlet. All curves fall approximately as  $O(N^{-1})$  (grey reference). At  $N \approx 200$  the error reaches  $\sim 10^{-3}$ ; the optimization runs use  $N_{\text{cell}} = 20$  (table 5.1), for which the error is  $\sim 10^{-2}$ .

$$s_{\text{IC-ox}} = d_{\text{IC-ox}}^2, \text{ consistent with equation (3.24) and } R_{\text{IC-ox}} = d_{\text{IC-ox}} / \sigma_{\text{IC-ox}} \text{ with } d_{\text{IC-ox}} = \sqrt{s_{\text{IC-ox}}}.$$

With piecewise-constant controls  $\mathbf{u}_i$  on  $[t_{i-1}, t_i]$ , implicit Euler gives

$$\begin{aligned} \mathbf{d}_{\text{Ni-mi},i} &= \mathbf{d}_{\text{Ni-mi},i-1} + \Delta t_i \dot{\mathbf{d}}_{\text{Ni-mi}}(\mathbf{d}_i, \mathbf{u}_i), \\ \mathbf{s}_{\text{IC-ox},i} &= \mathbf{s}_{\text{IC-ox},i-1} + \Delta t_i \dot{\mathbf{s}}_{\text{IC-ox}}(\mathbf{d}_i, \mathbf{u}_i), \end{aligned} \quad i = 1, \dots, N_t, \quad (4.18)$$

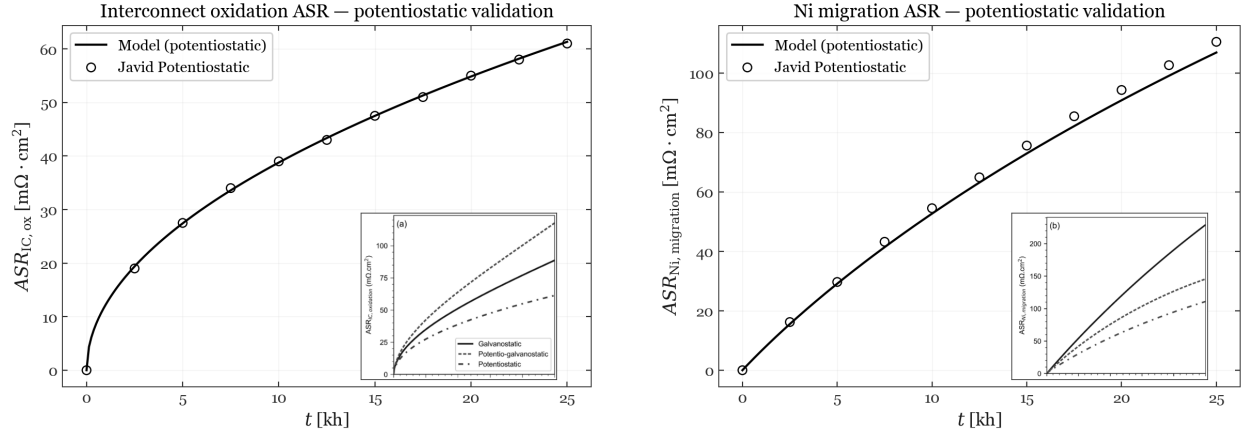
where  $\mathbf{d}_i$  stacks all per-cell degradation components at time  $t_i$ . The rates follow table 3.2:  $\dot{d}_{\text{Ni-mi},k} = c_1 T_k + c_2 i_k + c_3 T_k i_k + c_4$  depends on local temperature  $T_k$  and current density  $i_k$ , while  $\dot{s}_{\text{IC-ox},k} = k_{\text{mg}} \exp\{-E_{a,\text{ox}}/(RT_k)\}$  depends on  $T_k$  only.

Those FVM outputs are not explicit inputs to the ODE; they are obtained by running the steady-state FVM solver with control  $\mathbf{u}_i$  and degradation state  $\mathbf{d}_i$ . Because  $\mathbf{d}_i$  appears inside that solve, each time step is a coupled fixed-point problem: Guess  $\mathbf{d}_i$ , run FVM, evaluate the rates in equation (4.18), update  $\mathbf{d}_i$ , and iterate until convergence. Giving us the heavy computation of solving the steady state for each fixed-point iteration, leading to a large computational time.

In general, the problem can be viewed on a two-dimensional grid: Each spatial row (FVM cell) must be resolved repeatedly within a time step before advancing to the next column (time index).

#### 4.2.5 Validation of the ODE solver

With the steady-state FVM validated in the previous section, the next step is to check that the coupled degradation integrator in equation (4.18) reproduces expected long-term behaviour.



(a) Interconnect oxidation. At  $t = 25$  kh: Model  $61.3 \text{ m}\Omega \cdot \text{cm}^2$ , reference  $61.0 \text{ m}\Omega \cdot \text{cm}^2$ . RMSE =  $0.26 \text{ m}\Omega \cdot \text{cm}^2$ .

(b) Nickel migration. At  $t = 25$  kh: Model  $106.9 \text{ m}\Omega \cdot \text{cm}^2$ , reference  $110.5 \text{ m}\Omega \cdot \text{cm}^2$ . RMSE =  $2.45 \text{ m}\Omega \cdot \text{cm}^2$ .

Figure 4.5: Potentiostatic validation of ASR for (a) interconnect oxidation and (b) nickel migration. Black line: Implicit-Euler simulation. Open markers: Digitized reference data from [2].

The reference case follows the potentiostatic traces reported by Beyrami et al. [5] (figure. 8): Fixed cell voltage  $U = 1.285 \text{ V}$ , inlet temperature  $T_{\text{in}} = 750^\circ \text{C}$ , and fuel composition  $10\% \text{ H}_2 / 90\% \text{ H}_2\text{O}$ . At each time knot the FVM steady-state problem is solved, the degradation rates from table 3.2 are evaluated, and the states are updated with implicit Euler (equation (4.16)) until  $t = 25$  kh. This produces a history of how the degradation mechanisms develop over time. From these quantities, the area-specific resistances (ASR) can be computed and compared with the traces from Beyrami et al. in figure 4.5.

The interconnect-oxidation agrees over the 25 thousand hours horizon. Nickel migration follows the same trend but sits slightly below the reference curve. Given the digitization uncertainty and the simplified inlet specification relative to the original study, this level of agreement is considered sufficient for the degradation model to be used in the optimal-control setup in section 4.2.6.

#### 4.2.6 Solving the Optimal Control Problem

The optimal control problem (equation (4.1)) is solved using a nonlinear programming (NLP) solver. The problem is set up for single shooting so that an NLP solver can find the optimum. The control function  $\mathbf{u}$  is taken to be piecewise constant (as would also be the case in real applications).

$$\mathbf{u}^{(k)}(t) = \begin{cases} \mathbf{u}_i^{(k)} & t \in [t_i, t_{i+1}) \end{cases}$$

The problem is split into  $N_t$  subproblems, each with constant control input. From equation (4.1) the objective  $\varphi$  can be rewritten as:

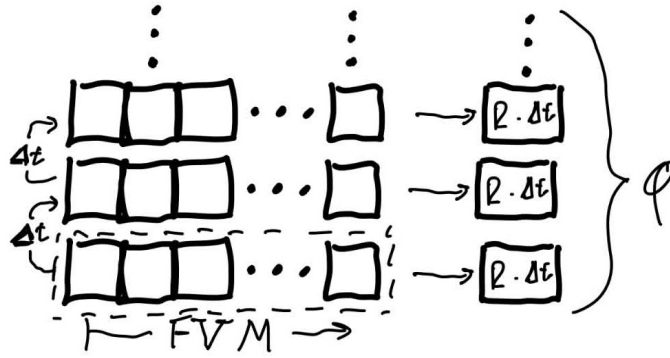


Figure 4.6: Overview of the single-shooting discretization. The horizon is divided into intervals of length  $\Delta t$  (vertical axis). At each interval a piecewise-constant control  $\mathbf{u}_i$  drives a steady-state FVM solve of the cell equations (spatial cells along the stack, dashed box). The resulting fields and degradation state feed the running objective rate  $R(\mathbf{u}, \mathbf{d})$ , which contribution  $R \Delta t$  is accumulated forward to give the total objective  $\varphi$ .

$$\begin{aligned}\varphi &= \int_0^T R(\mathbf{u}, \mathbf{d}) \, dt + \Psi(\mathbf{d}) \\ \Rightarrow \dot{\varphi} &= R(\mathbf{u}, \mathbf{d})\end{aligned}\tag{4.19}$$

Thus  $\varphi$  is obtained by solving the auxiliary differential equation with initial condition  $\varphi(0) = 0$  (numerical integration), concurrently with the state equations, and adding  $\Psi(\mathbf{d})$  at the final time when the degradation at  $t = T$  is known.

For the constraints in equation (4.1), which are equality constraints, they are used directly to compute degradation and molar flows, leaving only  $\mathbf{u}$  as a decision variable for the optimizer. The NLP solver also requires the gradient  $\nabla_{\mathbf{u}}\varphi$ ; adjoint gradient methods are used [22], [9]. Further details are in appendix 5; here it suffices that  $\nabla_{\mathbf{u}}\varphi$  is available.

### The full setup

The discretized problem solved is:

$$\begin{aligned}\max_{\mathbf{u}} \quad & \varphi_N + \Psi(\mathbf{d}_N) \\ \text{s.t.} \quad & \varphi_{i+1} = \varphi_i + \dot{\varphi}(\mathbf{u}_{i+1}, \mathbf{d}_{i+1}) \cdot \Delta t_i \quad \text{for } i = 1 \dots N-1 \\ & \mathbf{d}_{i+1} = \mathbf{d}_i + \dot{\mathbf{d}}(\mathbf{u}_{i+1}, \mathbf{d}_{i+1}) \cdot \Delta t_i \quad \text{for } i = 1 \dots N-1 \\ & \varphi_1 = 0, \\ & \mathbf{d}_1 = \mathbf{o}.\end{aligned}\tag{4.20}$$

as a single shooting problem, with  $\mathbf{d}$  computed from the constraint equations. A convex programming algorithm is used to find the optimal  $u$ ; the problem is generally convex. For this project the trust-constr method

---

from SciPy is used [27], a trust-region method that employs the computed gradient of  $\varphi$ .

# 5 Results and Findings

## 5.1 Introduction

With the full machinery in place, it is time to put it to work. The preceding chapters set up the model, the objective, and the solver; what is still missing before any results can be discussed is a list of the numerical settings used in the optimization runs. Those defaults are collected in table 5.1, table 5.3 and section 4.1.4. The first of the three tables, state the settings used by default, unless anything else is stated.

The objective itself is unchanged from equation (4.1): Maximize integrated revenue plus terminal salvage over a finite horizon. The prices and break levels are the concrete numbers plugged into the formulation. Unless stated otherwise, the runs use the settings in table 5.1. Literature values for stack geometry and flows [23], end-of-life limits [3], cell cost [4], and energy prices [1] are indicated in the reference column; entries marked *chosen* are project defaults.

Several entries in table 5.1 are marked as chosen rather than taken from the literature. Brief rationale is given here: The voltage and inlet-temperature bounds keep the optimizer within a range of physically plausible operating conditions for the unit-cell model. The number of FVM cells  $N_{\text{cell}}$  and the control segments  $N_t$  were set by the computational budget of the project: Finer discretizations were possible in principle, but the selected values keep each forward solve and outer iteration within tractable run times on the available hardware.

The 200-day horizon is chosen for the same practical reason. It is large enough that a non-constant control profile can show up within the optimization window, without the need of making too many Euler steps in the ODE solver, which requires more time to solve.

The parameter  $\alpha_\Psi$  in equation (4.4) determines how sharply the terminal penalty rises as the cell approaches its end-of-life degradation threshold. In this work, a value of  $\alpha_\Psi = 0.5$  is chosen: This means the penalty increases gently while the stack remains healthy, but becomes much steeper as the degradation nears the critical limit. With this choice, the optimizer remains sensitive to imminent failure, yet allows some flexibility for degradation earlier in the cell's lifetime.

This behavior is illustrated in figure 5.1, which plots the salvage penalty (for a single degradation parameter) for different values of

$\alpha_{\text{salvage}}$ . When

$\alpha_{\text{salvage}} = 1$ , the penalty grows linearly with degradation. As

$\alpha_{\text{salvage}}$  approaches zero, the penalty remains negligible until the breaking point is reached, after which the penalty jumps abruptly to its maximum. Selecting

$\alpha_{\text{salvage}} = 0.5$  strikes a balance: it penalizes all degradation to some degree, but especially emphasizes degradation close to failure, reflecting the greater importance of avoiding end-of-life conditions.

Through the chapter 3 scenarios are optimized and showed here, afterward a deeper analysis is carried out for sensitivity of relevant economic parameters for the scenarios where the analysis makes the most sense. An overview of the economic values given by the optimizers can be found in table 5.2.



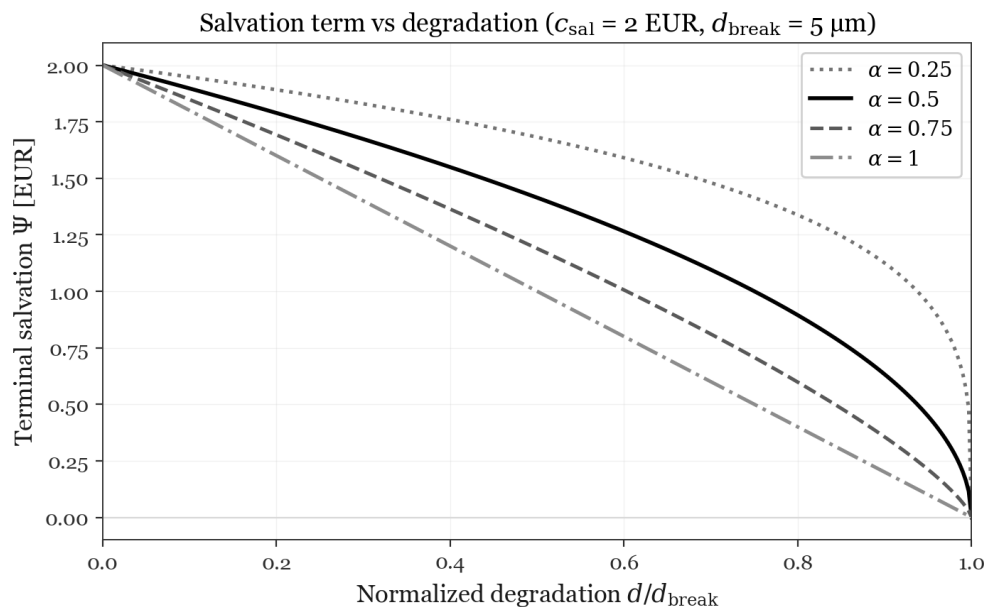


Figure 5.1: Effect of the salvage penalty sharpness parameter  $\alpha_\Psi$  in equation (4.4) for a single degradation variable. For  $\alpha_\Psi = 1$  (blue, linear) all degradation is penalized equally, while lower  $\alpha_\Psi$  (red, green) create a steeper penalty as end of life is approached, with near-zero penalty until close to the critical threshold. The value  $\alpha_\Psi = 0.5$  (green) is chosen as a compromise.

## 5.2 Baseline optimization result

As announced in section 5.1, the first result is a baseline in which both controls are held constant over the full horizon: A single cell voltage  $U$  and a single inlet temperature  $T_{\text{in}}$ , broadcast to all  $N_t = 5$  segments. The optimizer is run in potentiostatic-isothermal (pot-iso) mode, i.e. with fixed voltage and inlet temperature on every segment. This is the most restricted operating mode considered ( $U$  and  $T_{\text{in}}$  each collapse to one scalar decision variable).

All settings follow table 5.1. The NLP returns

$$U^* \approx 1.32 \text{ V}, \quad T_{\text{in}}^* \approx 751 \text{ }^\circ\text{C} \text{ (1024 K)}.$$

Figure 5.2 shows the replayed trajectory at these controls. Mean nickel-migration thickness grows from  $0.75 \text{ } \mu\text{m}$  to the  $5 \text{ } \mu\text{m}$  break limit; interconnect oxidation from  $1.5 \text{ } \mu\text{m}$  to about  $3.6 \text{ } \mu\text{m}$ . Running revenue accumulates over the 200-day window while nickel migration reaches the enforced break limit at  $t = T$ .

The accumulated revenue is 17.30 EUR and the terminal salvage term is  $\Psi \approx 1.01 \text{ EUR}$ . Because the degradation states are seeded from a 100-day burn-in (table 5.1), the salvage value at the horizon start is already  $\Psi_0 \approx 1.91 \text{ EUR}$  rather than the pristine-cell value 2.00 EUR; the corresponding salvage loss before the optimization period is  $\approx 0.09 \text{ EUR}$ . Over the 200-day window the cell loses a further  $\Psi_0 - \Psi \approx 0.90 \text{ EUR}$  of salvage value, giving a net economy  $J - \Psi_0 \approx 16.40 \text{ EUR}$  on the  $20 \text{ cm}^2$  cell.

The optimum operates exothermal rather than at thermoneutral conditions. The chosen voltage exceeds the

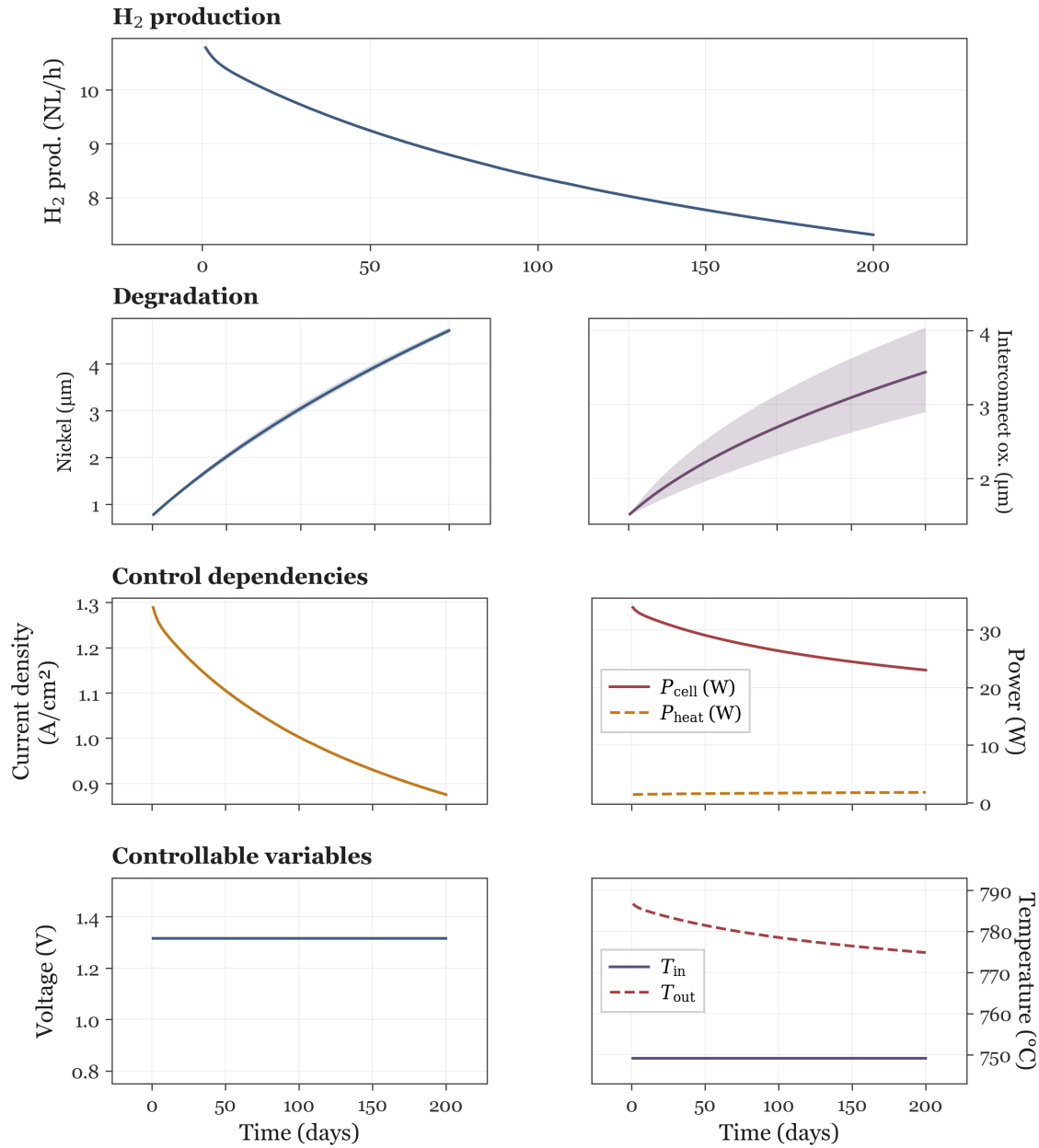


Figure 5.2: Baseline potentiostatic-isothermal optimum with no delivery contract ( $N_t = 5$ ; table 5.1). Single controls  $U^* \approx 1.32$  V and  $T_{\text{in}}^* \approx 751$  °C are held over 200 days. Exothermal operation ( $U^*$  above the thermoneutral voltage at  $T_{\text{in}}^*$ ) sustains positive  $P_{\text{heat}}$  and a rising temperature profile along the cell;  $\dot{n}_{\text{H}_2, \text{prod}}$  falls from  $\approx 11.3$  to 7.5 NL/h as nickel thickness approaches the break cap. Net economy 16.40 EUR on the 20 cm<sup>2</sup> cell.

| Setting                                       | Baseline value     | Unit               | Reference                     |
|-----------------------------------------------|--------------------|--------------------|-------------------------------|
| Horizon $T$                                   | 200                | days               | Chosen                        |
| Control segments $N_t$                        | 5                  | —                  | Chosen                        |
| $U$ bounds                                    | [0.8, 1.5]         | V                  | Chosen                        |
| $T_{\text{in}}$ bounds                        | [700, 900]         | °C                 | Chosen                        |
| FVM cells $N_{\text{cell}}$                   | 20                 | —                  | Chosen                        |
| Fuel / air flows                              | 24 / 50            | NL/h               | [14]                          |
| Initial $h_{\text{Ni-mi}} / h_{\text{IC-ox}}$ | 0.75 / 1.5         | $\mu\text{m}$      | Values after 100 days burn-in |
| Break $h_{\text{Ni-mi}} / h_{\text{IC-ox}}$   | 5 / 10             | $\mu\text{m}$      | [18]                          |
| $c_{\text{H}_2}$                              | 7.44               | EUR/kg             | [19]                          |
| $c_P$                                         | 0.079              | EUR/kWh            | [19]                          |
| $c_{\text{cell}}$                             | 1000               | EUR/m <sup>2</sup> | [14], [1]                     |
| $\alpha_\Psi$                                 | 0.5                | —                  | Chosen                        |
| Optimization solver                           | SLSQP/Trust-Constr | —                  | —                             |

Table 5.1: Baseline numerical settings for the optimization results in this chapter (unless a subsection states otherwise).

| Scen. | Description                                     | Key parameter varied                   | Avg. revenue [c/d] | Avg. salv. loss [c/d] | Avg. net econ. [c/d] |
|-------|-------------------------------------------------|----------------------------------------|--------------------|-----------------------|----------------------|
| 1     | Baseline run                                    | Current $I$                            | 8.65               | 0.45                  | 8.20                 |
| 2.1   | Constant delivery rate and tank storage         | $\kappa_{\text{sell}}$ , tank schedule | 6.18               | 0.17                  | 6.01                 |
| 2.2   | Constant delivery rate and limited tank storage | 6.15                                   | 0.17               | 5.98                  |                      |
| 3     | 24 h cyclic power price                         | Cyclic power price profile             | 6.24               | 0.19                  | 6.05                 |

Table 5.2: Overview of the three result scenarios analysed in this chapter. Integrated revenue, salvage loss ( $\Psi_0 - \Psi$ ), and net economy ( $J - \Psi_0$ ) from section 5.2, section 5.3, and section 5.4 are divided by the scenario horizon (200 days for scenarios 1–2, 24 h for scenario 3). Daily averages are reported in euro cents per day (c/d).

thermoneutral value at the selected inlet temperature, so both reaction and ohmic heating raise the local temperature from inlet to outlet. At  $t = T$ , the outlet reaches about 780 °C while the inlet remains near 751 °C. Figure 5.3 illustrates these effects along the channel at  $t = 0$ ,  $t = T/2$ , and  $t = T$ . By the end of the run, nickel migration emerges as the life-limiting degradation mechanism, with the nickel thickness within a few percent of the 5  $\mu\text{m}$  cap across all positions  $z$ . Interconnect oxidation, meanwhile, remains well below its 10  $\mu\text{m}$  cap and still shows a clear increase toward the warmer outlet. The almost uniform nickel profile at the end of life reflects two offsetting effects. The exothermal temperature field increases degradation toward the outlet, while the relatively even current-density distribution increases it toward the inlet. Together, these contributions produce a nearly uniform degradation pull along the cell. In contrast, interconnect oxidation continues to be more pronounced near the outlet, where temperatures are highest.

The next thing to do was to finally give the control more freedom and let the individual control segments take whatever value optimized the most. Here a somewhat trivial thing happened. The control did not change and the potentiostatic-isothermal operation is the best. Even though it is a boring result, it makes perfect

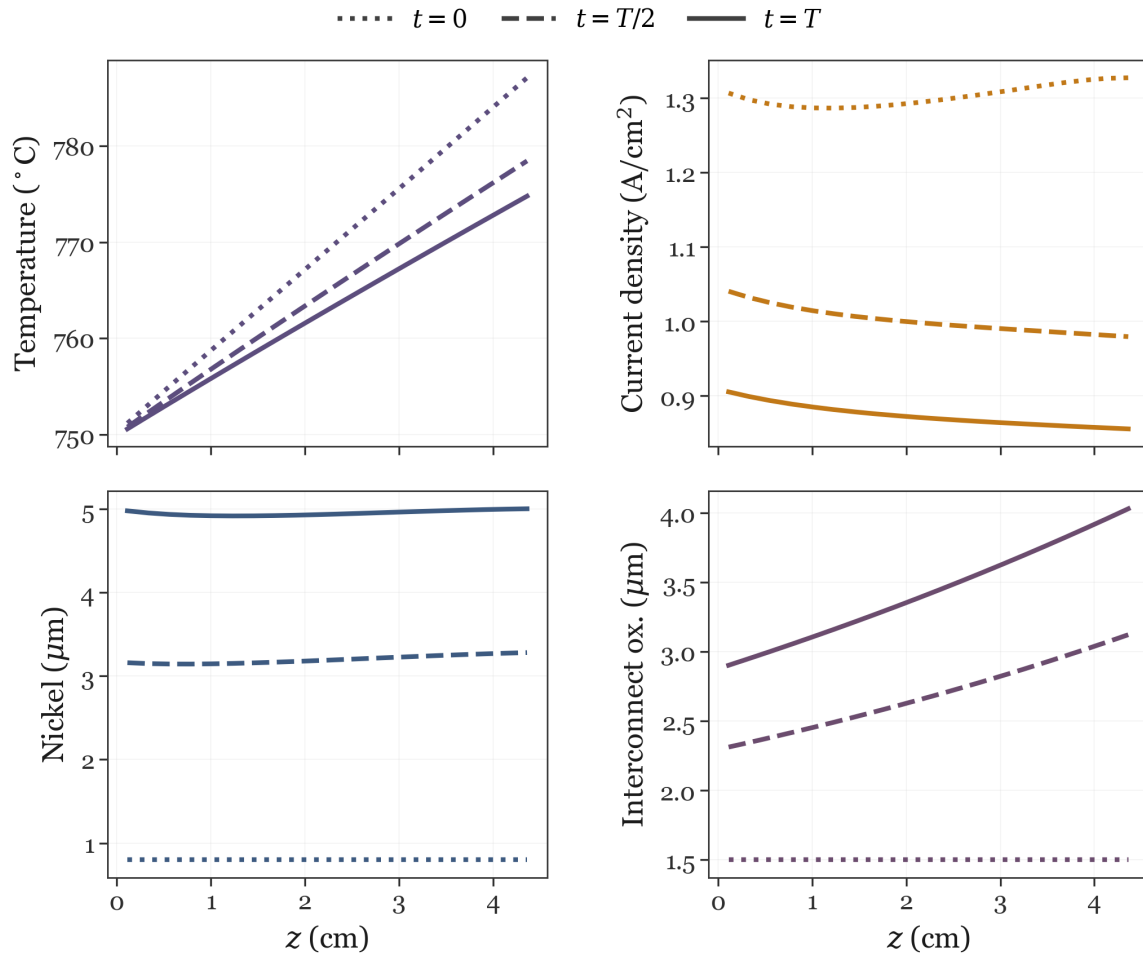


Figure 5.3: Spatial channel profiles at  $t = 0$ ,  $t = T/2$ , and  $t = T$  for the baseline pot-iso optimum (fine forward replay). Temperature rises along  $z$  under exothermal operation; nickel migration approaches the  $5 \mu\text{m}$  break limit almost uniformly at  $t = T$ , while interconnect oxidation stays below its cap with stronger outlet weighting.

| Setting                                                       | Value               | Unit            | Reference |
|---------------------------------------------------------------|---------------------|-----------------|-----------|
| Contractual delivery $\dot{n}_{\text{H}_2, \text{agreement}}$ | 6.5                 | NL/h            | Chosen    |
| Initial tank inventory $m_{\text{tank},0}$                    | 0                   | kg              | Chosen    |
| Spot purchase penalty $\kappa_{\text{buy}}$                   | 1.5                 | —               | Chosen    |
| Terminal liquidation credit $\kappa_{\text{sell}}$            | $\frac{1}{9}$       | —               | Chosen    |
| Tank inventory cap (case 1 / 2)                               | $\infty$ 13.5 grams | Chosen / Update |           |
| Updated variables                                             | Value               | Unit            | Reference |
| Control segments $N_t$                                        | 20                  | —               | Chosen    |

Table 5.3: Tank and delivery-agreement settings for the constant-delivery results (in addition to table 5.1).  $N_t$  is refined to 20 segments (10-day resolution) for the tank scenarios.

sense: The unrestricted baseline still favors steady potentiostatic-isothermal controls, with voltage and inlet temperature locked at the exothermal optimum that extracts revenue up to the nickel-migration break limit. So to make use of the dynamic flexibility of the model.

### 5.3 Constant delivery rate

The next result uses the storage-tank implementation from section 4.1.4 with a fixed contractual delivery rate. All other settings follow table 5.1; the additional tank parameters are listed in table 5.3.

The agreement rate  $\dot{n}_{\text{H}_2, \text{agreement}} = 6.5$  NL/h per cell sits below the unconstrained baseline production at  $t = 0$  (section 5.2), so surplus hydrogen can be stored rather than sold immediately. Scaled to a production setup of two stacks with about 150 cells each, the combined daily delivery is approximately 4.3 kg H<sub>2</sub>. The spot-purchase penalty  $\kappa_{\text{buy}} = 1.5$  fines under-delivery at 50% of the contract hydrogen price. The terminal liquidation credit  $\kappa_{\text{sell}} = \frac{1}{9}$  values leftover inventory at roughly 11% of the contract price, reflecting competition with lower-cost *black* hydrogen on an open market.

#### 5.3.1 Optimum at low liquidation credit

With  $\kappa_{\text{sell}} = \frac{1}{9}$ , terminal resale is so weak that the optimizer prefers to hoard surplus hydrogen rather than produce aggressively and liquidate at the horizon. Figure 5.4 shows the optimum at this liquidation credit: A single pot-iso pair

$$U^* \approx 1.289 \text{ V}, \quad T_{\text{in}}^* \approx 733.5 \text{ }^\circ\text{C},$$

held for all 200 days. Initial production ( $\approx 8.1$  NL/h) exceeds the agreement rate, so the tank fills monotonically to a peak of  $\approx 0.27$  kg around day 170. As degradation lowers production below 6.5 NL/h, the inventory buffer covers the shortfall; no spot purchases occur. Because unlimited storage is free in the model and  $\kappa_{\text{sell}}$  is small, the optimizer does not schedule galvanotactic intervals, but instead chooses the constant gentle operation minimizes salvage loss, but primarily minimizing the added resistance induced by degradation, and using the tank inventory to operate at lower production rates near the end.

The accumulated revenue is  $\approx 12.36$  EUR and the terminal salvage lost is  $\approx 0.34$  EUR; the terminal liquidation credit contributes only marginally. The net economy is  $\approx 12.02$  EUR on the 20 cm<sup>2</sup> cell over 200 days.

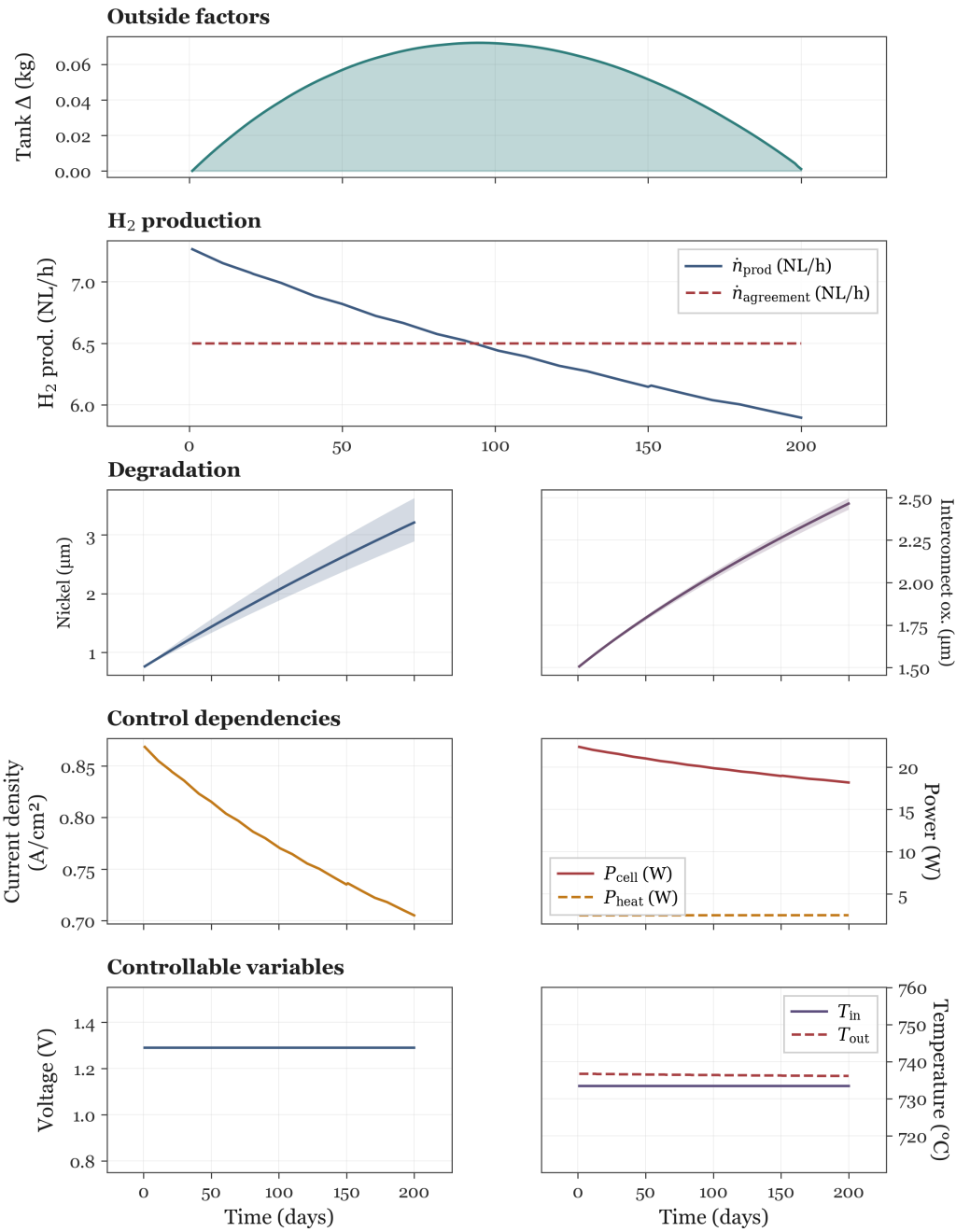


Figure 5.4: Constant-delivery optimum at  $\kappa_{\text{sell}} = \frac{1}{9}$  and  $\dot{n}_{\text{H}_2, \text{agreement}} = 6.5 \text{ NL/h}$  ( $N_t = 20$ ; table 5.3). Single controls  $U^* \approx 1.289 \text{ V}$ ,  $T_{\text{in}}^* \approx 750 \text{ }^{\circ}\text{C}$ ; tank inventory peaks at  $\approx 0.27 \text{ kg}$  (day 170) and remains near  $\approx 0.26 \text{ kg}$  at the horizon. Running revenue  $\approx 11.64 \text{ EUR}$ , terminal salvage  $\approx 1.50 \text{ EUR}$ , net economy  $\approx 13.36 \text{ EUR}$ .

### 5.3.2 Limited tank reserve via scheduled mode change

The unconstrained optimum in figure 5.4 fills storage to roughly 72 grams, which is more than many practical tanks would hold. Figure 5.5 illustrates an alternative from the same contract and  $\kappa_{\text{sell}}$ , with an added tank constraint, of maximum content to be  $\approx 13.5$  grams. The schedule has three phases:

1. **Gentle pot-iso** ( $U$ ,  $T_{\text{in}}$  slightly below the  $\kappa_{\text{sell}} = \frac{1}{9}$  optimum) fills the tank more slowly and reduces early degradation stress.
2. **Galvanostatic window** (fixed  $U$ , stepped  $T_{\text{in}}$  tuned to track  $\dot{n}_{\text{H}_2, \text{agreement}}$ ) holds delivery while inventory approaches the cap.
3. **Closing pot-iso** (return to  $U^*$ , hold  $T_{\text{in}}$  at the galvanotactic endpoint) lets production track degradation while inventory stays bounded.

Operationally it is preferable when inventory capacity is finite or carrying reserve hydrogen has off-model costs, even though the economic objective may differ slightly from figure 5.4. However, the important takeaway from this result is that the solver chooses to run at galvanotactic and potentiostatic operation, and adjusts by temperature.

The accumulated revenue is approximately 12.30 EUR, with a terminal salvage loss of about 0.34 EUR, resulting in a net revenue of roughly 11.96 EUR. Notably, the main penalty for deviating from the gentle potentiostatic-isothermal strategy appears as reduced revenue rather than a larger salvage loss. This reduction stems from increased resistance as degradation accelerates. Both strategies ultimately deliver the same amount of hydrogen and reach similar final cell states, operating at a constant thermoneutral voltage—where hydrogen production relative to power cost is maximized. However, during galvanostatic operation, the cell experiences faster degradation and higher resistance, which in turn increases the power cost per unit of hydrogen produced.

## 5.4 24-hour cycle with varying power price

The final scenario shortens the horizon to one day and replaces the flat power price with a time-varying profile. The cosine form in table 5.4 is a stylized stand-in for solar-driven day-night spreads: Electricity is cheapest around midday and most expensive around midnight, with a relative swing  $a = 0.9$  giving roughly a  $19\times$  ratio between the midnight maximum and the midday minimum. Together with the contractual delivery setup from section 5.3, this case tests whether the optimizer shifts production toward low-price periods while still meeting the agreement.

Terminal liquidation is disabled ( $\kappa_{\text{sell}} = 0$ ), and the cyclic inventory constraint equation (4.9) is enforced so that  $m_{\text{tank}}(0) = m_{\text{tank}}(T)$ . The day can therefore be studied as a repeating operating window without end-of-horizon sell-off. The updated settings are listed in table 5.4.

To help the optimizer converge to a sensible solution continuity penalties has been added to the objective, see appendix 6 for more details. The weights are defined in table 5.4, and the reasoning behind the choices is discussed in section 6.4.2.

Using these settings, temperature remains the primary free control. The optimal trajectory is shown in figure 5.6. Production is highest when the power price is low and reduced when prices rise at night. The response is qualitatively as expected for load shifting: The optimizer modulates inlet temperature rather than exploiting both voltage and temperature.

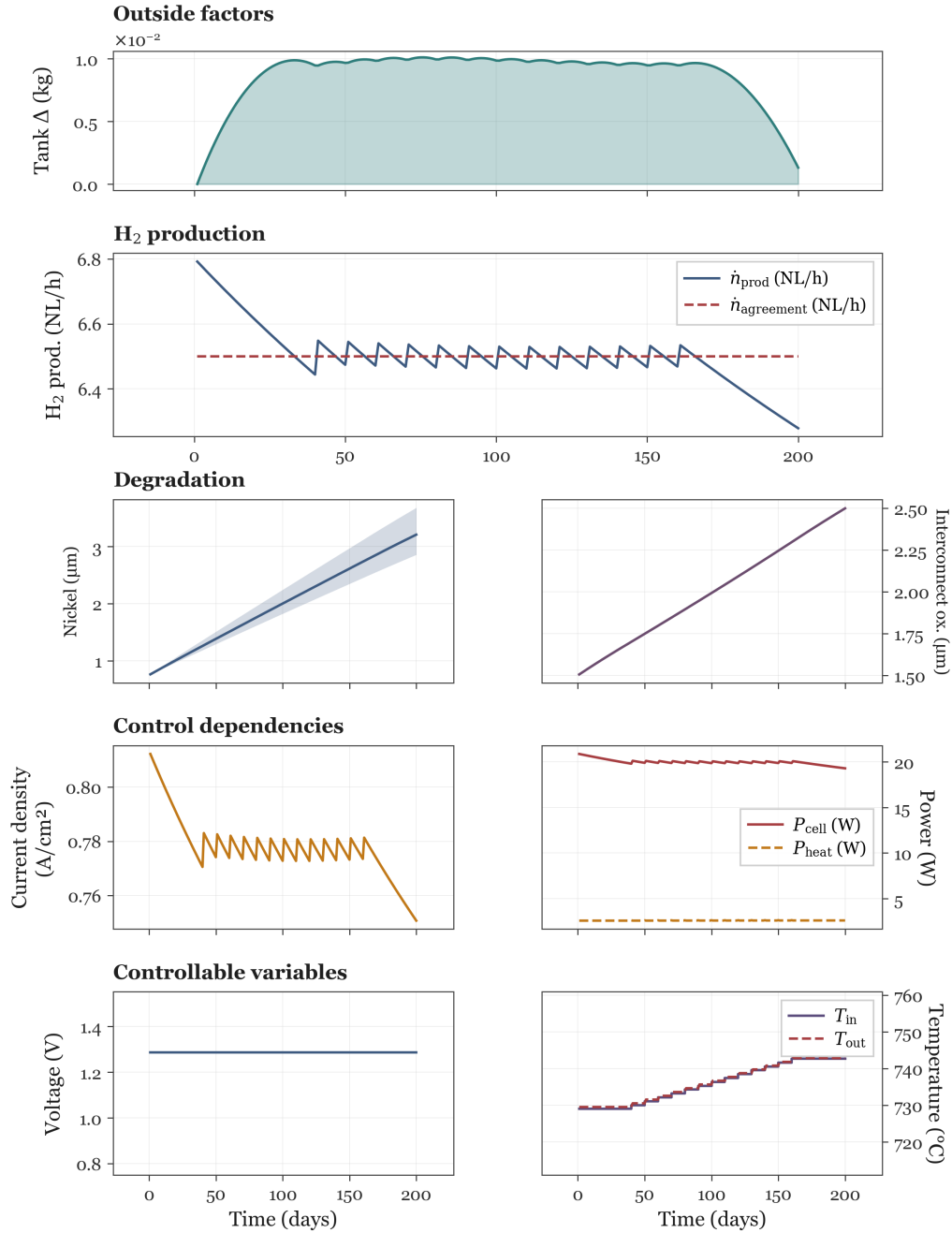


Figure 5.5: Limited capacity tank replay at  $\kappa_{\text{sell}} = \frac{1}{9}$  and  $\dot{n}_{\text{H}_2, \text{agreement}} = 6.5 \text{ NL/h}$  (table 5.3). Reduced pot-iso fill (segments 0-3;  $U \approx 1.289 \text{ V}$ ,  $T_{\text{in}} \approx 735^\circ \text{C}$ ), galvanostatic middle via stepped  $T_{\text{in}}$  at  $U \approx 1.289 \text{ V}$  (segments 4-10, ramp to  $\approx 748^\circ \text{C}$ ), and closing pot-iso hold at that same  $T_{\text{in}}$  (segments 11-19).



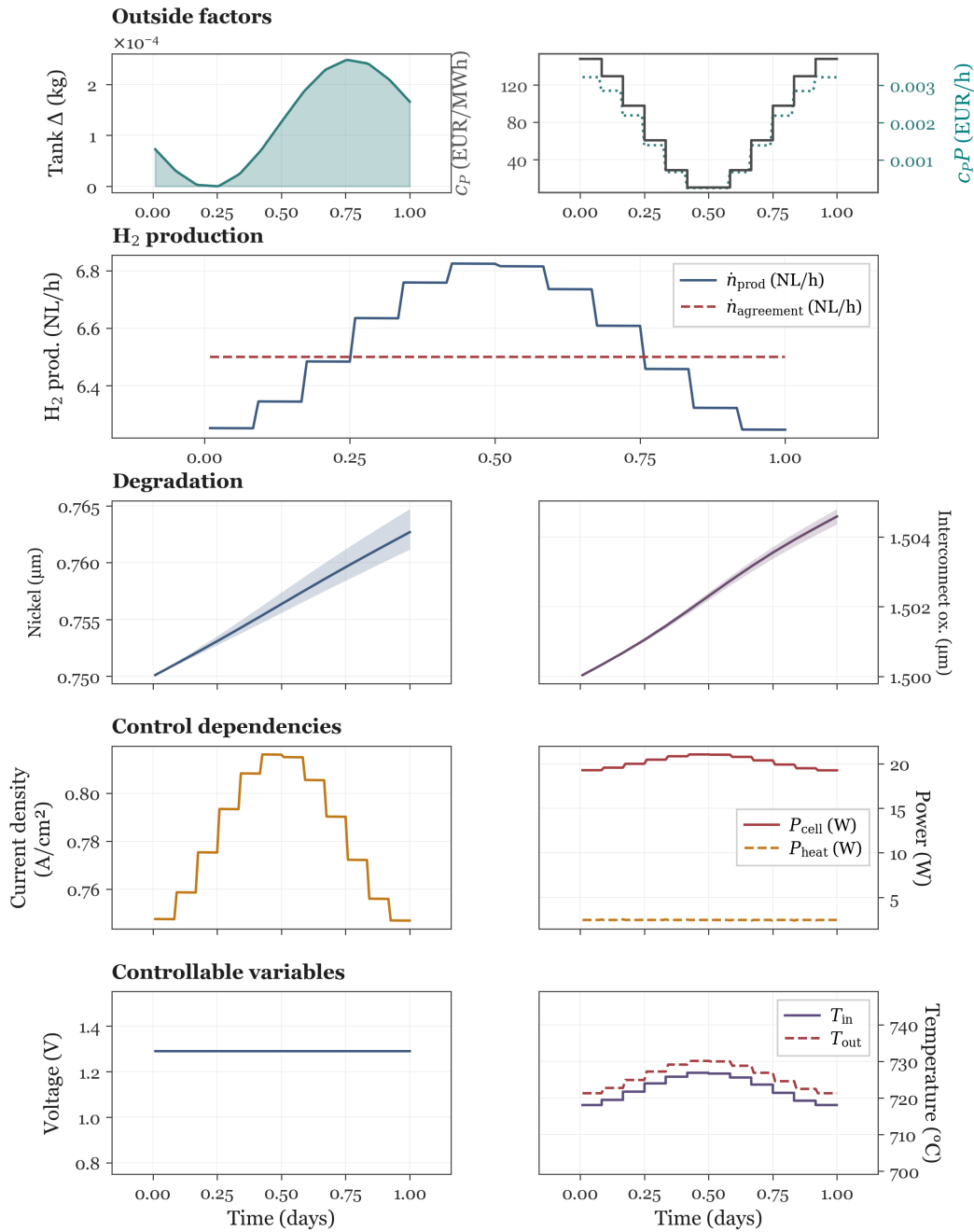


Figure 5.6: 24-hour cyclic optimum under a daily cyclic power price ( $N_t = 12$ ; table 5.4). Fixed  $U \approx 1.289 \text{ V}$  ( $w_U = 0.1$ );  $T_{\text{in}}$  is modulated ( $w_T = 10^{-5}$ ) to shift production toward low-price midday segments ( $c_P$  and  $c_P P$  in **Outside factors**). Peak  $\dot{n}_{\text{H}_2, \text{prod}} \approx 6.8 \text{ NL/h}$  versus  $\approx 6.2 \text{ NL/h}$  overnight; the agreement rate  $6.5 \text{ NL/h}$  is met on average via cyclic tank use ( $\kappa_{\text{sell}} = 0$ ). Running revenue  $\Phi \approx 0.062 \text{ EUR}$ , terminal salvage  $\Psi \approx 1.91 \text{ EUR}$ .

| Setting                                     | Value                                                                                      | Unit      | Reference                                |
|---------------------------------------------|--------------------------------------------------------------------------------------------|-----------|------------------------------------------|
| Temperature continuity penalty $w_T$        | $10^{-5}$                                                                                  | —         | Chosen<br>(appendix 6,<br>section 6.4.2) |
| Voltage continuity penalty $w_U$            | 0.1                                                                                        | —         | Chosen<br>(appendix 6)                   |
| Updated variable                            | Value                                                                                      | Unit      | Reference                                |
| Horizon $T$                                 | 24                                                                                         | h         | Chosen                                   |
| Control segments $N_t$                      | 12                                                                                         | —         | Chosen                                   |
| Terminal liquidation $\kappa_{\text{sell}}$ | 0                                                                                          | —         | Chosen                                   |
| Initial tank inventory $m_{\text{tank},0}$  | $m_{\text{tank},T}$                                                                        | —         | Periodic (4.1.4)                         |
| Power price $c_{P,k}$                       | $\bar{c}_P(1 - a \cos \theta_k)$ ,<br>$\theta_k = 2\pi(h_k - 12 \text{ h})/(24 \text{ h})$ | EUR/(W·s) | Chosen                                   |
| Mean power price $\bar{c}_P$                | 0.079                                                                                      | EUR/kWh   | [19]                                     |
| Relative price swing $a$                    | 0.9                                                                                        | —         | Chosen                                   |

Table 5.4: Updated model settings for the 24-hour cyclic scenario. No terminal sale of hydrogen is allowed ( $\kappa_{\text{sell}} = 0$ ), and the initial tank content is enforced as a periodic boundary. Twelve control segments are used for the day. The piecewise-constant power price on segment  $k$  follows a single cosine cycle over 24 h, with  $h_k$  the hour-of-day at the segment midpoint; the minimum  $c_{P,k} = \bar{c}_P(1 - a)$  occurs at midday ( $h_k = 12 \text{ h}$ ) and the maximum  $\bar{c}_P(1 + a)$  at midnight, modelling lower prices when solar supply is high ( $a = 0.9$  gives  $\approx 19 \times$  min/max spread). The continuity weights  $w_U$  and  $w_T$  enter the soft penalty appendix 6, including the last→first segment wrap consistent with the repeating day; with potentiostatic voltage only  $w_T$  is effective.

The economic decomposition of the optimized run is as follows: Integrated running revenue over the 24-h window is  $\Phi \approx 0.062 \text{ EUR}$  and the terminal salvage term is  $\Psi \approx 1.91 \text{ EUR}$ , giving a total objective  $J \approx 1.97 \text{ EUR}$ . Because the degradation states are seeded from the same 100-day burn-in as the long-horizon cases, the salvage value at the start of the day is already  $\Psi_0 \approx 1.91 \text{ EUR}$ ; the additional salvage loss over the first day is therefore only  $\approx 0.002 \text{ EUR}$ , so the net economy from operation is effectively  $\approx 0.060 \text{ EUR}$  for the  $20 \text{ cm}^2$  cell.

This outcome is primarily shaped by two modelling choices: the cyclic tank constraint, which ensures no terminal credit for inventory and enforces production/storage balance, and the use of continuity penalties. For  $w_T = 10^{-5}$ , the optimizer finds an intermediate solution with a daily cyclic temperature swing and load-shifting behavior. As  $w_T$  is increased ( $w_T \gg 0$ ), the solution approaches potentiostatic-isothermal operation with nearly flat  $T_{\text{in}}$ ; as  $w_T \rightarrow 0$ , the solution becomes increasingly piecewise-constant, allowing sharper temperature changes between control segments, leading to a cell that is only turned on during low power price hours.

## 5.5 Sensitivity analysis

### 5.5.1 Liquidation sensitivity

The constant-delivery result at  $\kappa_{\text{sell}} = \frac{1}{9}$  leaves negligible terminal inventory, whereas the unconstrained baseline favors higher current and temperature–behaviour that corresponds formally to  $\kappa_{\text{sell}} = 1$ . A sweep over the terminal liquidation credit was therefore carried out to identify when the optimizer prefers to drive the cell toward the nickel break limit rather than meet delivery through gentler operation. The outcome is

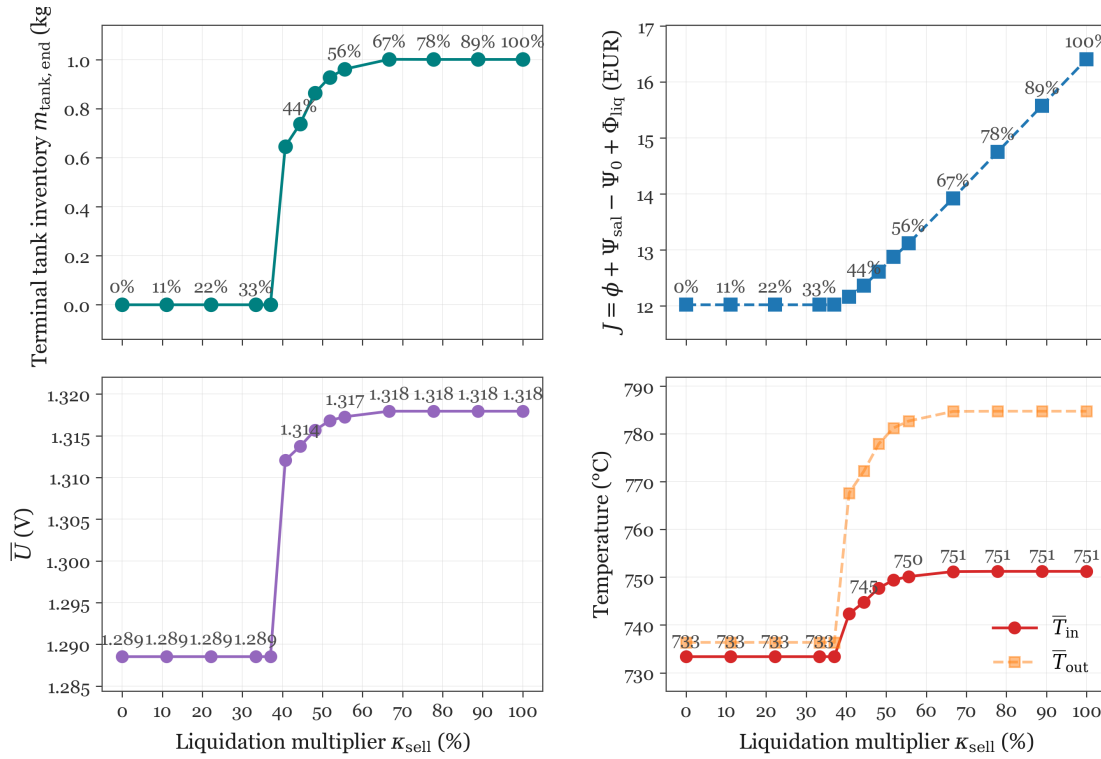


Figure 5.7: Terminal liquidation credit  $\kappa_{\text{sell}}$  sensitivity for the constant-delivery tank scenario ( $\dot{n}_{\text{H}_2, \text{agreement}} = 6.5 \text{ NL/h}$ , pot-iso controls, no tank cap). Top panels: Terminal tank inventory and net objective  $J = \phi + \Psi_{\text{sal}} - \Psi_0 + \Phi_{\text{liq}}$ ; bottom panels: Mean voltage and inlet/outlet temperature. For  $\kappa_{\text{sell}} \lesssim 37\%$  the optimizer leaves the tank empty and holds moderate  $U$  and  $T_{\text{in}}$ . Above  $\approx 40\%$  it hoards surplus hydrogen and drives the cell toward the nickel break limit.

shown in figure 5.7. For clarity, every point in the figure was obtained with potentiostatic-isothermal controls ( $U$  and  $T_{\text{in}}$  constant over the horizon), which is the mode the solver selects when no tank-cap constraint is imposed.

Figure 5.7 exhibits a clear transition: beyond a threshold, the marginal cost of additional production exceeds the value of terminal resale, and the optimizer drains the tank instead of filling it. This transition lies between  $\kappa_{\text{sell}} \approx 37\%$  and  $\kappa_{\text{sell}} \approx 40\%$ , corresponding to an effective resale price of roughly 2.92-3.16 cent/kWh. Above the breakpoint the optimum moves rapidly toward the degradation limit. Immediately below it, a narrow band remains in which modest over-production is still marginally worthwhile without triggering break-limited operation. In that region, rising resistance makes further production only weakly attractive relative to power cost, and the salvage term contributes to the trade-off: the liquidation credit is not yet high enough to justify the extra degradation.

### 5.5.2 Power price sensitivity

Following this sweep,  $\kappa_{\text{sell}} = 40\%$  was selected for the remaining one-parameter studies. The power price  $c_P$  was varied next. Because terminal liquidation acts as a post-horizon hydrogen price, lowering  $\kappa_{\text{sell}}$  is qualitatively similar to raising  $c_P$ ; the  $c_P$  sweep therefore mirrors the liquidation response in reverse. Figure 5.8 confirms this expectation.

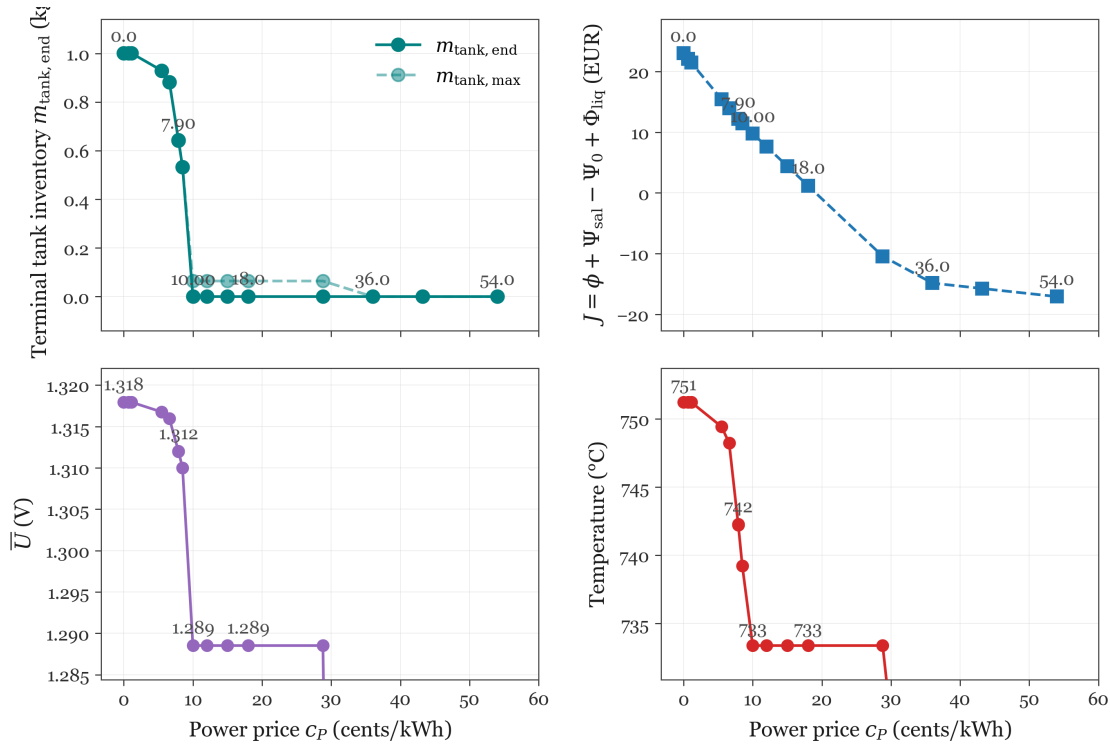


Figure 5.8: Power price  $c_P$  sensitivity at fixed  $\kappa_{\text{sell}} = 40.74\%$  and  $\dot{n}_{\text{H}_2, \text{agreement}} = 6.5 \text{ NL/h}$  (pot-iso controls). Top panels: Terminal and peak tank inventory and net objective  $J$ ; bottom panels: Mean voltage and inlet temperature. At the baseline price ( $\approx 7.9 \text{ cent/kWh}$ , starred) the optimizer hoards  $\approx 0.53 \text{ kg}$  by running the cell hard. As  $c_P$  rises, inventory falls and  $J$  decreases while production is kept alive down to  $\approx 29 \text{ cent/kWh}$ . Above  $\approx 36 \text{ cent/kWh}$  electrolysis shuts down and contract hydrogen is purchased at spot.

Two distinct regimes appear. In the first, the optimum shifts from break-limited operation toward an empty terminal tank while still maintaining production to satisfy the delivery obligation-electrolysis remains cheaper than spot purchase even as inventory is run down. In the second, at sufficiently high  $c_P$ , production is abandoned entirely: the cell idles at minimum temperature and zero current, and all contractual hydrogen is bought on the spot market.

### 5.5.3 salvage price sensitivity

The sensitivity to the salvage price  $c_{\text{sal}}$  was examined at the same reference point ( $\kappa_{\text{sell}} \approx 40\%$ ), where terminal inventory also begins to rise. Figure 5.9 shows the result.

The  $c_{\text{sal}}$  response again separates into two steps. At low salvage prices the optimum is dominated by resistance growth: degradation raises ohmic losses and the terminal salvage term is secondary. As  $c_{\text{sal}}$  increases, salvage loss weighs more heavily until additional production is no longer justified relative to the value destroyed in the stack. At the highest prices examined, spot purchase becomes cheaper than operating the cell.

### 5.5.4 Delivery rate sensitivity

Finally, the contractual delivery rate  $\dot{n}_{\text{H}_2, \text{agreement}}$  was varied to test which control adjusts when terminal liquidation forces the tank empty. Results in section 5.4 and section 5.3 suggest that, under potentiostatic

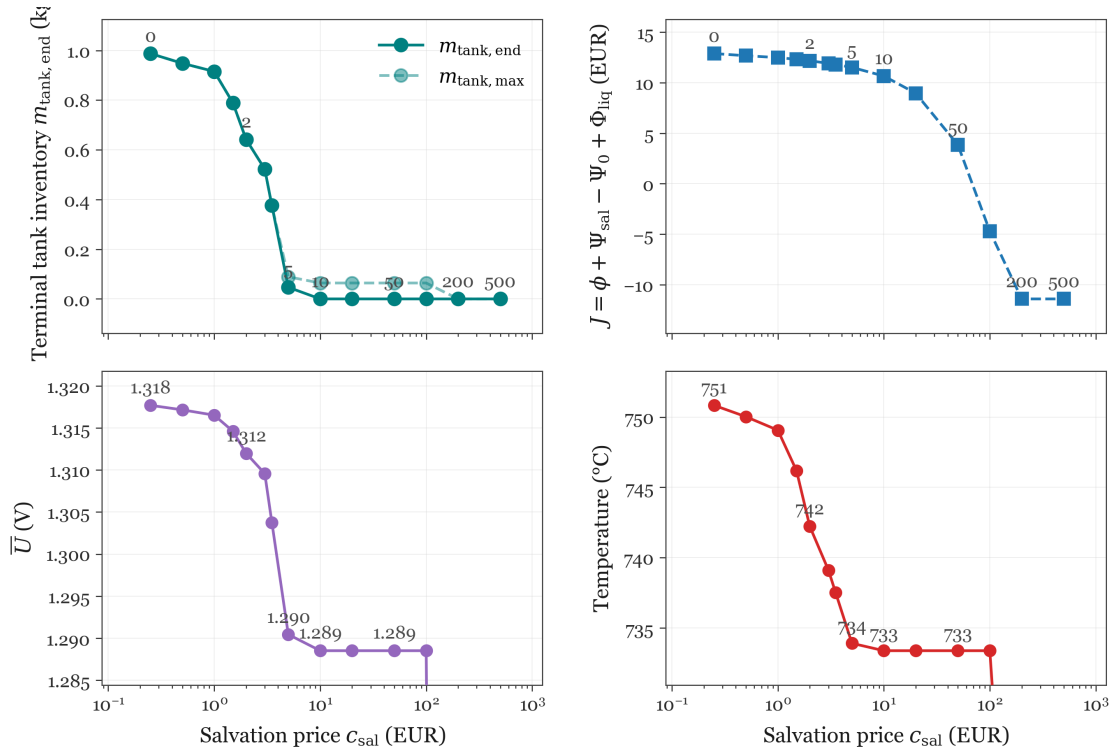


Figure 5.9: salvage price  $c_{\text{sal}}$  sensitivity at fixed  $\kappa_{\text{sell}} = 40.74\%$  and  $\dot{n}_{\text{H}_2, \text{agreement}} = 6.5 \text{ NL/h}$  (pot-iso controls). Top panels: Terminal and peak tank inventory and net objective  $J$ ; bottom panels: Mean voltage and inlet temperature. At the baseline price (2 EUR, starred) the optimizer hoards  $\approx 0.64 \text{ kg}$  by running the cell hard. As  $c_{\text{sal}}$  rises, inventory falls and  $J$  decreases; above  $\approx 10 \text{ EUR}$  the optimum reverts to gentle pot-iso operation with negligible terminal inventory.

operation, temperature alone should modulate production while voltage stays fixed. Figure 5.10 confirms this pattern at  $\kappa_{\text{sell}} = 0$ .

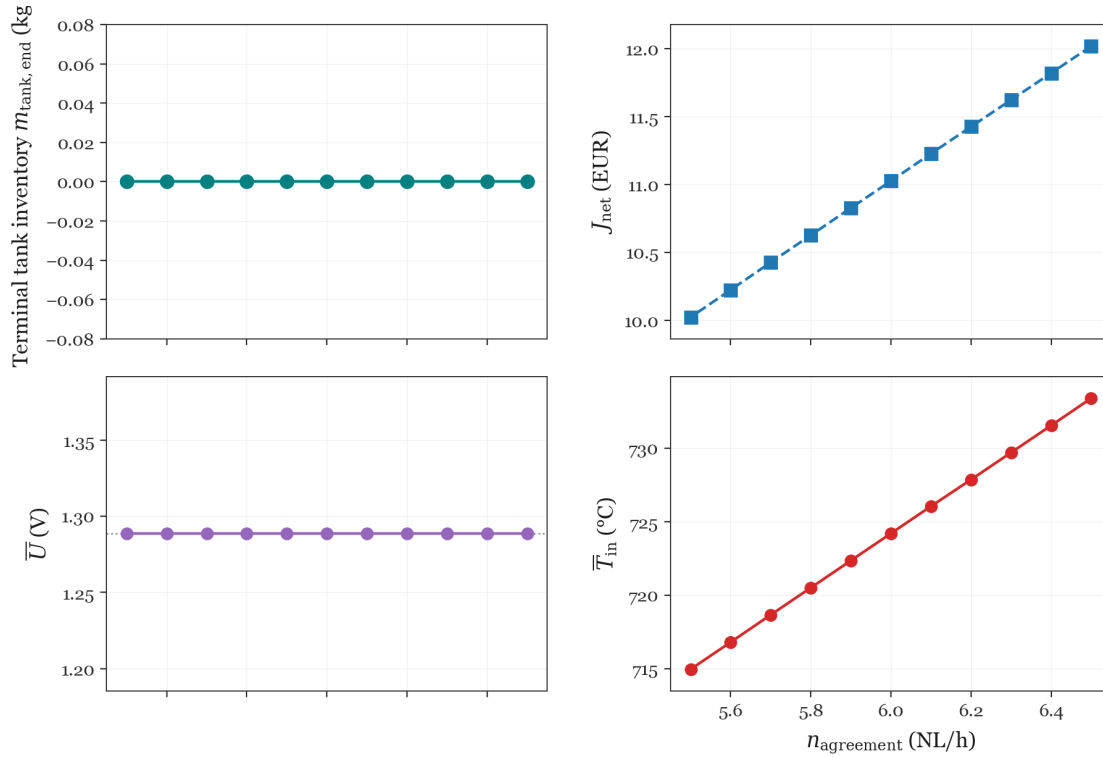


Figure 5.10: Contractual delivery rate sensitivity at  $\kappa_{\text{sell}} = 0$  and pot-iso controls ( $N_t = 20$ ).  $U$  is fixed to the 6.5 NL/h optimum ( $\approx 1.289$  V);  $T_{\text{in}}$  is reduced for lower agreement rates so that  $m_{\text{tank, end}} = 0$ . Top panels: Terminal tank inventory and net objective  $J_{\text{net}}$ ; bottom panels: Mean voltage and inlet temperature.  $J_{\text{net}}$  rises almost linearly from  $\approx 10.0$  to  $\approx 12.0$  EUR as  $\dot{n}_{\text{H}_2, \text{agreement}}$  increases from 5.5 to 6.5 NL/h. Voltage stays flat while  $\bar{T}_{\text{in}}$  falls from  $\approx 734$  to  $\approx 715$  °C.

# 6 Discussion and Conclusion

## 6.1 Introduction

The results from section 5.1 to section 5.4 show that the optimizer generally favors potentiostatic-isothermal operation, adjusting temperature as needed to meet delivery or shift production to lower-cost periods. In the unconstrained baseline (section 5.2), exothermal operation is optimal but limited by the salvage term and nickel break cap. Otherwise, delivery contracts and price profiles dominate, with the salvage term having less impact. The following sections interpret these findings, discuss parameter choices, explain modelling assumptions, and outline limitations and future directions.

## 6.2 Interpretation of optimal operating strategies

Taken together, the results indicate that extra control freedom is used only when contractual delivery or time-varying prices force a departure from potentiostatic-isothermal operation. The subsections below follow that logic.

### 6.2.1 Baseline without delivery obligations

The first question is how much of the available control freedom is actually used. For the unrestricted baseline (section 5.2), the answer is very little: Without a delivery floor, maximizing  $\int R dt + \Psi$  favours steady efficient production, and the optimal mode remains potentiostatic-isothermal. The additional decision variables are largely redundant in this scenario.

The baseline optimum is nevertheless exothermal:  $U^* \approx 1.32 \text{ V}$  exceeds the thermoneutral voltage at  $T_{\text{in}}^* \approx 751^\circ\text{C}$ , so heat release raises the temperature along the channel (roughly  $751$  to  $780^\circ\text{C}$  at  $t = T$ ; figure 5.3). Revenue is maximized up to the nickel-migration break limit at  $5 \mu\text{m}$ ; interconnect oxidation stays near  $3.6 \mu\text{m}$  and is not life-limiting. Because the chosen voltage is effectively pinned at this exothermal operating point,  $U$  is also constant across the other long-horizon cases considered. A repeat study with voltage fixed potentiostatic and only one remaining degree of freedom, either piecewise galvanostatic current with inferred temperature, or piecewise isothermal  $T_{\text{in}}$  with inferred current, would likely reproduce the same qualitative conclusions while reducing the decision space.

### 6.2.2 Effect of a fixed delivery contract

A second question concerns how the control variables are parametrized within each segment. In the present setup, both  $U$  and  $T_{\text{in}}$  are held constant over each control interval, simply because the NLP formulation is structured this way (section 4.2.6). In the fixed-delivery scenario (section 5.3), the production tends to fluctuate around the delivery agreement; in those periods, it would actually be more natural to hold the current constant—i.e., operate galvanostatically—instead of voltage or temperature. A natural next step would be to give the optimizer the freedom to decide, for each interval, whether to use potentiostatic, galvanostatic, or isothermal operation.

Even without this implementing, the current results already reveal when galvanostatic operation becomes relevant, and show how tank inventory enables or enforces these switches between operating modes.

### 6.2.3 Temperature jumps and practical implementation

Piecewise-constant optimal controls can imply large instantaneous changes in  $T_{\text{in}}$  between segments. Such jumps are difficult to implement on hardware and are not charged directly in the base formulation. In the 24-hour run, a squared continuity soft penalty was added retrospectively (appendix 6). A more physical alternative would be to include the auxiliary preheat power required to move from one inlet set-point to the next within a segment duration, or to model thermal cycling as an additional degradation driver.

## 6.3 Implied lifetime and proximity to break limits

Across the long-horizon runs, current densities are relatively high and the modelled degradation states approach the chosen break limits within 200 days (table 5.1). In the baseline replay (section 5.2), mean nickel-migration thickness reaches the  $5\text{ }\mu\text{m}$  cap at  $t = T$  while interconnect oxidation remains well below its  $10\text{ }\mu\text{m}$  limit. The window is far shorter than stack lifetimes reported in the SOEC literature. Beyrami et al. predict lifetimes above 50,000 h at  $0.6\text{ A/cm}^2$  when component failure criteria are enforced [3], and later embed stack lifetimes of 4–5 years into plant-scale economic optimization [4]. Giridhar et al. obtain replacement schedules on the order of 2 to 5 years depending on electricity price and operating mode [13].

The gap is partly expected. This thesis maximizes revenue plus salvage over a fixed 200-day horizon rather than full lifetime, and the delivery scenarios include galvanostatic periods that accelerate the modelled degradation mechanisms. It nevertheless suggests that one or more modelling choices may be too permissive. The fitted nickel-migration rate law (section 3.2.2) is a simple empirical relation and may overpredict thickness growth at the currents seen in optimization.

### 6.3.1 Revenue-Salvage balance in the objective

A complementary explanation involves the way the scalar objective is formulated. In the baseline case (section 5.2), the cell still reaches its break limit, yet the salvage term is not driven to zero. This is a consequence of decisions made early in the project: specifically, the choice of  $\Psi$  was intended to provide a simple trade-off between maximum and mean degradation. As a result, the balance between ongoing revenue and the terminal salvage term may not be fully aligned. If either  $c_{\text{cell}}$  or  $\alpha_{\Psi}$  in equation (4.4) underestimate the true cost of approaching end-of-life, the optimizer may select aggressive (exothermal) temperature and current profiles that literature-based lifetime studies would typically avoid.

In this baseline, monotonic heating along the channel results in nearly uniform nickel-migration growth (figure 5.3), so failure occurs almost uniformly rather than at a single hotspot—but the  $5\text{ }\mu\text{m}$  cap is still eventually reached at  $t = T$ . Meanwhile, ohmic losses increase linearly with resistance from nickel migration and interconnect oxidation, but the degradation rates themselves tend to slow as the states evolve. This means that a small increase in temperature can generate disproportionately more revenue for a given increment in resistance. Since the salvage term is bounded, it does not increase sharply enough throughout state space to fully offset this trend, unless  $\alpha_{\Psi}$  is set extremely high.

One alternative is to replace the hard break limits, used in this thesis, with a failure probability that scales revenue at each instant. A schematic form is given in equation (6.1).

$$\int_0^T (1 - P_{\text{fail}}(U, T, I, \mathbf{x})) R dt + \mathbb{E}[\tau_{\text{life}} | \mathbf{x}_T] \mathbb{E}[R | \mathbf{x}], \quad (6.1)$$



where  $P_{\text{fail}}$  is the probability that the cell reaches a failure criterion given the controls and degradation state  $\mathbf{x}$ , and  $\mathbb{E}[\tau_{\text{life}} \mid \mathbf{x}_0]$  is the expected remaining lifetime from the initial state.

Rather than stopping the simulation at a hard cap, revenue would be discounted by the instantaneous risk of failure, and the second term would reward expected production over life. In principle,  $P_{\text{fail}}$  could rise sharply once degradation states pass critical thresholds, giving a stronger, and nonlinear, penalty for operating too aggressively than the linear resistance increase combined with the bounded salvage term. Such a formulation would also ease the transition from cell-scale models to stack- and plant-scale objectives.

Such a model would require much more data than is currently available. However, an approximate model might be feasible with a comprehensive literature review on the probability of failure for each cell component, as this information is likely partially known. From there, it could be possible to construct a probabilistic model, though this would require substantial additional work. With such a model, it would then be possible to consider whether to optimize for expected revenue or for a different objective, such as a lower quantile of performance, depending on the desired risk tolerance.

Another, simpler approach to this problem, would be to keep the current structure, but use A salvage term that multiplies the maximum degradation by the mean degradation. This would yield a squared salvage term that simultaneously weighs both mean and maximum degradation.

## 6.4 Justification of project-specific settings

The results chapter marks several inputs as *chosen* in table 5.1, table 5.3, and table 5.4. Settings for the horizon, control segments, finite-volume mesh, operating bounds, and  $\alpha_{\Psi}$  are already motivated in section 5.1 and the grid-independence study (section 4.2.3), which covers the baseline case (table 5.1). The subsections below focus on the delivery and cyclic scenarios.

### 6.4.1 Delivery setup

Table 5.3 lists both delivery scenarios: Case~1 uses an uncapped tank ( $\infty$ ), which isolates the contract and penalty structure from inventory-cap effects; case~2 imposes a maximum inventory of 13.5 g, chosen as a practical reserve level after the unconstrained optimum in figure 5.4 fills storage to roughly 72 g.

The contractual delivery rate and the multipliers  $\kappa_{\text{buy}}$  and  $\kappa_{\text{sell}}$  are strongly coupled to the optimal control. Exploratory runs in section 5.3 guide the reported values. If  $\dot{n}_{\text{H}_2, \text{agreement}}$  is raised above the unconstrained production rate at the baseline optimum (section 5.2), the solver eventually drives the cell to the break limits, because sustaining that throughput is too aggressive. If the agreement rate is set below the baseline terminal production rate ( $\approx 7.5 \text{ NL/h}$  at  $t = T$  in section 5.2), the solution depends mainly on  $\kappa_{\text{sell}}$ : With  $\kappa_{\text{sell}} = 1$ , the optimum coincides with the unrestricted baseline (including terminal tank liquidation); with  $\kappa_{\text{sell}} = 0$ , inlet temperature is reduced so that  $m_{\text{tank}}(T) = 0$ . Intermediate values of  $\kappa_{\text{sell}}$  interpolate between these extremes. The interpolation is shown in figure 5.7, where it is clear that a breaking point exists near  $\kappa_{\text{sell}} \approx 33\%$ : Below this value the optimizer hoards hydrogen in the tank (figure 5.4); above it, production aligns with the unrestricted baseline (section 5.2). Figure 5.5 shows how the capped case forces a three-phase schedule that respects the 13.5 g limit while still meeting the contract.

Together, the entries in table 5.3 were selected because they expose informative tank behaviour at  $\dot{n}_{\text{H}_2, \text{agreement}} = 6.5 \text{ NL/h}$ : Hoarding at low  $\kappa_{\text{sell}}$  with unlimited storage (figure 5.4) and a practical capped schedule with galvanostatic tracking (figure 5.5).

### 6.4.2 Cyclic setup

For the 24-hour case (table 5.4), the horizon and  $N_t$  were scaled to one day with twelve two-hour segments. Terminal liquidation is set to  $\kappa_{\text{sell}} = 0$  so that the periodic inventory constraint from section 4.1.4 governs end-of-day inventory rather than a sell-off term.

The continuity penalty (appendix 6) was added because exploratory NLP runs without it often converged to degenerate solutions with the same objective value but large, unphysical segment-to-segment jumps in  $T_{\text{in}}$ . The reported weight  $w_T = 10^{-5}$  keeps that term small enough for load shifting to remain worthwhile. With the inlet bounds  $[700, 900]^\circ\text{C}$  (200 K window), a single 10 K step (5% of that window) contributes  $w_T(\Delta T_{\text{in}})^2 \approx 0.001$  EUR to  $P_{\text{cont}}$ , i.e.  $\approx 20\%$  of the mean running revenue per two-hour segment ( $\approx 0.0051$  EUR). For the reported optimum, the total penalty is  $P_{\text{cont}} \approx 3.3 \times 10^{-4}$  EUR ( $\approx 0.5\%$  of the daily  $\Phi$ ), and typical adjacent steps ( $|\Delta T_{\text{in}}| \approx 1.9\text{--}2.3$  K) cost  $\lesssim 1\%$  of mean segment revenue each. The cosine power-price profile provides a simple, interpretable day–night swing; the inlet-temperature amplitude was checked in the sweep in section 5.4.

## 6.5 Influence of modelling assumptions

The assumptions collected in section 3.3 are introduced to keep the average cell model tractable for gradient-based optimal control. They are not neutral with respect to the results in section 5.2–section 5.4: The simplifications below directly shape which operating modes appear optimal and how aggressively the optimizer can run the cell.

### 6.5.1 Average cell model and scale

Items 14–16 (section 3.1.1) mean our model works with a single averaged channel, so the optimized voltage ( $U$ ) and inlet temperature ( $T_{\text{in}}$ ) are the same everywhere in the system. This is different from what you would see if you looked at a more detailed model where conditions can vary in different parts of the stack [21, 23]. The model also simplifies things by scaling up delivery and using a single, lumped value for heating power ( $P_{\text{heat}}$ ), which skips details like the way gas is routed through manifolds or heat is lost in exchangers. This makes the model more likely to favor temperature changes as a control strategy in some cases (section 5.3 and section 5.4). However, this simplification by itself is not what would produce odd or unrealistic results.

### 6.5.2 Electrochemical and thermal simplifications

Negligible concentration overpotential and Butler-Volmer kinetics (section 3.3, items 4, 8) keep the current-voltage relation smooth in the parameter range fitted by [17]. When explicit concentration terms are retained instead, mass-transport limitations add a current-dependent voltage loss: Mao et al. report that concentration overpotential from porous-electrode transport *cannot be neglected under high operating currents*, and that if steam supply falls below the electrochemical consumption rate the polarization curve warps upward [18]. Parametric SOEC models show the same sensitivity to steam fraction, porosity, and tortuosity [19]. Gas-phase conversion can likewise introduce flow-dependent overpotentials when compositions are non-uniform [21]. The present model omits these contributions, so the optimizer can raise voltage and current without that additional mass-transport penalty, consistent with the exothermal baseline at  $U^* \approx 1.32$  V (section 5.2, figure 4.3).

Dropping axial heat conduction (section 3.3, item 5) makes the along-channel temperature rise depend mainly on advection and local ohmic/activation heat release. Stack-scale models with resolved thermal fields

instead report non-uniform temperature distributions and outlet hot spots that can amplify stress risks [23]. That distinction matters for the argument that exothermal operation homogenises nickel-migration growth (figure 5.3, section 6.3): Conductive spreading would redistribute heat and could shift both the temperature profile and the preferred operating point away from the present exothermal optimum [24].

### 6.5.3 Degradation scope and time integration

Restricting dynamic degradation to nickel migration and interconnect oxidation, while fixing nickel agglomeration through  $\lambda_c \approx 0.55$  (section 3.3, items 9–10), narrows which failure modes the optimizer anticipates. The baseline replay reaches the nickel cap at  $t = T$  while interconnect oxidation stays secondary (section 5.2), so conclusions about exothermal preference and uniform migration are conditional on that mechanism ranking. If interconnect oxidation were dominant or agglomeration evolved on the optimization horizon, thermoneutral or lower-current strategies might look preferable (section 6.7).

Spatially resolved degradation states (item 11) are necessary for the observed along-channel profiles, but the fitted nickel-migration rate and hard break limits remain empirical choices. Together with implicit Euler time stepping (item 13), they set how quickly the optimizer can approach end-of-life within the 200-day window (section 6.3). A slower or threshold-sharper degradation law would likely reduce current density and temper the exothermal optimum without changing the broader scheduling logic seen in the delivery and cyclic cases.

## 6.6 Model and numerical limitations

The results above are useful for qualitative comparison of operating modes, but several modelling and numerical choices bound how far the trajectories should be extrapolated. How the physical assumptions in section 3.3 feed into the optimal strategies is discussed in section 6.5; the subsections below focus on overall scope, discretization, and solver effects.

### 6.6.1 Physical scope and mechanism coverage

The average cell model is a single one-dimensional channel with quasi-steady electrochemistry and temperature (section 3.3, table 3.1, section 3.1.1). Flow rates, pressure, and geometry are fixed; only  $U$  and  $T_{\text{in}}$  are optimized. That formulation is appropriate for asking how the stack-average cell responds to price and delivery incentives, but it does not resolve stack-to-stack thermal coupling, manifold maldistribution, heat-exchanger transients, or plant-level auxiliary loads beyond the lumped  $P_{\text{heat}}$  bookkeeping (see section 6.5).

Degradation is limited to nickel migration and interconnect oxidation (table 3.2). Other ageing paths—cathode delamination, chromium poisoning, seal leakage—are not modelled dynamically. Nickel agglomeration enters only as a fixed multiplier  $\lambda_c \approx 0.55$  on the exchange current (section 3.2.1), so any further cathode loss at high current is not captured. Together with the fitted nickel-migration rate law (section 3.2.2), this narrow mechanism set likely contributes to the short implied lifetimes discussed in section 6.3.

### 6.6.2 Spatial and temporal discretization

Spatial resolution and the scenario-dependent values of  $N_t$ ,  $T$ , and the continuity weights are discussed in section 6.4. In brief, grid-independence analysis (figure 4.4) indicates that the budget-limited mesh in table 5.1 accepts spatial error on the order of  $10^{-2}$ , which is adequate for comparing control modes but not for quantitative lifetime forecasting.

In time, controls are piecewise constant on  $N_t$  segments (section 4.2.6) and degradation is advanced with

implicit Euler on the same grid. The segment count varies between scenarios, so finer horizons do not automatically use finer control resolution.

### 6.6.3 Nonlinear programming and solution quality

The optimal control problem is transcribed as a single-shooting NLP solved with SciPy's trust-region method and adjoint gradients (section 4.2.6, appendix 5). Gradient checks support the implementation, but the problem is not guaranteed to be convex in practice. The delivery scenario, with tank limit (section 5.3), returned more distinct local minima with equal objective value (without considering continuity penalty), which shows that reported trajectories need not be unique without extra constraints such as minimum inventory, continuity cost, or an operating-reserve preference.

Hard break limits on degradation states stop the forward integration when thresholds are reached. That keeps the solver well posed but introduces a non-smooth boundary that can interact with the bounded salvage term (equation (4.4)). Finally, the rolling horizon  $T$  is finite and scenario-dependent (200 days versus 24 hours), so terminal salvage can dominate short windows even when running revenue drives the qualitative scheduling pattern (section 5.4).

## 6.7 Conclusions

A consistent theme across the results is potentiostatic-isothermal operation: voltage held near thermoneutral and inlet temperature kept constant (section 5.2, section 5.3). This is the gentlest control mode and the preferred point for efficient hydrogen production. When a fixed delivery obligation forces the control toward galvanostatic operation (section 5.3, figure 5.5), the preferred response is to adjust temperature only and keep voltage near thermoneutral (figure 5.10, section 5.4). When degradation and cell break limits become binding (section 5.2), exothermal operation is preferable: it spreads nickel migration more uniformly along the channel rather than concentrating it at the inlet (figure 5.3, section 6.3).

This last point depends strongly on the break limits assumed in the model (table 5.1). Nickel migration is the critical degradation mechanism here (section 3.2.2); if interconnect oxidation were life-limiting instead, the optimizer would likely favour thermoneutral operation to spread oxidation more evenly (figure 5.3, section 6.3).

The sensitivity analysis shows that the salvage term, as currently defined using the cell-area price (equation (4.4)), has limited influence on the optimum. An effect appears only in a narrow window where hydrogen price, power price, and other economic parameters align (figure 5.9, figure 5.7, figure 5.8). A much larger salvage weight could dominate the objective, but such weighting lacks physical or economic justification, so this aspect remains inconclusive. An alternative salvage formulation, as proposed in section 6.3, is a natural direction for future work.

## 6.8 Use of generative AI tools

Generative AI tools were used throughout this thesis. Aside from occasional clarification questions to ChatGPT [20], the main tool was Cursor [2], an AI-assisted code editor.

The tools were used first for code debugging and script generation. The core optimal-control implementation was written manually, with autocomplete and agent assistance from Cursor; subsequent debugging was also carried out with Cursor. Many of the bash submission scripts for optimization sweeps on the DTU HPC

cluster [11], together with plotting scripts and gradient test scripts, were generated by Cursor and checked manually before use.

Cursor was also used as a writing aid for the thesis. It helped improve the language of individual sections, check grammar, and draft supplementary prose in the mathematically heavy appendices (appendix 1, appendix 5); all equations and derivations were done manually, with the generated text serving as supporting exposition.

### **6.8.1 Discussion**

The use of generative AI in academic work is a pressing matter, and familiarity with such tools is increasingly expected in graduate projects [26]. In that sense, competent use of today's AI assistants can be seen as part of completing a modern Master's thesis. Here, the tools accelerated both coding and writing substantially. Without them, the project would likely be less complete, in part because a hand fracture slowed typing during a central period of the work. Coding agents helped keep development moving in that interval.

That period also illustrates the limits of the approach: Cursor was at times relied on more heavily than was ideal, which introduced defects that required intensive manual debugging later. Even so, the time spent correcting those issues was still less than writing and debugging the same code one-handed would have taken.

In summary, all modelling choices, physical interpretation, optimization results, and conclusions remain the author's responsibility. Generated code and text were reviewed, tested where applicable (section 4.2.3, appendix 7), and revised before inclusion. Generative AI sped up routine tasks but did not replace the scientific judgment underlying this thesis.

## 7 Future Work

The sections below list further-work topics that were considered and discussed during this project, and suggest how to move forward and obtain stronger results from analyses built on the framework of this thesis. They extend the discussion in section 6.6–section 6.7 without repeating its conclusions.

### 7.1 Modelling and degradation

- Extend the degradation state beyond nickel migration and interconnect oxidation (table 3.2), or calibrate the fitted rate laws (section 3.2.2) against long-duration experiments at the current densities seen in optimization (section 5.2).
- Replace the fixed agglomeration multiplier  $\lambda_c$  with a dynamic state when operating far from the calibration point (section 3.2.1, section 3.3).
- Include auxiliary preheat power or a thermal-cycling degradation term so that segment-to-segment temperature changes carry an explicit cost (section 6.2, appendix 6).
- Move from a single channel to stack- or plant-scale models with shared thermal and gas-management constraints (section 6.6; cf. [23]).

### 7.2 Objectives and economics

- Implement the risk-aware objective sketched in equation (6.1), or compare the present  $\int R dt + \Psi$  form directly with LCOH- and efficiency-based objectives from the literature (section 6.3; [4, 8, 13]).
- Break objective degeneracies in the delivery case with a minimum-inventory penalty, reserve constraint, or preference for smoother tank utilisation (section 6.6, section 5.3).
- Test sensitivity to  $\kappa_{\text{buy}}$ ,  $\kappa_{\text{sell}}$ ,  $\alpha_\Psi$ , and alternative terminal treatments for tank inventory (section 5.3, section 5.4, figure 5.7).

### 7.3 AC:DC optimization

The short-horizon and periodic-control ideas below connect to electrothermally balanced operation and dynamic SOEC response discussed by Skaftø et al. [24].

- Develop an objective function specifically designed for very short time-horizon optimizations, in which the cell’s mechanical and electrochemical responses cannot be assumed to be instantaneous (section 5.4). This would require accounting for the finite dynamic response of the cell, capturing any lag between input changes and their physical effects.
- Extend the degradation modelling framework to include not only irreversible but also reversible degradation mechanisms (section 3.2.1). By doing so, it becomes possible to study phenomena where the cell’s performance may partially recover under certain operating conditions, enabling a more nuanced optimization of lifetime and efficiency.
- Formulate a single-period control trajectory that is completely unconstrained in shape, allowing the optimization procedure to discover non-standard or unexpected control profiles that might yield improved performance (section 4.2.6). This unconstrained formulation could help identify new strategies that would be otherwise overlooked under more restrictive parameterizations.
- Establish realistic constraints for the periodic control function in order to ensure that the resulting

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schedules are physically feasible and implementable in practice (appendix 6, section 6.4.2). These constraints could include bounds on ramp rates, limits on the magnitude of control variables, or conditions that enforce smooth transitions between periods, reflecting actual system limitations and safety requirements.



## 8 References

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# 1 Deriving PDEs for species balances

Following [21], eqs. [A-3]–[A-5]:

$$\frac{PV}{RT} \frac{\partial x_{\text{H}_2\text{O}}}{\partial t} = \underbrace{A_{int} N_{in, \text{H}_2\text{O}}}_{\text{Inlet}} + \underbrace{A_{cell} N_{a, \text{H}_2\text{O}}}_{\text{Cell contribution}} - \underbrace{A_{int} N_{out, \text{H}_2\text{O}}}_{\text{Outlet}}$$

$$N_{in/out, \text{H}_2\text{O}} = v_{fuel} \mu_{in/out, \text{H}_2\text{O}}$$

$$N_{a, \text{H}_2\text{O}} = \nu_{\text{H}_2\text{O}} \frac{i}{2F}$$

The equations are also required in terms of the concentration  $\mu_{\text{H}_2\text{O}}$ , defined by:

$$\mu_{\text{H}_2\text{O}} = \frac{n_{\text{H}_2\text{O}}}{V}$$

Substituting these relations and the ideal gas law, then discretizing, gives

$$\frac{\partial \mu_{\text{H}_2\text{O}}}{\partial t} = \frac{A_{cell}}{V} \nu_{\text{H}_2\text{O}} \frac{i}{2F} + \frac{A_{int}}{V} v_{fuel} (\mu_{\text{H}_2\text{O}}(z - \Delta z) - \mu_{\text{H}_2\text{O}}(z))$$

With  $A_{cell} = w\Delta z$  and  $V = \Delta z A_{int}$ , isolating the time derivative yields

$$\begin{aligned} \frac{\partial \mu_{\text{H}_2\text{O}}}{\partial t} &= \frac{w\Delta z}{A_{int}\Delta z} \nu_{\text{H}_2\text{O}} \frac{i}{2F} + \frac{A_{int}}{A_{int}\Delta z} v_{fuel} (\mu_{\text{H}_2\text{O}}(z - \Delta z) - \mu_{\text{H}_2\text{O}}(z)) \\ &= \frac{w}{A_{int}} \nu_{\text{H}_2\text{O}} \frac{i}{2F} + v_{fuel} \frac{(\mu_{\text{H}_2\text{O}}(z - \Delta z) - \mu_{\text{H}_2\text{O}}(z))}{\Delta z} \end{aligned}$$

Letting  $\Delta z \rightarrow 0$  yields

$$\frac{\partial \mu_{\text{H}_2\text{O}}}{\partial t} = \frac{w}{A_{int}} \nu_{\text{H}_2\text{O}} \frac{i}{2F} - v_{fuel} \frac{\partial \mu_{\text{H}_2\text{O}}}{\partial z}$$

or, since  $N_{\text{H}_2\text{O}} = v_{fuel} \mu_{\text{H}_2\text{O}}$ ,

$$\frac{\partial \mu_{\text{H}_2\text{O}}}{\partial t} = \frac{w}{A_{int}} \nu_{\text{H}_2\text{O}} \frac{i}{2F} - \frac{\partial N_{\text{H}_2\text{O}}}{\partial z}$$

The same steps for  $\text{H}_2$  and  $\text{O}_2$  give

$$\frac{\partial \mu_i}{\partial t} = \frac{w}{A_{int}} \nu_i \frac{i}{2F} - \frac{\partial N_i}{\partial z}, \quad \text{for } i \in \{\text{H}_2\text{O}, \text{H}_2, \text{O}_2\}$$

with  $\nu_i = (1, -1, -\frac{1}{2})^\top$ .

## 1.1 Unit check (H<sub>2</sub>O)

A unit check on the H<sub>2</sub>O equation gives

$$\frac{\partial \mu_{\text{H}_2\text{O}}}{\partial t} = \frac{w}{A_{int}} \frac{i}{2F} - v_{fuel} \frac{\partial \mu_{\text{H}_2\text{O}}}{\partial z}$$

**Left side units:**  $-\frac{\partial \mu_{\text{H}_2\text{O}}}{\partial t}: [\text{mol} \cdot \text{m}^{-3} \cdot \text{s}^{-1}]$

**Right side units:**  $-\nu_{\text{H}_2\text{O}} \frac{w}{A_{int}} \frac{i}{2F}: -\nu_{\text{H}_2\text{O}}$  is dimensionless.  $-\frac{w}{A_{int}}: [m]/[m^2] = [m^{-1}]$   $-\frac{i}{2F}: [A \cdot \text{m}^{-2}]/[C \cdot \text{mol}^{-1}]$

$$[m^{-1}] \cdot [A \text{ m}^{-2}]/[C \text{ mol}^{-1}] = [A \text{ m}^{-3}]/[C \text{ mol}^{-1}]$$

Next, since  $1 \text{ A} = 1 \text{ C/s}$ :

$$= [C \text{ s}^{-1} \text{ m}^{-3}]/[C \text{ mol}^{-1}] = [\text{mol} \text{ s}^{-1} \text{ m}^{-3}]$$

So the total units are  $[\text{mol} \text{ m}^{-3} \text{ s}^{-1}]$

- $v_{fuel} \frac{\partial \mu_{\text{H}_2\text{O}}}{\partial z}: [m \cdot \text{s}^{-1}] \cdot \frac{[\text{mol} \cdot \text{m}^{-3}]}{[m]} = [\text{mol} \cdot \text{m}^{-3} \cdot \text{s}^{-1}]$  (since  $\partial \mu / \partial z$  has units concentration per length, i.e.  $[\text{mol} \cdot \text{m}^{-3}]/[m] = [\text{mol} \cdot \text{m}^{-4}]$ ).

Both terms on the right match the left-hand side,  $[\text{mol} \cdot \text{m}^{-3} \cdot \text{s}^{-1}]$ .

## 2 Deriving PDEs for energy balances

Starting from the thermal energy equation with  $\mathcal{S} = \{\text{H}_2\text{O}, \text{O}_2, \text{H}_2\}$ :

$$\rho \frac{\partial h}{\partial t} = \frac{\partial p}{\partial t} + \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (h_i N_i) + Q_T$$

With  $\frac{\partial p}{\partial t} = \frac{\partial p}{\partial z} = 0$  and  $h \approx c_p T$ , this becomes

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda_T \nabla T) - \sum_{i \in \mathcal{S}} \nabla \cdot (c_p T N_i) + Q_T$$

Neglecting thermal conductivity,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} c_{p,i} \nabla \cdot (T N_i) + Q_T$$

In one dimension,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} c_{p,i} \frac{\partial T N_i}{\partial z} + Q_T$$

With  $N_i = v_i \mu_i$ ,

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i c_{p,i} \frac{\partial T \mu_i}{\partial z} + Q_T$$

### 2.1 Source term

The volumetric source follows from the cell power,

$$P = i A_{cell} U,$$

which is the current times the cell voltage. Dividing by the volume gives  $Q_T$ :

$$Q_T = \frac{P}{V} = \frac{A_{cell}}{V} i U$$

With  $A_{cell} = w dz$  and  $V = A_{int} dz$ ,

$$Q_T = \frac{w}{A_{int}} i U$$

The full equation is therefore

$$\rho c_p \frac{\partial T}{\partial t} = - \sum_{i \in \mathcal{S}} v_i c_{p,i} \frac{\partial T \mu_i}{\partial z} + \frac{w}{A_{int}} i U$$

## 2.2 Steady-state limit

Expanding the advective term and setting  $\partial_t T = 0$  gives

$$0 = - \sum_{i \in \mathcal{S}} v_i c_{p,i} \left( \frac{\partial T}{\partial z} \mu_i + T \frac{\partial \mu_i}{\partial z} \right) + \frac{w}{A_{int}} i U$$

Substituting  $h_i = c_{p,i} T$  and the steady-state species balance  $v_i \frac{\partial \mu_i}{\partial z} = \frac{w}{A_{int}} \nu_i \frac{i}{n_e F}$  yields

$$0 = - \sum_{i \in \mathcal{S}} \left( v_i c_{p,i} \frac{\partial T}{\partial z} \mu_i + \frac{w}{A_{int}} \frac{i}{n_e F} \nu_i h_i \right) + \frac{w}{A_{int}} i U$$

Factoring out the common terms,

$$0 = - \frac{\partial T}{\partial z} \sum_{i \in \mathcal{S}} v_i c_{p,i} \mu_i - i \frac{w}{A_{int}} \sum_{i \in \mathcal{S}} \nu_i h_i + i \frac{w}{A_{int}} U$$

With

$$\sum_{i \in \mathcal{S}} v_i c_{p,i} \mu_i = C_p, \quad U_{th} = \frac{1}{n_e F} \sum_{i \in \mathcal{S}} \nu_i h_i$$

this gives

$$0 = i \frac{w}{A_{int}} (U - U_{th}) - C_p \frac{\partial T}{\partial z}$$

# 3 Computing the finite-volume discretization

## 3.1 Species balances

From equation (3.5), equation (3.6) and equation (3.7), integration over a cell volume gives

$$\int_{\Omega_k} \frac{\partial \mu_i}{\partial t} dV = \int_{\Omega_k} \frac{w}{A_{int}} N_{c,i} dV - \int_{\Omega_k} \frac{\partial N_i}{\partial z} dV$$

The time derivative can be taken outside the integral, and integrating the concentration over a volume gives the moles in that volume:

$$\int_{\Omega_k} \frac{\partial \mu_i}{\partial t} dV = \partial_t \int_{\Omega_k} \mu_i dV = \frac{\partial n_i}{\partial t} = \dot{n}_i$$

Using the approximation in section 4.2.2 with  $V_k = A_{int} \Delta z$ ,

$$\int_{\Omega_k} \frac{w}{A_{int}} N_{c,i} dV \approx \frac{w}{A_{int}} N_{c,i}(z_k) V_k = w N_{c,i}(z_k) \Delta z$$

For the flux term, integration over  $z$  is performed first, then over the cross-sectional area:

$$\begin{aligned} \int_{A_{int}} \int_{z_{k-\frac{1}{2}}}^{z_{k+\frac{1}{2}}} \frac{\partial N_i}{\partial z} dz dA &\approx \int_{A_{int}} \left( N_i(z_k) - N_i(z_{k-1}) \right) dA \\ &= \left( N_i(z_k) - N_i(z_{k-1}) \right) A_{int} \end{aligned}$$

Each cell therefore satisfies

$$|\Omega_k| \frac{\partial \mu_i}{\partial t}(z_k) = \dot{n}_i(z_k) \approx w N_{c,i} \Delta z - A_{int} (N_i(z_k) - N_i(z_{k-1}))$$

## 3.2 Energy balance

The steady-state form is discretized first, then the transient term.

### 3.2.1 Steady state

Starting from equation (3.18):

$$0 = i \frac{w}{A_{int}} (U - U_{th}) - C_p \frac{\partial T}{\partial z}$$



The same procedure as for the species balances gives

$$0 = \int_{\Omega_k} i \frac{w}{A_{int}} (U - U_{th}) - \int_{\Omega_k} C_p \frac{\partial T}{\partial z}$$

As before,

$$\int_{\Omega_k} C_p \frac{\partial T}{\partial z} dV \approx C_p (T(z_k) - T(z_{k-1})) A_{int}$$

Collecting terms,

$$0 \approx iw(U - U_{th})\Delta z - C_p(T(z_k) - T(z_{k-1}))A_{int}$$

### 3.2.2 Transient term

For the time derivative,

$$\int_{\Omega_k} \rho c_p \frac{\partial T}{\partial t} = \int_{\Omega_k} Q_T - v \sum_{i \in \mathcal{S}} \int_{\Omega_k} c_{p,i} \frac{\partial T \mu_i}{\partial z}$$

On the left-hand side,  $\rho$  integrated over the cell volume is the cell mass  $m_k$ , so

$$\int_{\Omega_k} \rho c_p \frac{\partial T}{\partial t} \approx m_k c_p \frac{\partial T}{\partial t}(z_k)$$

The source term follows the same pattern as above,  $\int_{\Omega_k} Q_T = Q_T(z_k) A_{int} \Delta z$ . For the advective flux, one species is shown explicitly; the others are analogous:

$$\int_{\Omega_k} c_{p,i} \frac{\partial T \mu_i}{\partial z} \approx c_{p,i} (T(z_k) \mu_i(z_k) - T(z_{k-1}) \mu_i(z_{k-1})) A_{int}$$

The discrete energy balance is therefore

$$m_k c_p \frac{\partial T}{\partial t}(z_k) \approx Q_T(z_k) A_{int} \Delta z - \sum_{i \in \mathcal{S}} v_i c_{p,i} (T(z_k) \mu_i(z_k) - T(z_{k-1}) \mu_i(z_{k-1})) A_{int}$$

## 4 Relevant gradients for the numerical solver

### 4.1 Molar balance

$$\text{Jacobian}_\mu[x] = \begin{bmatrix} \frac{1}{\mu^f} & -\frac{1}{\mu^f} & 0 \\ -\frac{1}{\mu^f} & \frac{1}{\mu^f} & 0 \\ 0 & 0 & \frac{\mu_{in,N_2}}{(\mu_{in,N_2} + \mu_{O_2})} \end{bmatrix}$$

Then

$$\nabla_\mu f(x) = (\text{Jacobian}_\mu[x])^\top \nabla_x f$$

#### 4.1.1 Nernst Equation

Gradient for molar fractions:  $\nabla_x$

$$\nabla_x \eta_{nernst} = \frac{RT}{2F} \begin{bmatrix} -\frac{1}{x_{H_2O}} \\ \frac{1}{x_{H_2}} \\ \frac{1}{2x_{O_2}} \end{bmatrix}$$

Gradient for temperature:  $\nabla_T$

$$\nabla_T \eta_{nernst} = \frac{\nabla_T(\Delta G)}{2F} + \frac{R}{2F} \log \left\{ \frac{x_{H_2} \sqrt{x_{H_2}}}{x_{H_2O} \sqrt{x_0}} \right\}$$

$$\nabla_T(\Delta G) = -0.0062T + -49.119$$

#### 4.1.2 Activation overpotential

Gradient for molar fractions:  $\nabla_x$

$$\nabla_x (\text{Butler-Volmer}_\zeta) = \nabla_x (i_0^\zeta) \left( \exp \left[ \alpha_\zeta \frac{n_e F}{RT} \eta_{act}^\zeta \right] - \exp \left[ - (1 - \alpha_\zeta) \frac{n_e F}{RT} \eta_{act}^\zeta \right] \right)$$

$$\nabla_x (i_0^a) = \gamma_a T \exp \left\{ -\frac{E_{act}^a}{RT} \right\} \begin{bmatrix} 0 \\ 0 \\ m_{O_2} \left( \frac{x_{O_2}}{x_0} \right)^{m_{O_2}-1} \end{bmatrix}$$

$$\nabla_x (i_0^f) = \lambda_f \gamma_f T \exp \left\{ -\frac{E_{act}^f}{RT} \right\} \begin{bmatrix} m_{H_2} \left( \frac{x_{H_2O}}{x_0} \right)^{m_{H_2O}-1} \left( \frac{x_{H_2}}{x_0} \right)^{m_{H_2}} \\ m_{H_2O} \left( \frac{x_{H_2O}}{x_0} \right)^{m_{H_2O}} \left( \frac{x_{H_2}}{x_0} \right)^{m_{H_2}-1} \\ 0 \end{bmatrix}$$

**Gradient for temperature:**  $\nabla_T$ 

$$\begin{aligned}\nabla_T(\text{Butler-Volmer}_\zeta) &= \nabla_T(i_0^\zeta) \left( \exp\left[\alpha_\zeta \frac{n_e F}{RT} \eta_{act}^\zeta\right] - \exp\left[-(1 - \alpha_\zeta) \frac{n_e F}{RT} \eta_{act}^\zeta\right] \right) \\ &\quad + i_0^\zeta \left( -\alpha_\zeta \frac{n_e F}{RT^2} \eta_{act}^\zeta \exp\left[\alpha_\zeta \frac{n_e F}{RT} \eta_{act}^\zeta\right] - (1 - \alpha_\zeta) \frac{n_e F}{RT^2} \eta_{act}^\zeta \exp\left[-(1 - \alpha_\zeta) \frac{n_e F}{RT} \eta_{act}^\zeta\right] \right) \\ \nabla_T(i_0^\zeta) &= \left(1 - \frac{E_{act}^\zeta}{RT}\right) \lambda_\zeta \gamma_\zeta \exp\left\{-\frac{E_{act}^\zeta}{RT}\right\} \prod_{i \in \mathcal{S}} \left(\frac{x_i}{x_0}\right)^{m_i}\end{aligned}$$

**Gradient for activation overpotential:**  $\nabla_{\eta_{act}^\zeta}$ 

$$\nabla_{\eta_{act}^\zeta}(\text{Butler-Volmer}_\zeta) = i_0^\zeta \left( \alpha_\zeta \frac{n_e F}{RT} \exp\left[\alpha_\zeta \frac{n_e F}{RT} \eta_{act}^\zeta\right] + (1 - \alpha_\zeta) \frac{n_e F}{RT} \exp\left[-(1 - \alpha_\zeta) \frac{n_e F}{RT} \eta_{act}^\zeta\right] \right)$$

## 4.2 Degradation

### 4.2.1 Ohmic degradation

**Gradient for molar fractions:**  $\nabla_x$ 

$$\nabla_x R_{ohm} = \mathbf{0}$$

**Gradient for temperature:**  $\nabla_T$ 

$$\nabla_T R_{ohm} = \left(1 - \frac{E_{a,ohm}}{RT}\right) \frac{1}{B_{ohm}} \exp\left\{\frac{E_{a,ohm}}{RT}\right\} + \sum_{d \in \mathcal{D}_{ohm}} \nabla_T R_d$$

$$\nabla_T R_{\text{Ni-mi}} = d_{\text{Ni-mi}} \frac{T - E_{\text{act,elec}}/R}{\sigma_{\text{elec,o}} T \exp\left\{-\frac{E_{\text{act,elec}}}{RT}\right\}}$$

## 4.3 Temperature equations

### 4.3.1 Specific heat

**Gradient for molar fractions:**  $\nabla_\mu$ 

$$\nabla_\mu C_p = \begin{bmatrix} c_{p,\text{H}_2\text{O}} \\ c_{p,\text{H}_2} \\ c_{p,\text{O}_2} \end{bmatrix}$$

**Gradient for temperature:**  $\nabla_T$ 

$$\nabla_T C_p = \sum_{i \in \mathcal{S}} \mu_i \nabla_T c_{p,i}$$

### 4.3.2 Thermaneutral voltage

**Gradient for molar fractions:**  $\nabla_\mu$ 

$$\nabla_\mu U_{th} = \mathbf{0}$$

---

**Gradient for temperature:**  $\nabla_T$

$$\nabla_T U_{th} = \frac{1}{n_e F} \sum_{i \in \mathcal{S}} \nu_i \nabla_T h_i$$

## 5 NLP Gradients

This appendix collects the gradients used in the NLP formulation of equation (4.1). The discrete chain rule for the stage model is stated first, followed by the partial derivatives that enter it and the linear systems for sensitivities of the spatial FVM model and the slow trajectory in time.

### 5.1 Discrete chain rule for the stage model

Between shooting stages, the integrated objective contribution  $\phi$  and the degradation variables  $d$  satisfy a discrete sensitivity chain. For a scalar stage index  $i$  and a control component  $u_k$ ,

$$\frac{d}{du_k} \phi_{i+1} = \nabla_{u_i} \dot{\phi} \cdot \nabla_{u_k} u_i + \nabla_{d_i} \dot{\phi} \frac{d}{du_k} d_i + \frac{d}{du_k} \phi_i,$$

with the analogous relation for the degradation state,

$$\frac{d}{du_k} d_{i+1} = (\nabla_{u_i} \dot{d}_i) \nabla_{u_k} u_i + (\nabla_{d_i} \dot{d}_i) \frac{d}{du_k} d_i + \dots,$$

where the trailing terms collect any additional dependence of the stage map on  $u_k$  (for example through implicit stage equations).

The partial gradients of the instantaneous rates are

$$\begin{aligned} \nabla_u \dot{\phi} &= c_{H_2} \nabla_u \frac{\partial n_{H_2}}{\partial t} - c_P (u \nabla_u I - I), \\ \nabla_d \dot{\phi} &= c_{H_2} \nabla_d \frac{\partial n_{H_2}}{\partial t} - c_P u \nabla_d I - \left[ c_{i, \text{utilization}} \frac{1}{d_{i, \text{critical}}} \right]_{i \in \mathcal{D}_{\text{fail}}}, \\ \nabla_\phi \dot{\phi} &= 0, \\ \nabla_u \dot{d} &= c_2 \nabla_u i + c_3 \nabla_u T + c_4 (\nabla_u i T + i \nabla_u T), \\ \nabla_d \dot{d} &= c_2 \nabla_d i + c_3 \nabla_d T + c_4 (\nabla_d i T + i \nabla_d T), \\ \nabla_\phi \dot{d} &= 0. \end{aligned}$$

Note that  $\nabla_\zeta I = \frac{A_{\text{cell}}}{N_{\text{cell}}} \sum_{j=1}^{N_{\text{cell}}} \nabla_\zeta i_j$  for  $\zeta \in \{u, d, \phi, \dots\}$ .

The remaining ingredients are the gradients of the current density with respect to cell voltage and local degradation state; these are given below. Further electrochemical partials are listed in appendix 4.

### 5.2 Gradient with respect to cell voltage: $\nabla_u i$

The current is expressed in terms of fuel- and air-side activation branches in the same notation as appendix 4. With scalars  $\nabla_{\eta_{\text{act}}^f}$  (Butler-Volmer<sub>f</sub>) and  $\nabla_{\eta_{\text{act}}^a}$  (Butler-Volmer<sub>a</sub>) denoting  $\partial i / \partial \eta_{\text{act}}^f$  and  $\partial i / \partial \eta_{\text{act}}^a$  at the operating point,

$$\nabla_u i = \frac{1}{R_{\text{ohm}} + \frac{1}{\nabla_{\eta_{\text{act}}^f}(\text{Butler-Volmer}_f)} + \frac{1}{\nabla_{\eta_{\text{act}}^a}(\text{Butler-Volmer}_a)}}.$$

### 5.3 Gradient with respect to degradation: $\nabla_d i$

Similarly, with  $\nabla_d R_{\text{ohm}}$  the partial of the ohmic resistance with respect to the local degradation variable,

$$\nabla_d i = -i \frac{\nabla_d R_{\text{ohm}}}{R_{\text{ohm}} + \frac{1}{\nabla_{\eta_{\text{act}}^f}(\text{Butler-Volmer}_f)} + \frac{1}{\nabla_{\eta_{\text{act}}^a}(\text{Butler-Volmer}_a)}}.$$

### 5.4 Sensitivity of the spatial FVM discretization

At each time level the converged spatial state satisfies a residual of flux form. With cell index  $k$  along the stack, let  $Q(u_k)$  denote the cellwise source residual and  $F(u_k)$  the numerical flux between neighbours. The source term is scaled by the cell thickness  $\Delta z$  and the flux difference by the interface area  $A_{\text{int}}$ , so the semi-discrete steady relation is

$$0 = \Delta z Q(u_k) - A_{\text{int}} (F(u_k) - F(u_{k-1})).$$

Differentiating with respect to a parameter  $p$  (a control coefficient, a boundary value, etc.) and writing  $\partial u_k / \partial p$  for the sensitivity of the converged state gives the familiar block recursion

$$\begin{aligned} & \left( \Delta z \frac{\partial Q}{\partial u_k}(u_k) - A_{\text{int}} \frac{\partial F}{\partial u_k}(u_k) \right) \frac{\partial u_k}{\partial p} + A_{\text{int}} \frac{\partial F}{\partial u_{k-1}}(u_{k-1}) \frac{\partial u_{k-1}}{\partial p} \\ & = A_{\text{int}} \left( \frac{\partial F}{\partial p}(u_k) - \frac{\partial F}{\partial p}(u_{k-1}) \right) - \Delta z \frac{\partial Q}{\partial p}(u_k). \end{aligned}$$

Stacking all cells, this is the linear system

$$J_{\text{fvm}} \frac{\partial \mathbf{u}}{\partial p} = b_{\text{fvm}}^{(p)},$$

with the block-lower-bidiagonal Jacobian and right-hand side

$$J_{\text{fvm}} = \begin{bmatrix} \Delta_1 & & & & \\ A_{\text{int}} \frac{\partial F}{\partial u_1}(u_1) & \Delta_2 & & & \\ & \ddots & & \ddots & \\ & & A_{\text{int}} \frac{\partial F}{\partial u_{N-1}}(u_{N-1}) & \Delta_N & \end{bmatrix},$$

with  $\Delta_k = \Delta z \frac{\partial Q}{\partial u_k}(u_k) - A_{\text{int}} \frac{\partial F}{\partial u_k}(u_k),$

$$b_{\text{fvm}}^{(p)} = \begin{bmatrix} A_{\text{int}} \left( \frac{\partial F}{\partial p}(u_1) - \frac{\partial F}{\partial p}(u_{\text{in}}) \right) - \Delta z \frac{\partial Q}{\partial p}(u_1) \\ A_{\text{int}} \left( \frac{\partial F}{\partial p}(u_2) - \frac{\partial F}{\partial p}(u_1) \right) - \Delta z \frac{\partial Q}{\partial p}(u_2) \\ \vdots \\ A_{\text{int}} \left( \frac{\partial F}{\partial p}(u_N) - \frac{\partial F}{\partial p}(u_{N-1}) \right) - \Delta z \frac{\partial Q}{\partial p}(u_N) \end{bmatrix}$$

The same block structure matches the flux assembly used in appendix 3.

## 5.5 Sensitivity of the ODE system

For this section the problem is written in general form:

$$\begin{aligned} \max_u \quad & \left\{ \text{Objective} = \int_0^T \Phi(X, d, u) dt + \Psi(X, d, u) \right\} \\ \text{s.t.} \quad & 0 = R_{\text{fvm}}(X, d, u) \\ & \dot{d} = f(X, d, u) \end{aligned}$$

The method of adjoint gradients [22], [9] is used, with the following discretization ( $t_0 = 0 < t_1 \dots t_i \dots t_N = T$ ):

$$\begin{aligned} X(t_i) &\approx X_i, \quad d(t_i) \approx d_i, \quad u(t_i) \approx u_i. \\ R_{\text{fvm},i} &= R_{\text{fvm}}(X_{i+1}, d_{i+1}, u_i), \\ D_i &= d_{i+1} - f(X_{i+1}, d_{i+1}, u_i) \Delta t - d_i \\ \Phi_i &= \Phi(X_{i+1}, d_{i+1}, u_i) \Delta t \approx \int_{t_i}^{t_{i+1}} \Phi(X, d, u) dt \end{aligned}$$

The combined residual for the FVM constraints and the ODE defect is then:

$$R_i = \begin{bmatrix} D_i \\ R_{\text{fvm},i} \end{bmatrix}$$

The state variables are combined as

$$w_i = \begin{bmatrix} X_i \\ d_i \end{bmatrix}$$

The Jacobian of the residual with respect to the internal and control variables is

$$\begin{aligned} \frac{\partial R_i}{\partial w_i} &= \begin{bmatrix} \mathbf{0} & \frac{\partial D_i}{\partial d_i} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} & \frac{\partial R_i}{\partial w_{i+1}} &= \begin{bmatrix} \frac{\partial D_i}{\partial X_{i+1}} & \frac{\partial D_i}{\partial d_{i+1}} \\ \frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}} & \frac{\partial R_{\text{fvm},i}}{\partial d_{i+1}} \end{bmatrix} & \frac{\partial R_i}{\partial u_i} &= \begin{bmatrix} \frac{\partial D_i}{\partial u_i} \\ \frac{\partial R_{\text{fvm},i}}{\partial u_i} \end{bmatrix} \\ & & \frac{\partial \Phi_i}{\partial w_{i+1}} &= \begin{bmatrix} \frac{\partial \Phi_i}{\partial X_{i+1}} \\ \frac{\partial \Phi_i}{\partial d_{i+1}} \end{bmatrix} & & \frac{\partial \Phi_i}{\partial u_i} \end{aligned}$$

For implementation these blocks are spelled out in more detail:

$$\begin{aligned} \frac{\partial R_i}{\partial w_i} &= \begin{bmatrix} \mathbf{0} & -\mathbf{I} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \\ \frac{\partial R_i}{\partial w_{i+1}} &= \begin{bmatrix} -\Delta t \frac{\partial f}{\partial X_{i+1}}(X_{i+1}, d_{i+1}, u_i) & \mathbf{I} - \Delta t \frac{\partial f}{\partial d_{i+1}}(X_{i+1}, d_{i+1}, u_i) \\ \frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}}(X_{i+1}, d_{i+1}, u_i) & \frac{\partial R_{\text{fvm},i}}{\partial d_{i+1}}(X_{i+1}, d_{i+1}, u_i) \end{bmatrix} \\ \frac{\partial R_i}{\partial u_i} &= \begin{bmatrix} -\Delta t \frac{\partial f}{\partial u_i}(X_{i+1}, d_{i+1}, u_i) \\ \frac{\partial R_{\text{fvm},i}}{\partial u_i}(X_{i+1}, d_{i+1}, u_i) \end{bmatrix} \end{aligned}$$

The blocks  $\partial R_{\text{fvm},i} / \partial X_{i+1}$  coincide with  $J_{\text{fvm}}$  from the spatial sensitivity section (same flux assembly). At each time level:

$$\frac{\partial R_{\text{fvm},i}}{\partial X_{i+1}} = J_{\text{fvm}} \quad \frac{\partial R_{\text{fvm},i}}{\partial u_i} = \begin{bmatrix} \left| \begin{array}{c} b_{\text{fvm}}^{(u_1^{(U)})} \\ b_{\text{fvm}}^{(u_i^{(T_0)})} \end{array} \right| \end{bmatrix} \quad \frac{\partial R_{\text{fvm},i}}{\partial d_i} = b_{\text{fvm}}^{(d)}$$

With these blocks, the objective gradient with respect to  $u_i$  is

$$\nabla_{u_i} \text{Objective} = \frac{\partial \Phi_i}{\partial u_i} + \left( \frac{\partial R_i}{\partial u_i} \right)^\top \lambda_{i+1} \quad \text{for } i = 1, \dots, N$$

The adjoint variables  $\Lambda = (\lambda_k)_k$  satisfy

$$\begin{aligned} \left( \frac{\partial R_{N-1}}{\partial w_N} \right)^\top \lambda_N &= -\frac{\partial \Phi_{N-1}}{\partial w_N} - \frac{\partial \Psi}{\partial w_N}, \\ \left( \frac{\partial R_{i-1}}{\partial w_i} \right)^\top \lambda_i + \left( \frac{\partial R_i}{\partial w_i} \right)^\top \lambda_{i+1} &= -\frac{\partial \Phi_{i-1}}{\partial w_i}, \quad i = 1, \dots, N-1. \end{aligned}$$

Equivalently, one large block system can be assembled for  $\Lambda$ .



$$\begin{bmatrix} \frac{\partial R_0}{\partial w_1} & & & & \\ \frac{\partial R_1}{\partial w_1} & \frac{\partial R_1}{\partial w_2} & & & \\ & \ddots & \ddots & & \\ & & \frac{\partial R_{N-2}}{\partial w_{N-1}} & & \\ & & \frac{\partial R_{N-1}}{\partial w_{N-1}} & \frac{\partial R_{N-1}}{\partial w_N} & \end{bmatrix}^\top \Lambda = - \begin{bmatrix} \frac{\partial \Phi_0}{\partial w_1} \\ \frac{\partial \Phi_1}{\partial w_2} \\ \vdots \\ \frac{\partial \Phi_{N-2}}{\partial w_{N-1}} \\ \frac{\partial \Phi_{N-1}}{\partial w_N} + \frac{\partial \Psi}{\partial w_N} \end{bmatrix}$$

## 6 Squared continuity penalty

Piecewise-constant controls on  $N_t$  segments can introduce jumps between adjacent intervals. When both voltage and inlet temperature are optimized segment-wise, the optimizer may exploit those jumps unless they are penalized. The implementation adds an optional soft penalty to the running objective, proportional to the sum of squared differences between neighbouring segment values.

For segment index  $k = 0, \dots, N_t - 2$ , let  $U_k$  and  $T_{\text{in},k}$  denote the piecewise-constant controls on segment  $k$ . With non-negative weights  $w_U$  and  $w_T$  (CLI: `--continuity-weight-u`, `--continuity-weight-T`), the penalty is

$$P_{\text{cont}} = w_U \sum_{k=0}^{N_t-2} (U_{k+1} - U_k)^2 + w_T \sum_{k=0}^{N_t-2} (T_{\text{in},k+1} - T_{\text{in},k})^2.$$

It is added to the running economic cost before integration over the horizon. Setting  $w_U = w_T = 0$  disables the term (the default in table 5.1).

In **potentiostatic** operation ( $U_k \equiv U$ ), voltage is a single decision variable rather than a length- $N_t$  profile, so adjacent voltage differences vanish and the  $w_U$  term contributes nothing. Only the inlet-temperature sum is active in that mode.

The penalty is smooth and cheap to evaluate; its gradient with respect to the control knots is implemented analytically in the NLP (`code/src/scripts/nlp/continuity.py`). Saved run metadata reports  $P_{\text{cont}}$ ,  $P_U$ , and  $P_{T_{\text{in}}}$  as the two sums in appendix 6.

## 7 Gradient verification plots

The NLP solver uses adjoint gradients assembled from the partial derivatives in appendix 4 and appendix 5. Before running the optimization cases in section 5.1–section 5.4, analytical derivatives were checked against central finite-difference (FD) approximations. Standalone scripts in `code/src/standalone/` generate the figures below; additional component-level checks exist in `code/figures/testers/gradients_test/` but are omitted here.

### 7.1 End-to-end NLP objective

Figure 7.1 sweeps uniform cell voltage and inlet temperature on a single-shooting problem with  $N_t = 100$  time segments ( $\Delta t = 4320$  s), exercising the full adjoint chain over a long horizon. The adjoint totals  $\sum_j \partial f / \partial U_j$  and  $\sum_j \partial f / \partial T_{in,j}$  match the FD curves over the operating range.

### 7.2 Spatial FVM Jacobian

The steady-state cell residual  $r(\mu_{H_2}, \mu_{O_2}, T)$  must be differentiated when assembling the block-bidiagonal system in appendix 5. Figure 7.2 compares each Jacobian entry to a secant FD estimate; the curves agree on the test grid.

### 7.3 Electrochemical coupling

Current density enters both the degradation rates and the economic objective. Figure 7.3 and figure 7.4 verify  $\nabla_U i$  and  $\nabla_d i$  from appendix 5. Figure 7.5 checks the fuel-side Butler–Volmer branch and its partials w.r.t. temperature, composition, and activation overpotential.

### 7.4 Instantaneous economic rate

Figure 7.6 checks partial derivatives of the running economic rate  $\dot{\phi}$  w.r.t. outlet chemical potentials and the stack-outlet temperature that feed the stage objective in appendix 5.

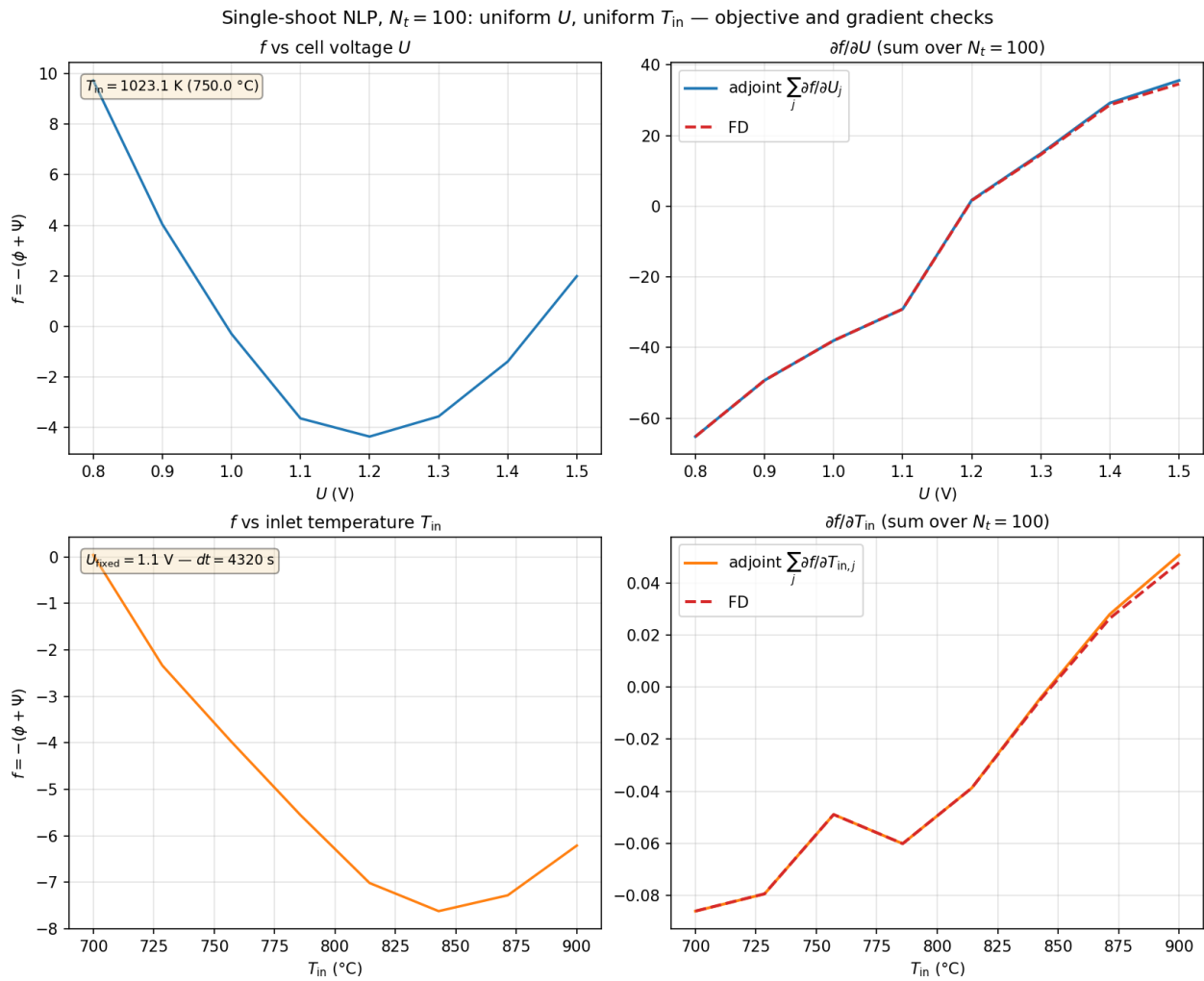


Figure 7.1: Single-shooting NLP gradient check at  $N_t = 100$ : Objective  $f = -(\phi + \psi)$  and its derivatives w.r.t. uniform cell voltage  $U$  (top) and inlet temperature  $T_{in}$  (bottom). Solid lines: Adjoint; dashed: Central FD with all segment controls perturbed together.

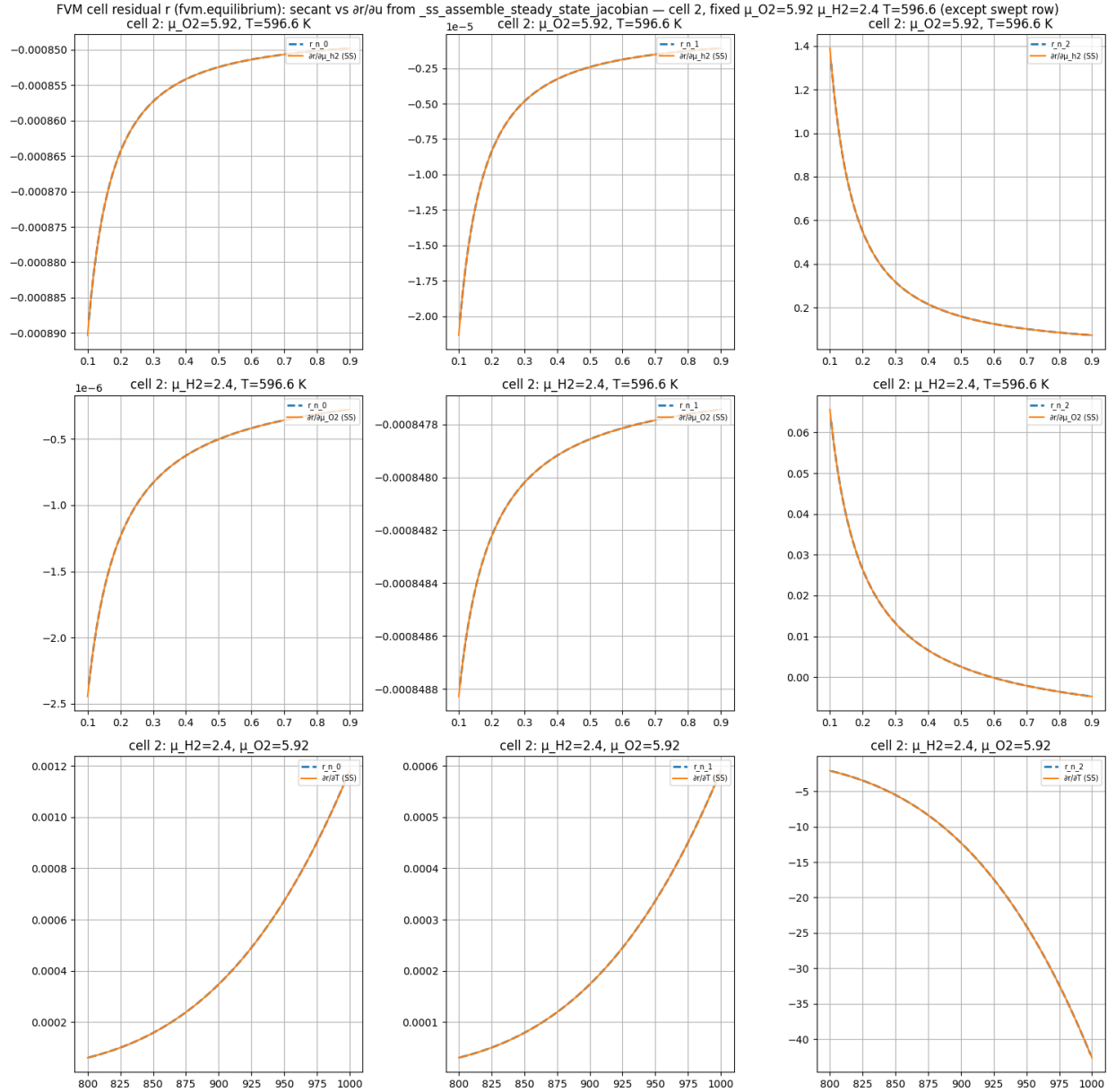


Figure 7.2: Steady-state FVM residual Jacobian vs. secant FD for the three residual components  $r_0$ ,  $r_1$ ,  $r_2$  and inputs  $\mu_{H_2}$ ,  $\mu_{O_2}$ , and  $T$ . Dashed: Numerical secant; solid: Analytical steady-state Jacobian.

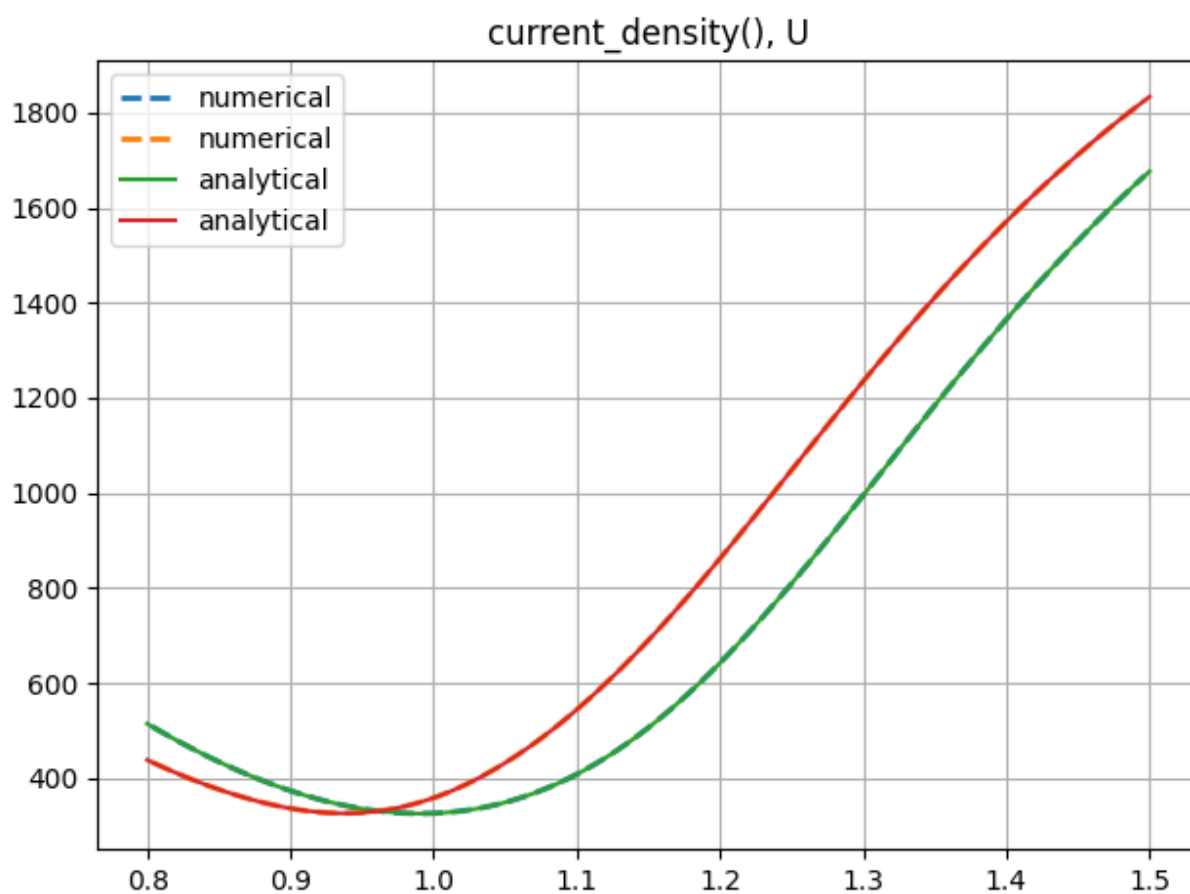


Figure 7.3: Analytical vs. numerical gradient of cell current density w.r.t. cell voltage  $U$ .

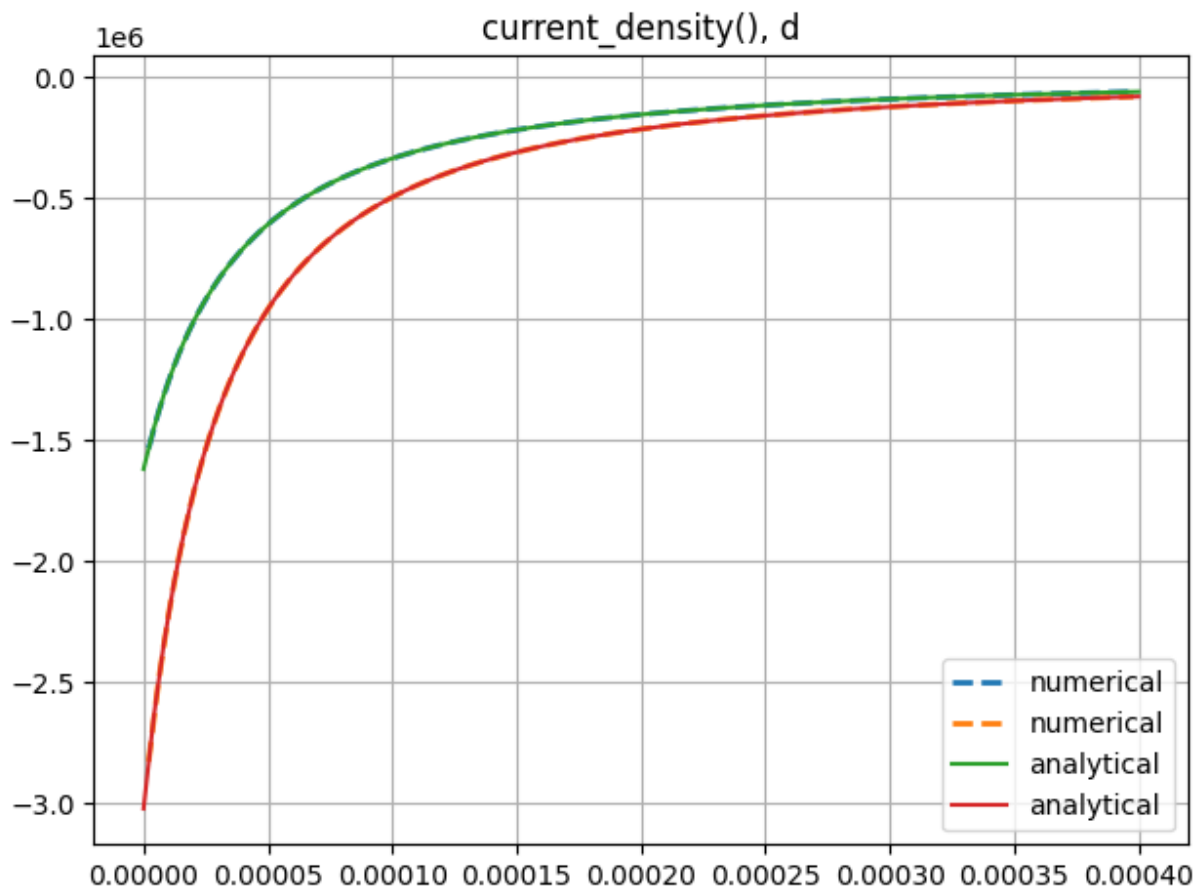


Figure 7.4: Analytical vs. numerical gradient of cell current density w.r.t. local degradation state  $d$ .

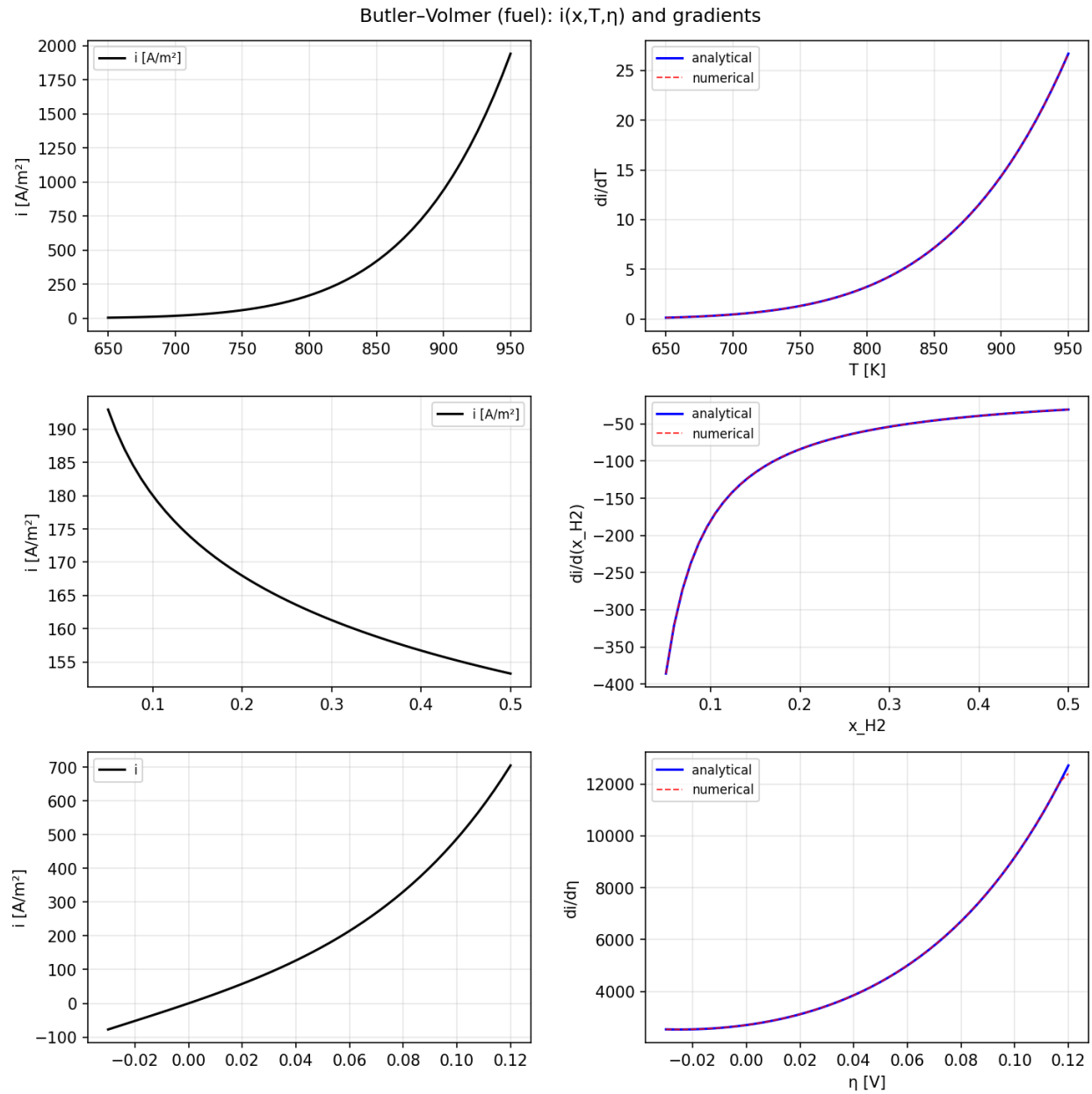


Figure 7.5: Fuel-side Butler–Volmer current  $i(x, T, \eta)$  and gradients w.r.t.  $T$ ,  $x_{H_2}$ , and activation overpotential  $\eta$ . Left column:  $i$ ; right column: Analytical (solid) vs. numerical (dashed) partial derivatives.



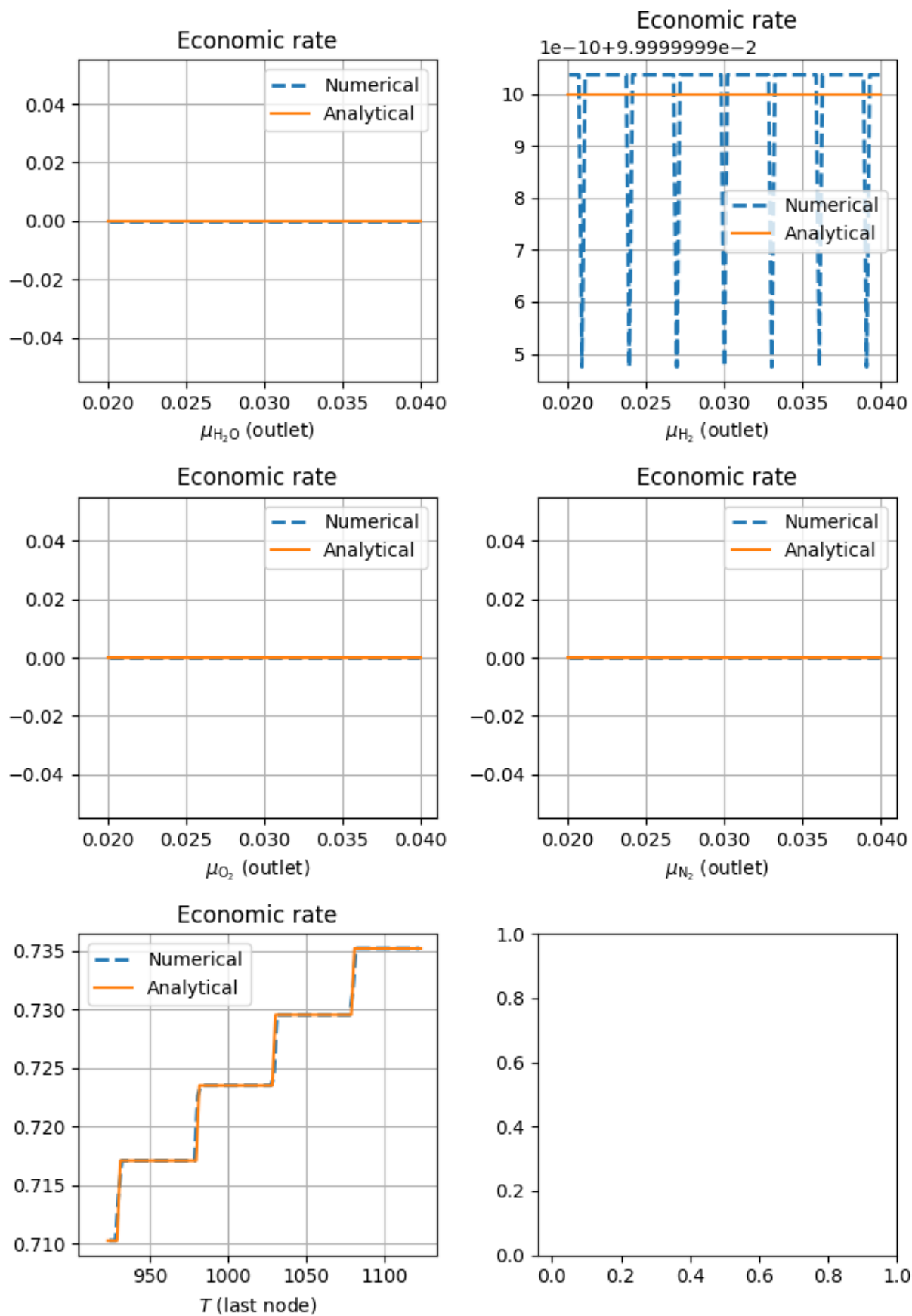


Figure 7.6: Analytical vs. numerical gradients of the instantaneous economic rate w.r.t. outlet species chemical potentials and the last-node temperature  $T$ .